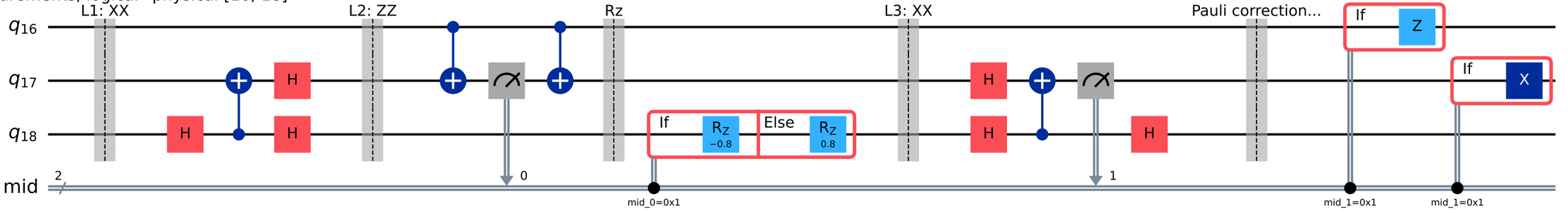
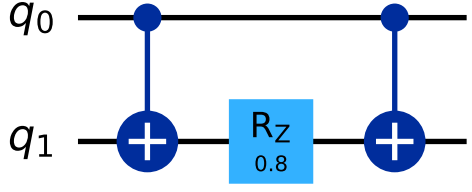
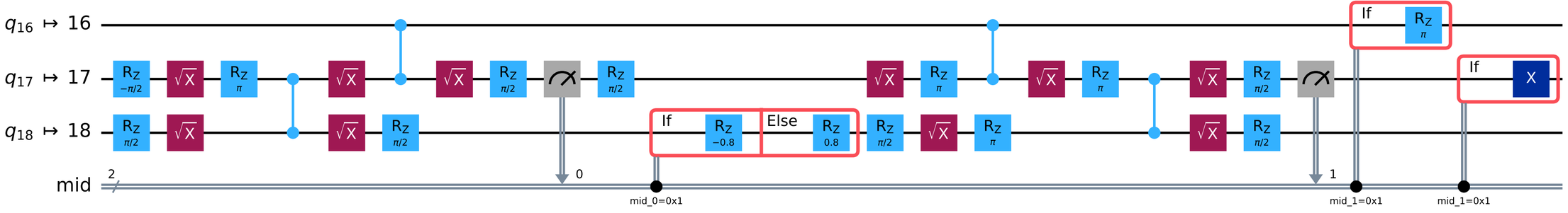
[Z2] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 4, 2 mid-circuit measurements, logical→physical [16, 18]



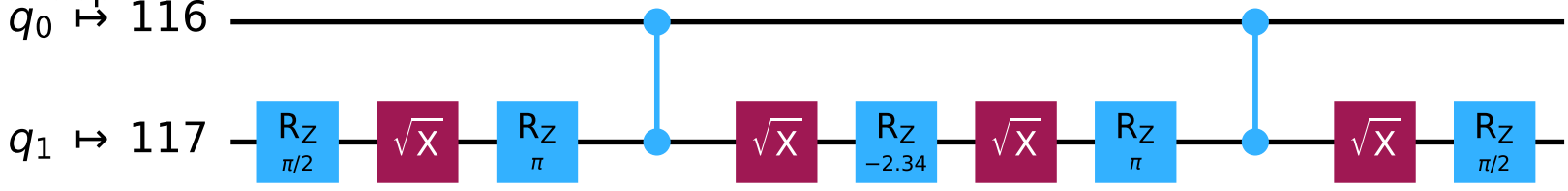
[Z2] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 2



[Z2] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:12 sx:10 cz:4 if_else:3 measure:2)
exp(-i θ P), θ=0.4: P = Z0 Z1 (2 logical qubits)
2q depth 4, total depth 21

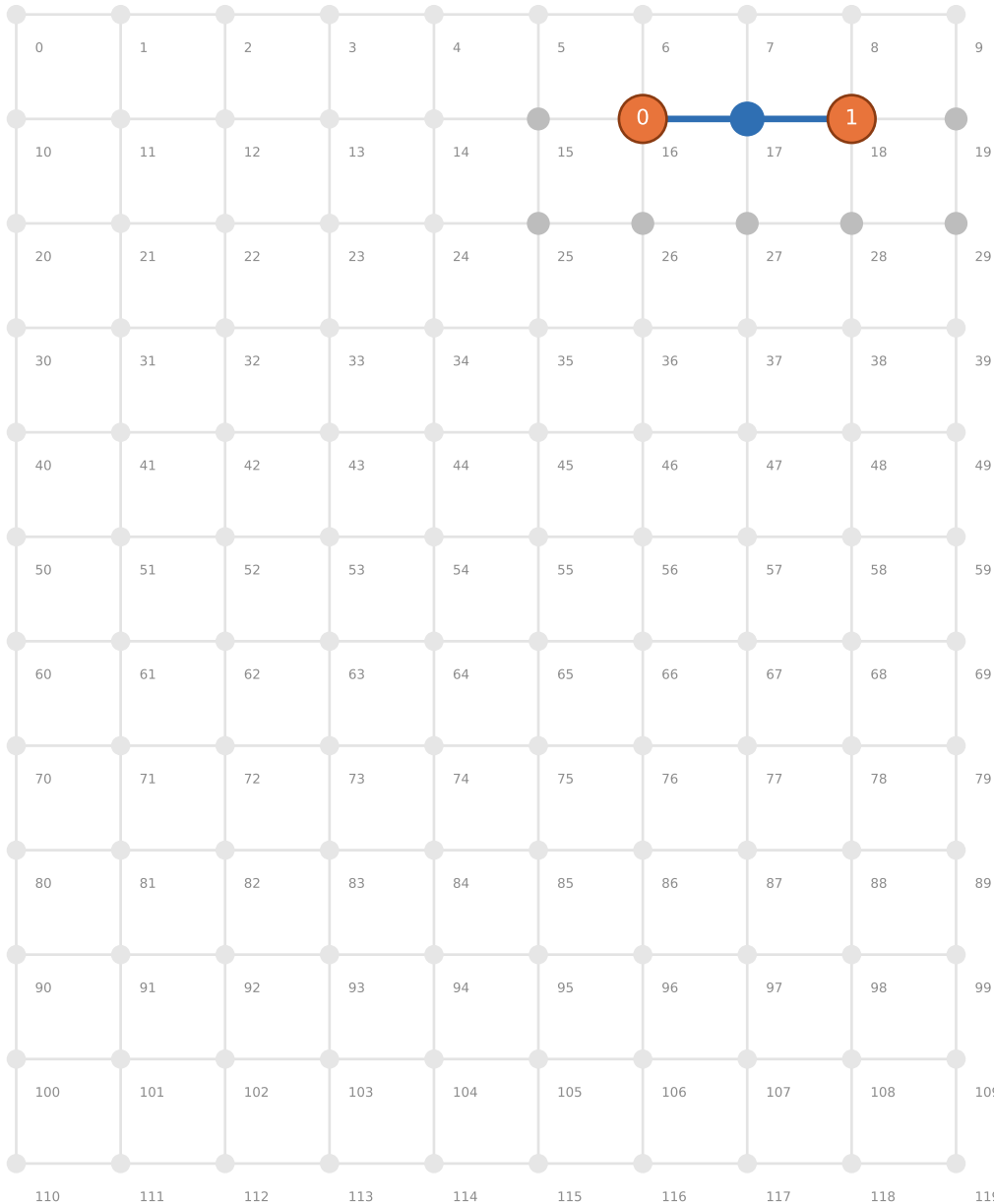


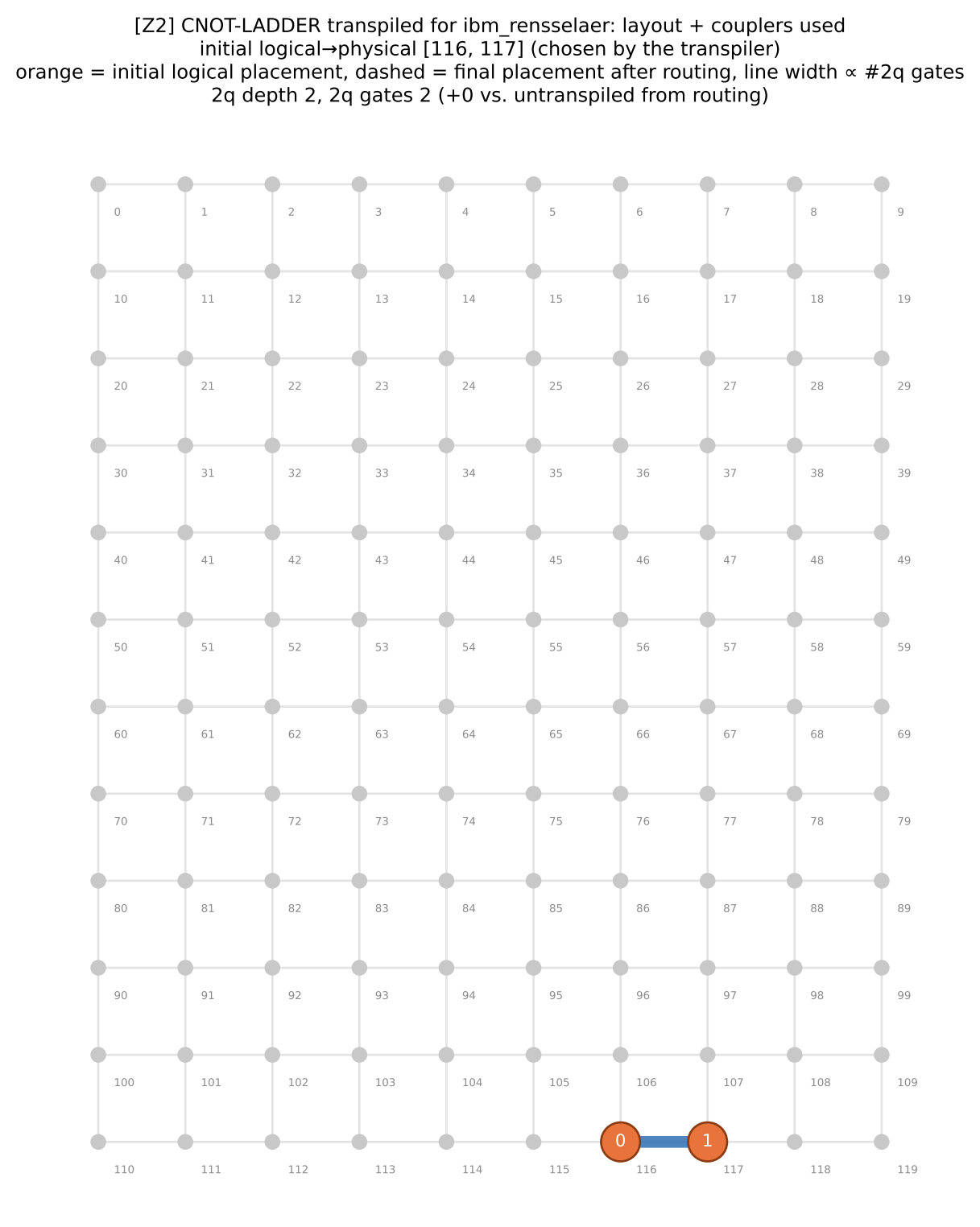
[Z2] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:5 sx:4 cz:2)
Global Phase: 0
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 2, total depth 11

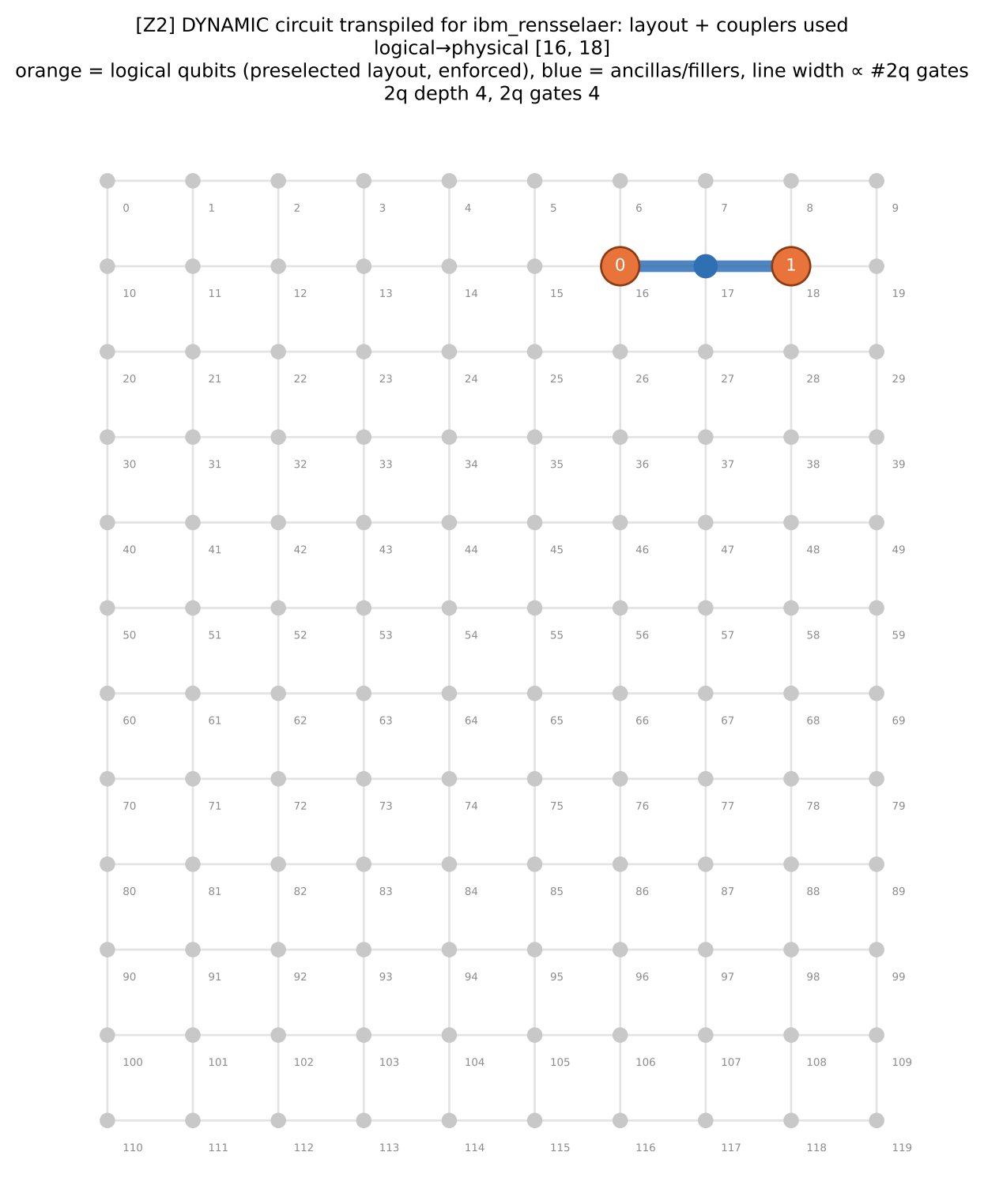


[Z2] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [16, 18]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

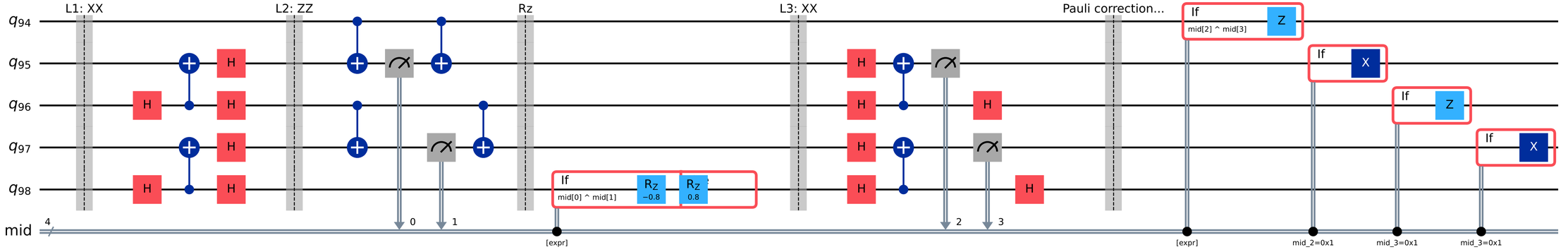
term 0: Z0 Z1
chain length 3, 1 ancillas, 0 |0> fillers



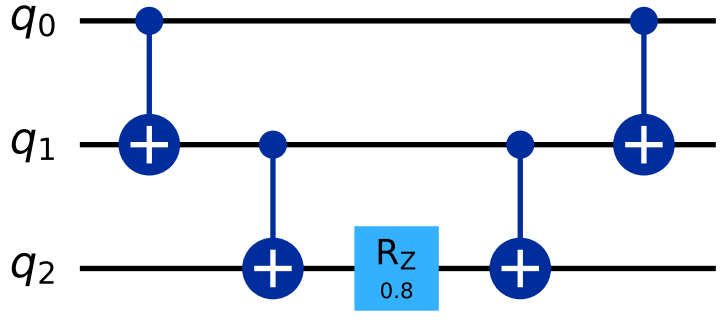

[Z2] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 2, 2q gates 2 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1


[Z2] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [16, 18]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 4
0
1
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

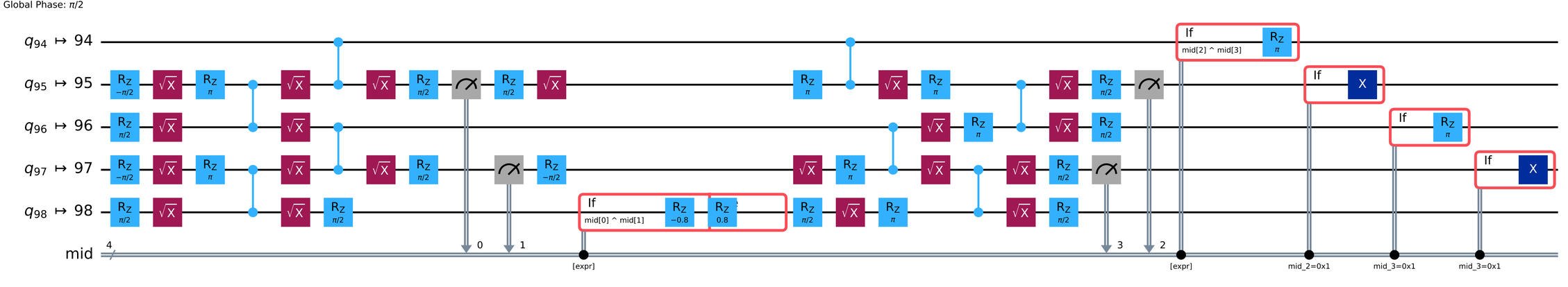
[Z3] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4, 4 mid-circuit measurements, logical→physical [94, 96, 98]



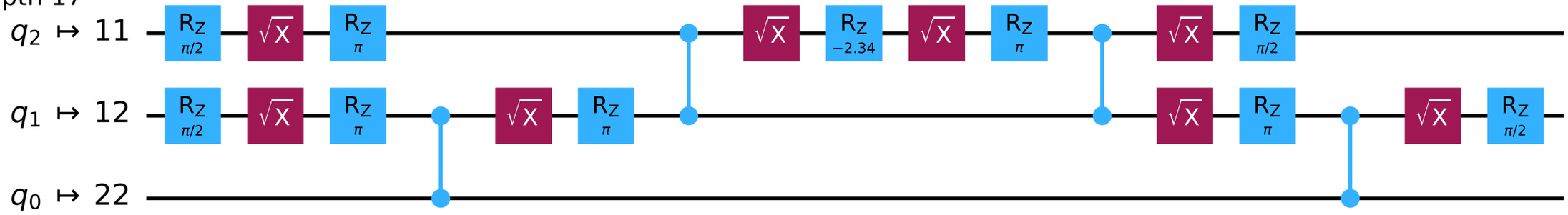
[Z3] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4



[Z3] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:21 sx:20 cz:8 if_else:5 measure:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 (3 logical qubits)
2q depth 4, total depth 22

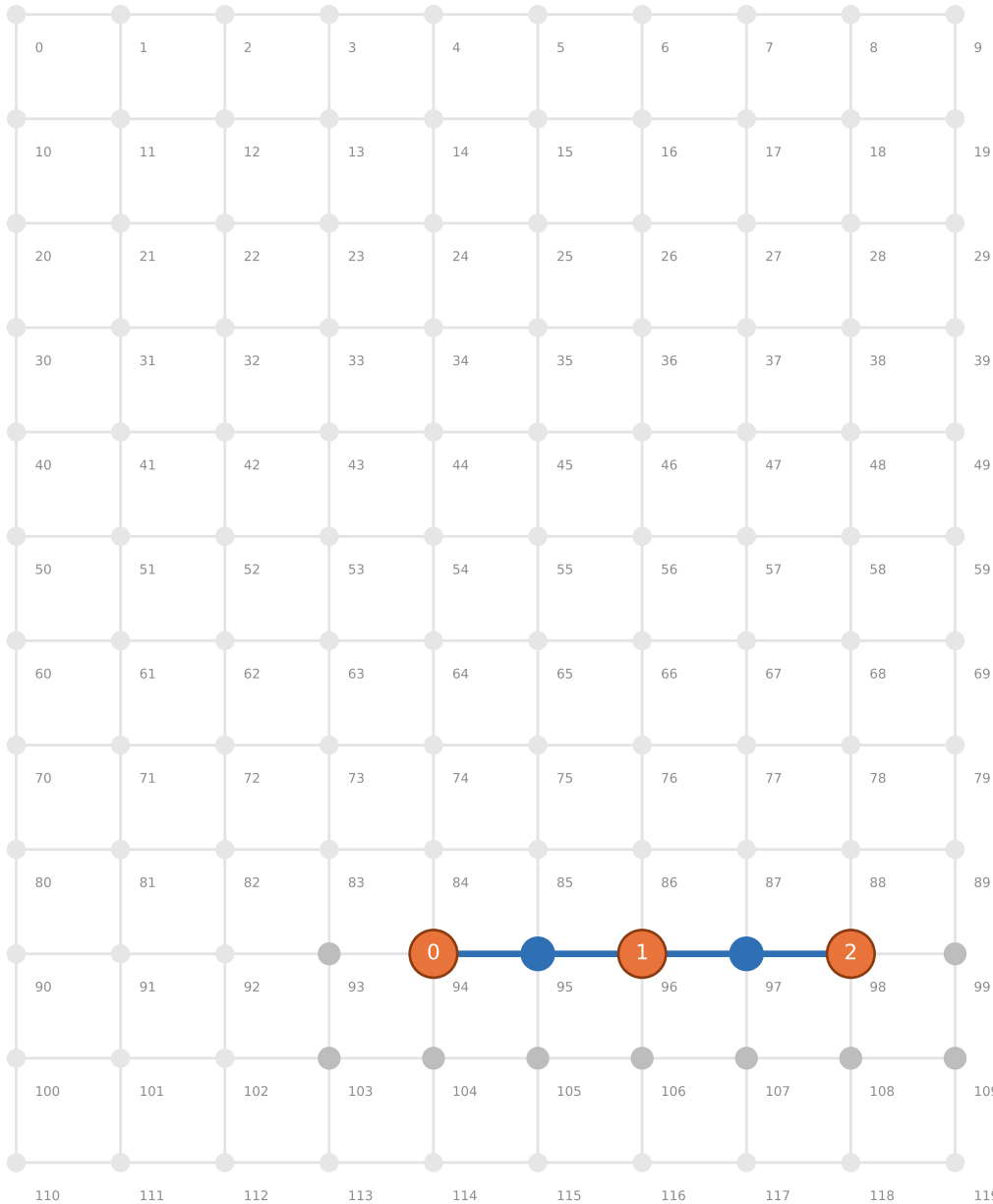


[Z3] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:10 sx:8 cz:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 (3 logical qubits)
2q depth 4, total depth 17

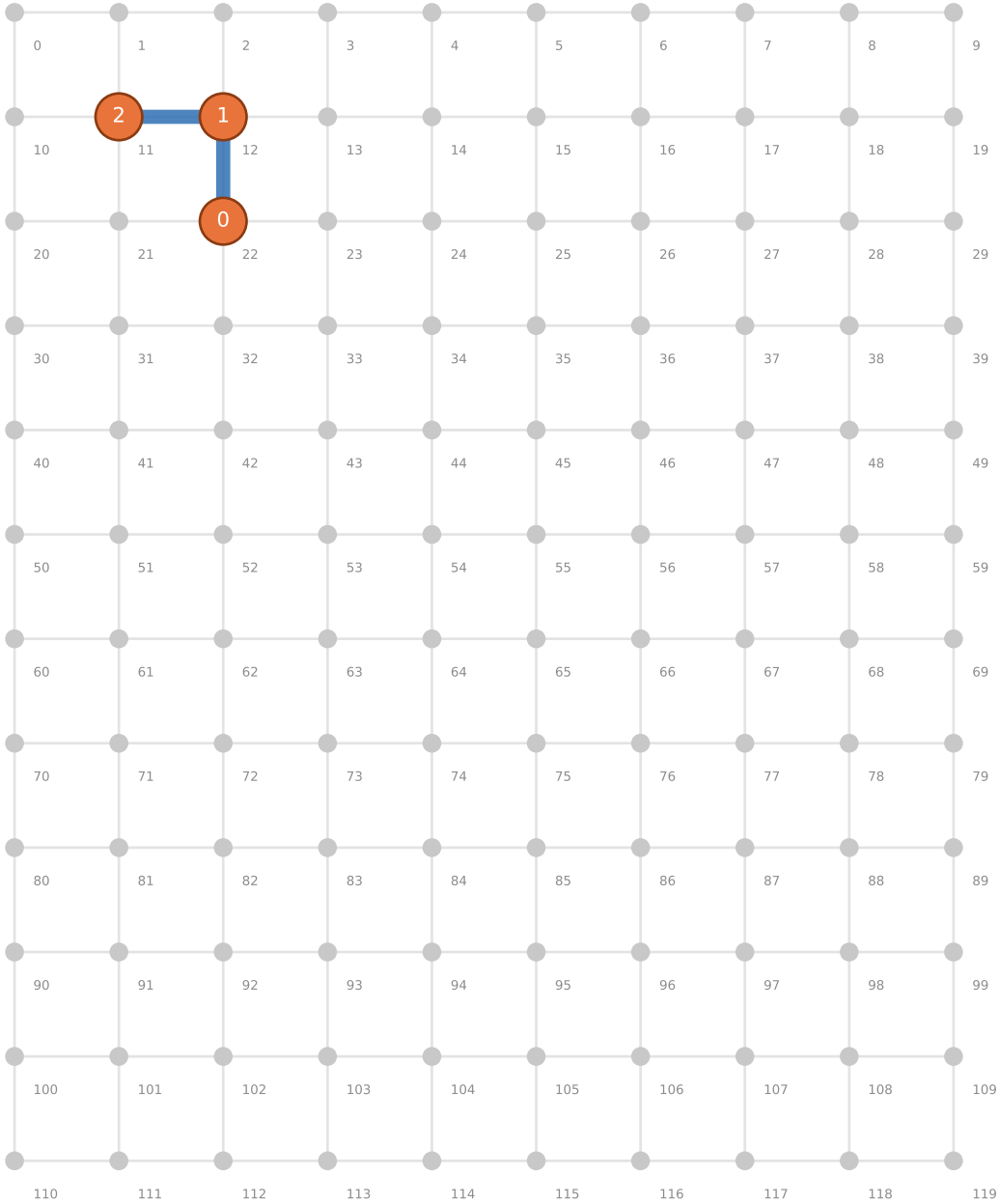


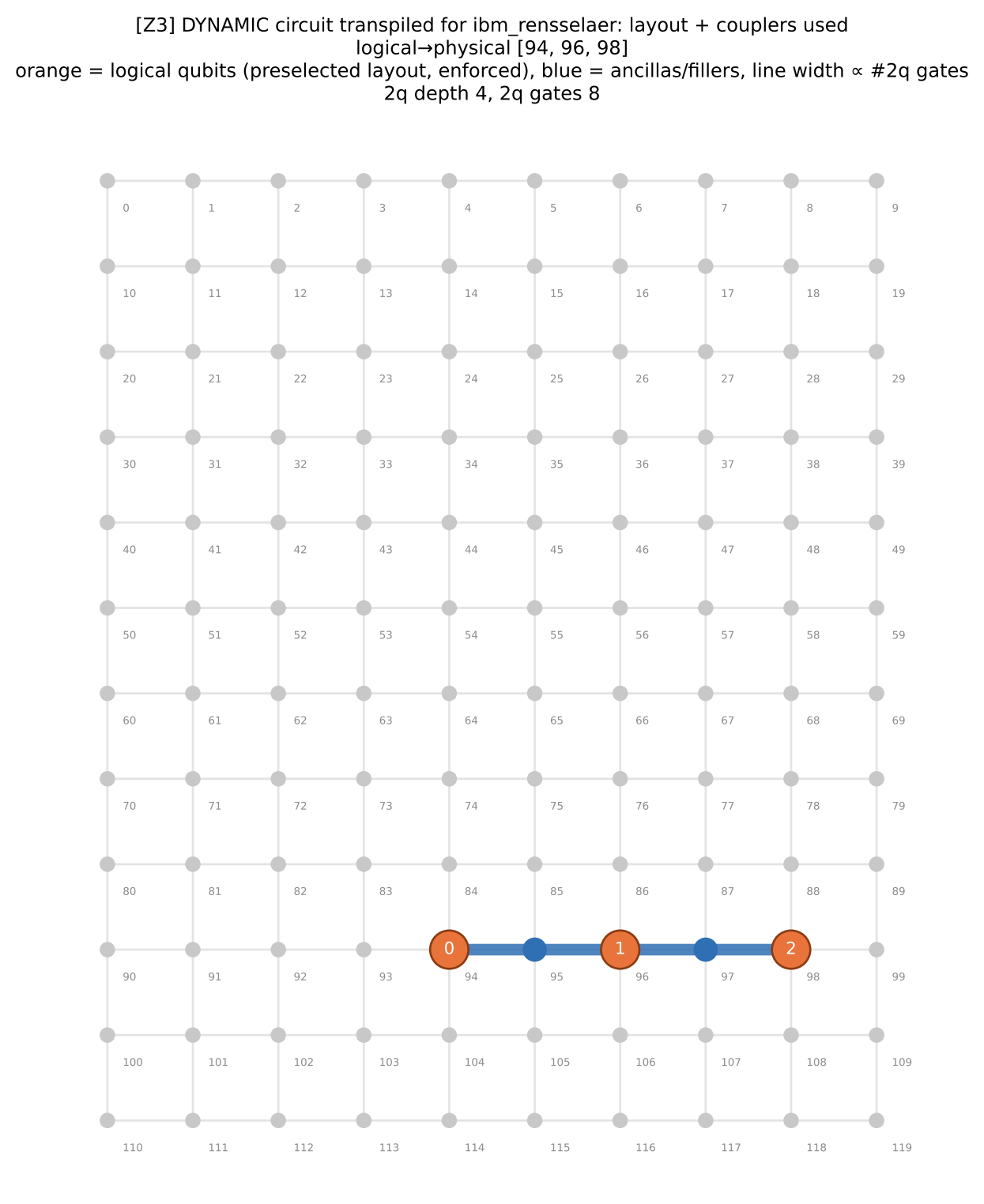
[Z3] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [94, 96, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2
chain length 5, 2 ancillas, 0 |0> fillers

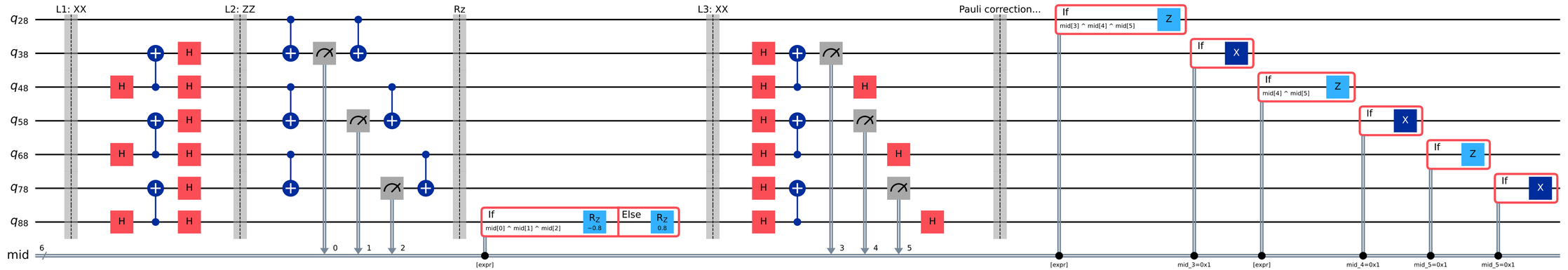


[Z3] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 12, 11] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 4, 2q gates 4 (+0 vs. untranspiled from routing)

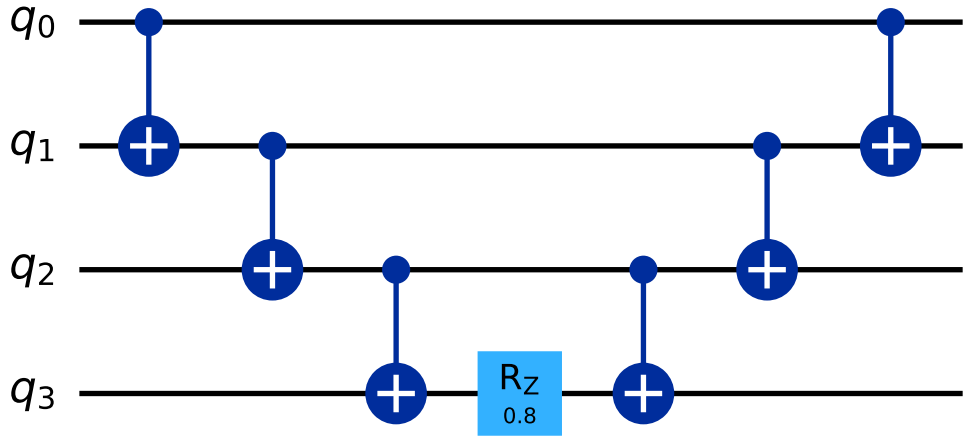



[Z3] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [94, 96, 98]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 8
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2

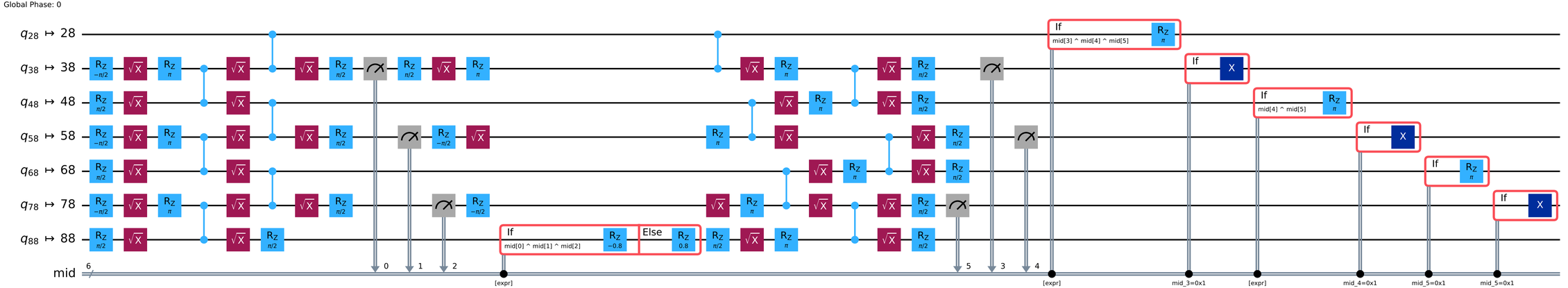
[Z4] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 4, 6 mid-circuit measurements, logical→physical [28, 48, 68, 88]



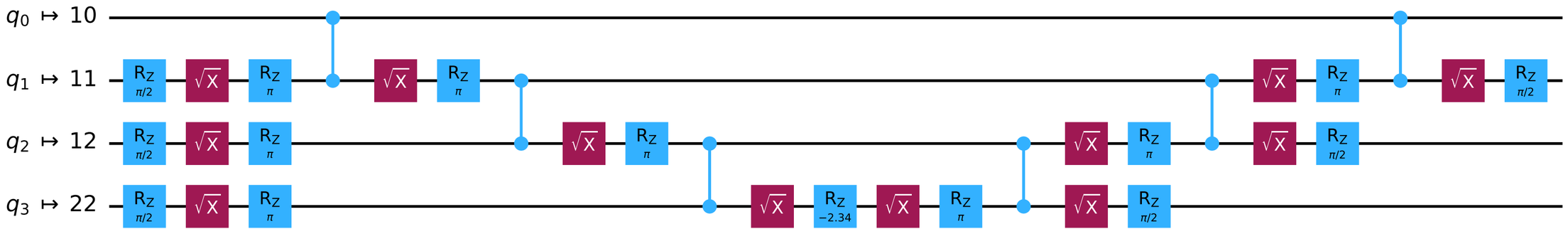
[Z4] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6



[Z4] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:30 sx:30 cz:12 if_else:7 measure:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 4, total depth 23

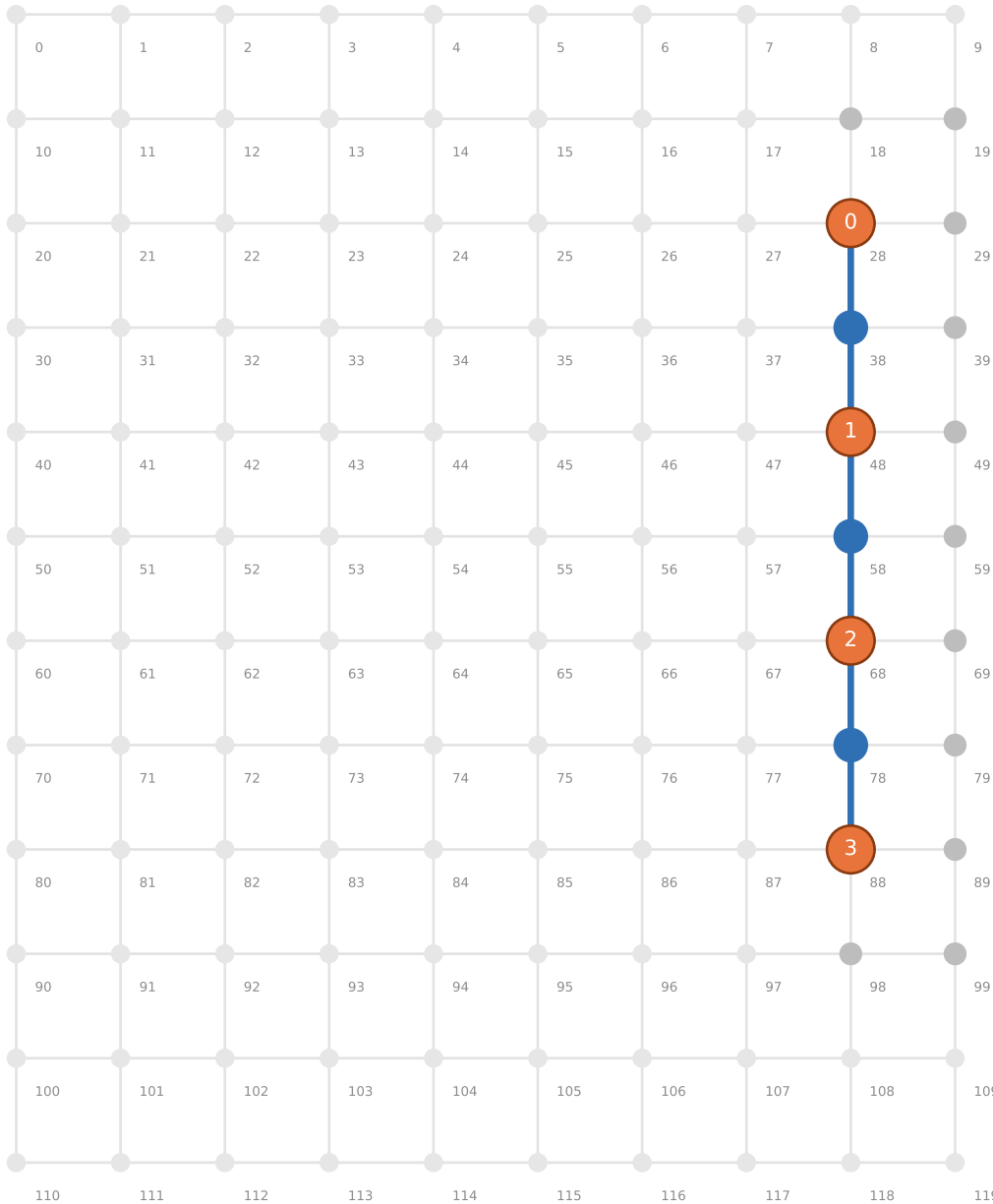


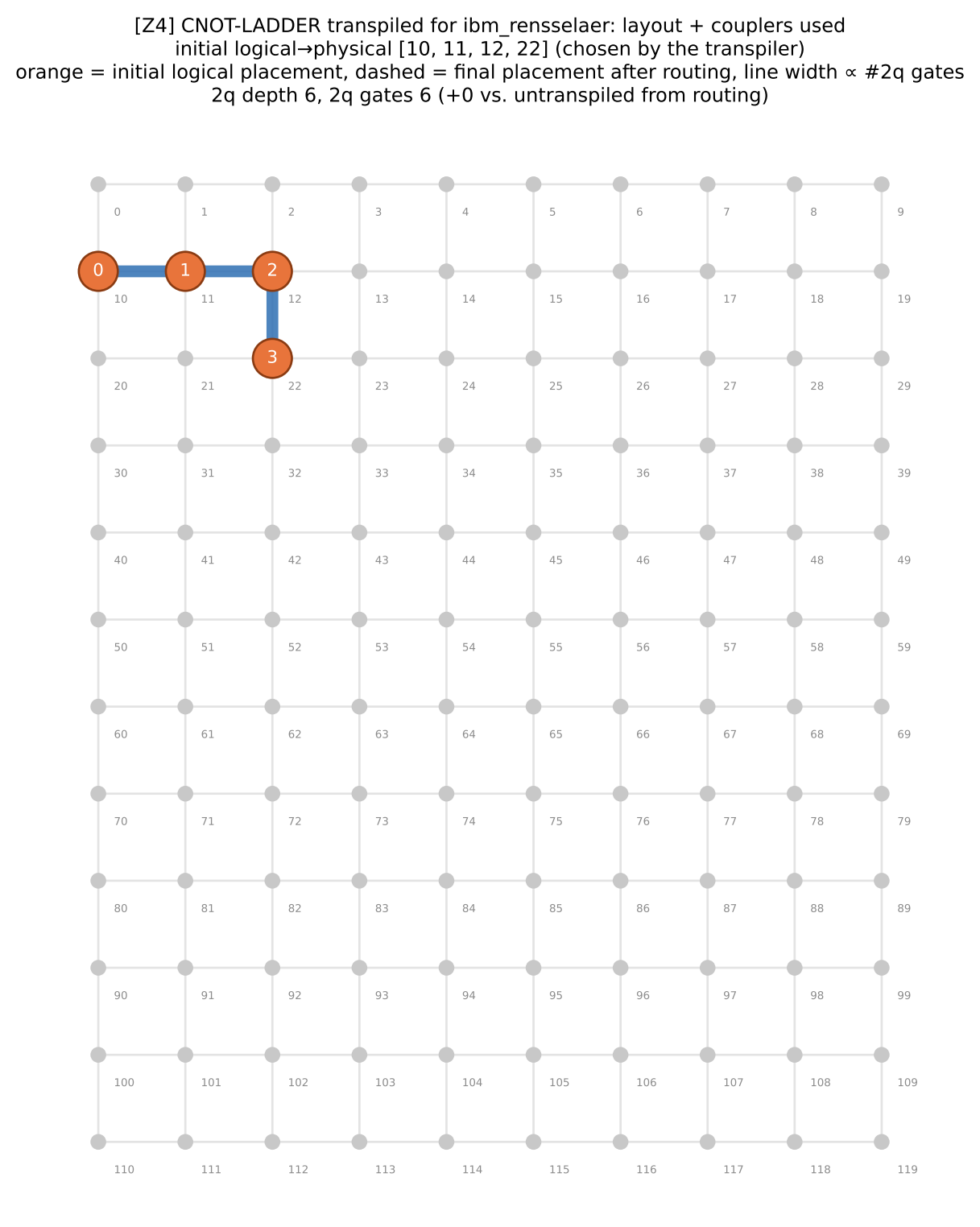
[Z4] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:15 sx:12 cz:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
Global Phase: 0
2q depth 6, total depth 23



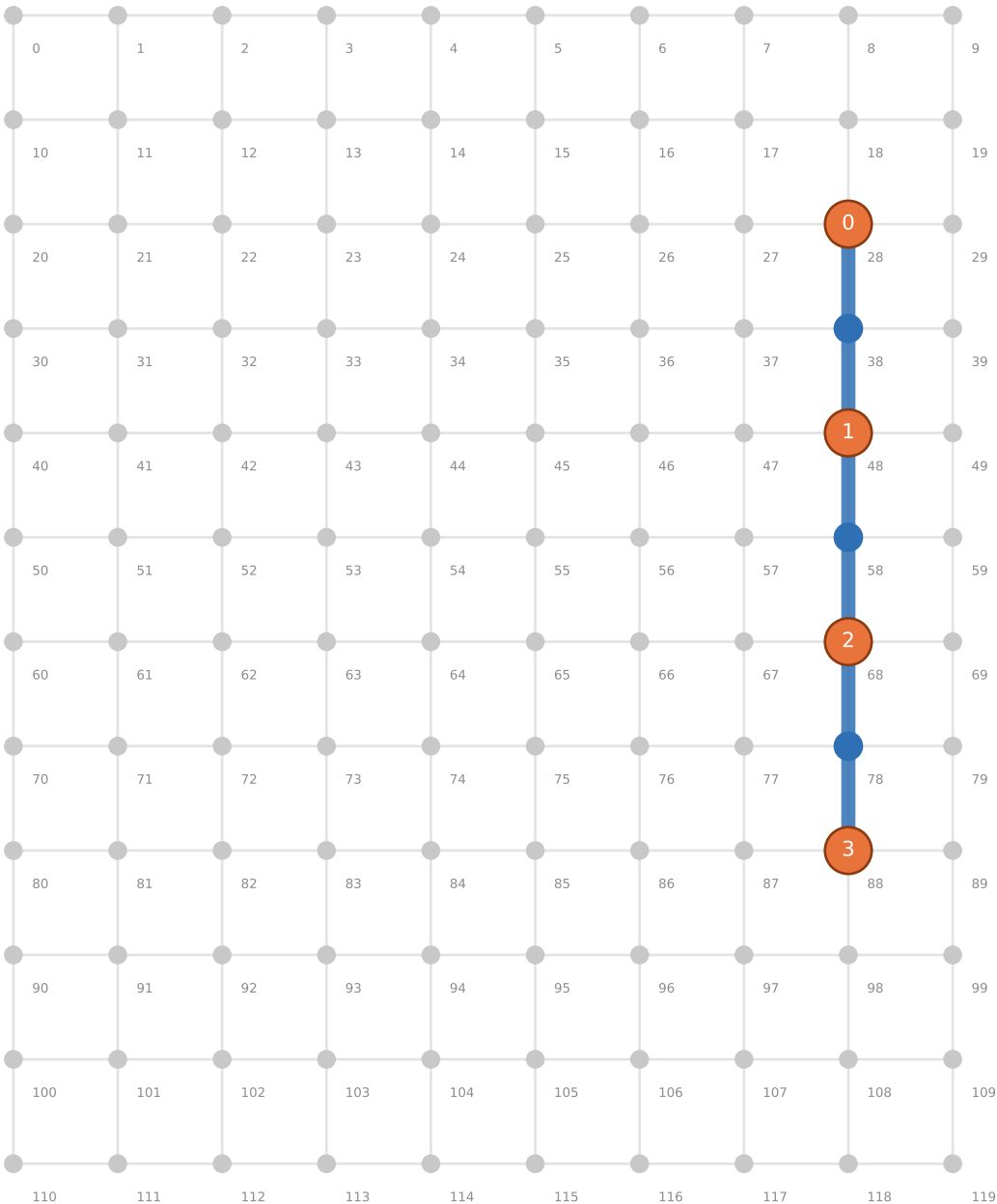
[Z4] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [28, 48, 68, 88]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2 Z3
chain length 7, 3 ancillas, 0 |0> fillers

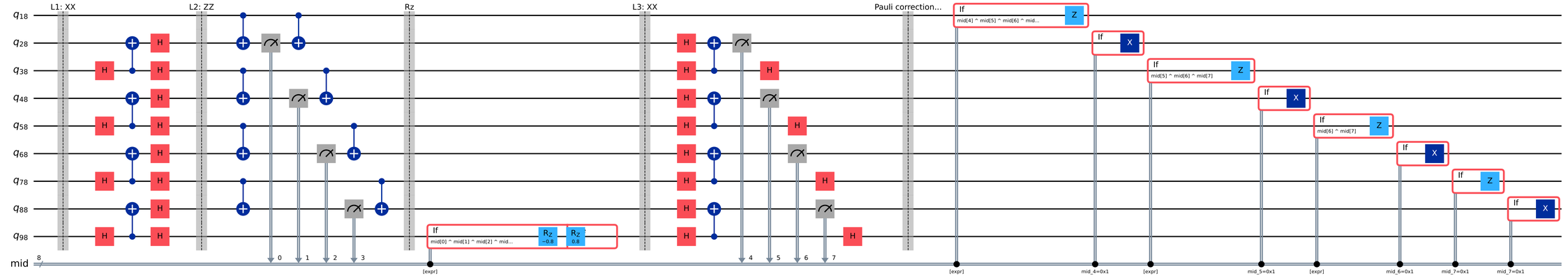



[Z4] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [10, 11, 12, 22] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
0
1
2
3
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

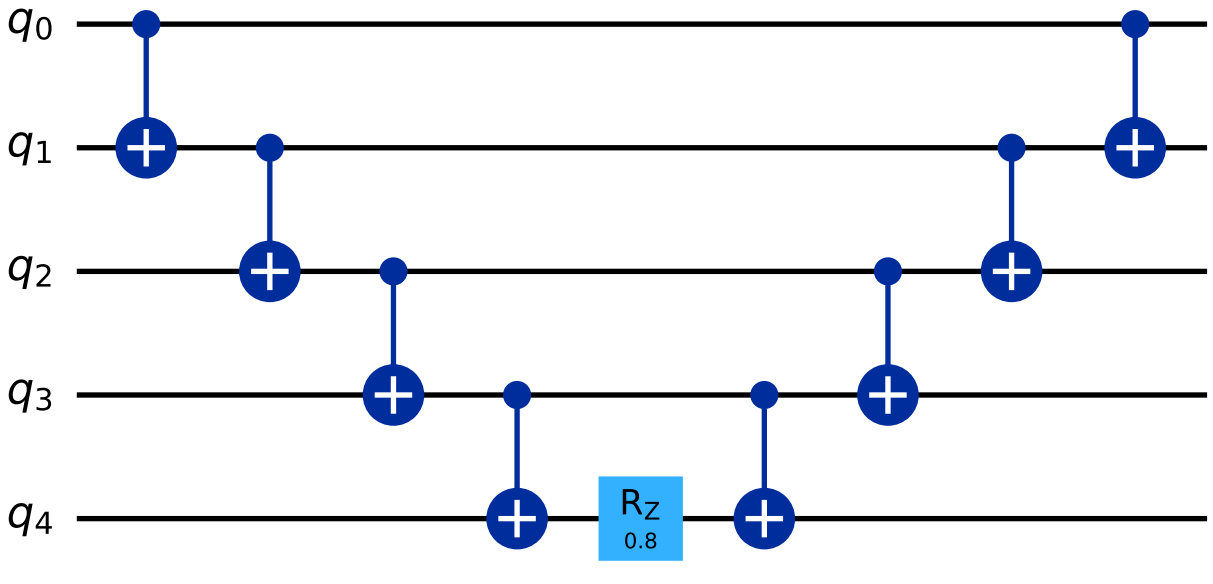
[Z4] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [28, 48, 68, 88]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 12



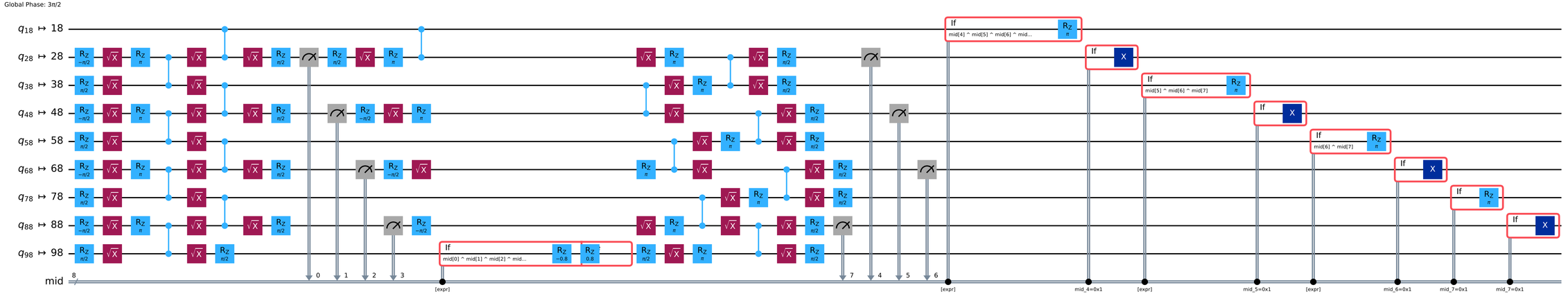
[Z5] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 4, 8 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



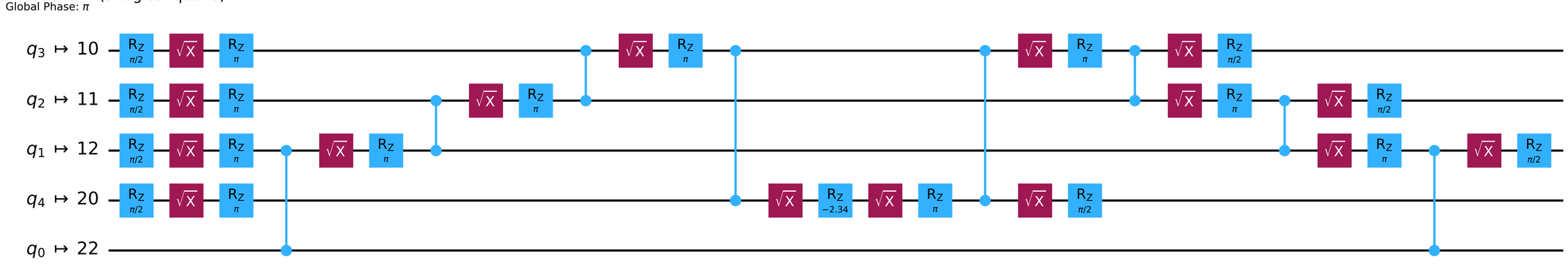
[Z5] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8



[Z5] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:40 rz:39 cz:16 if_else:9 measure:8)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 4, total depth 24

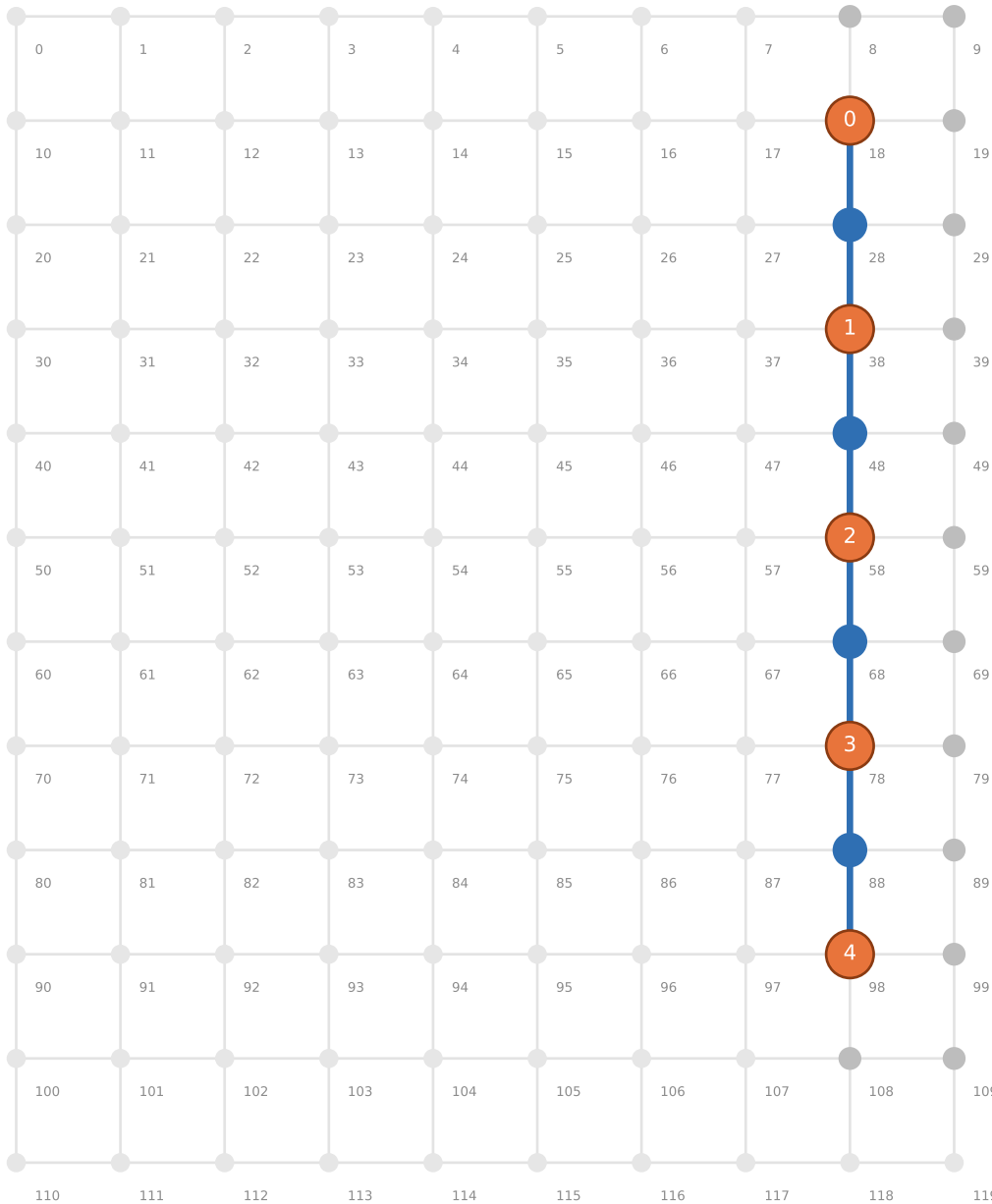


[Z5] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:20 sx:16 cz:8)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8, total depth 29

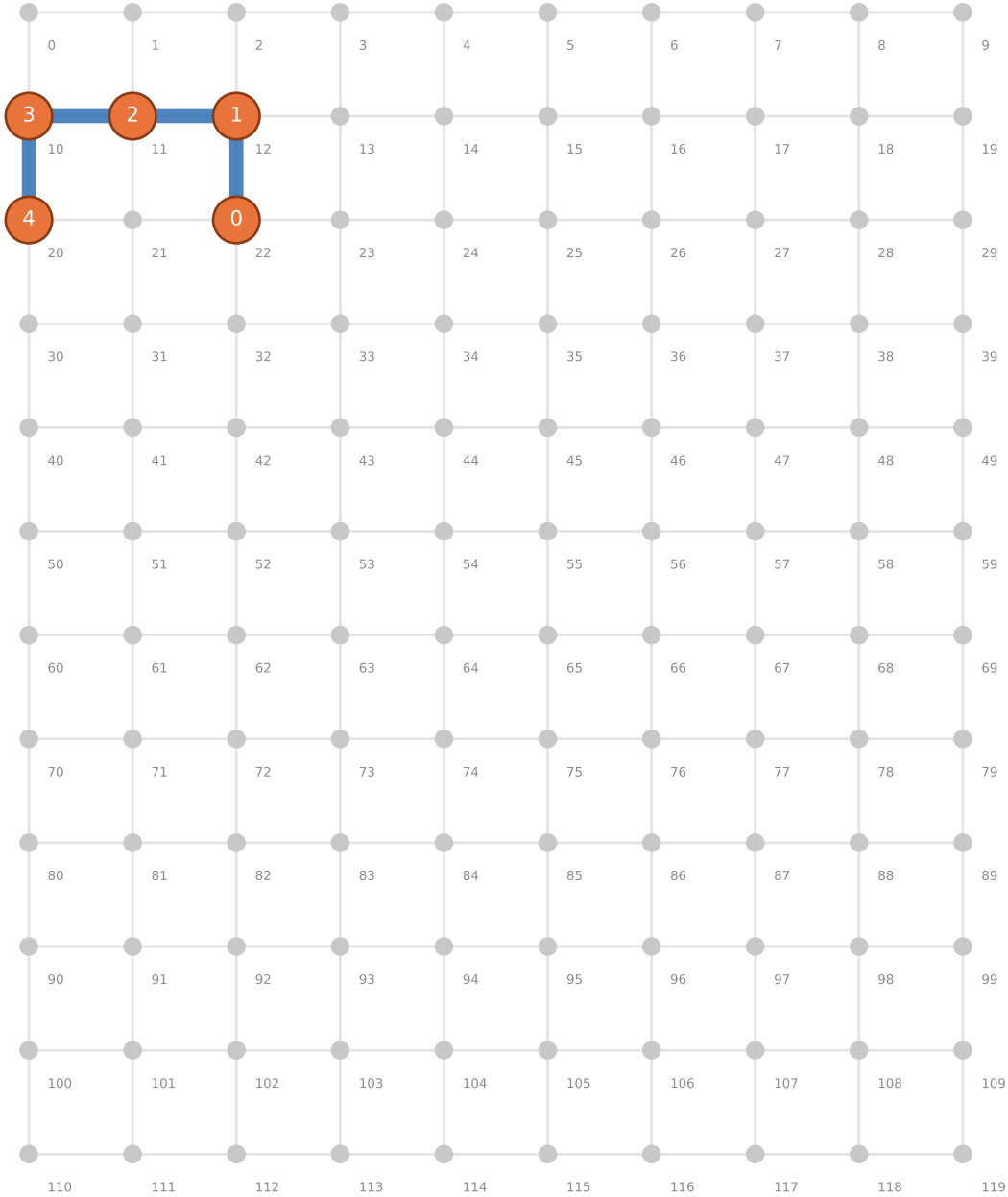


[Z5] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

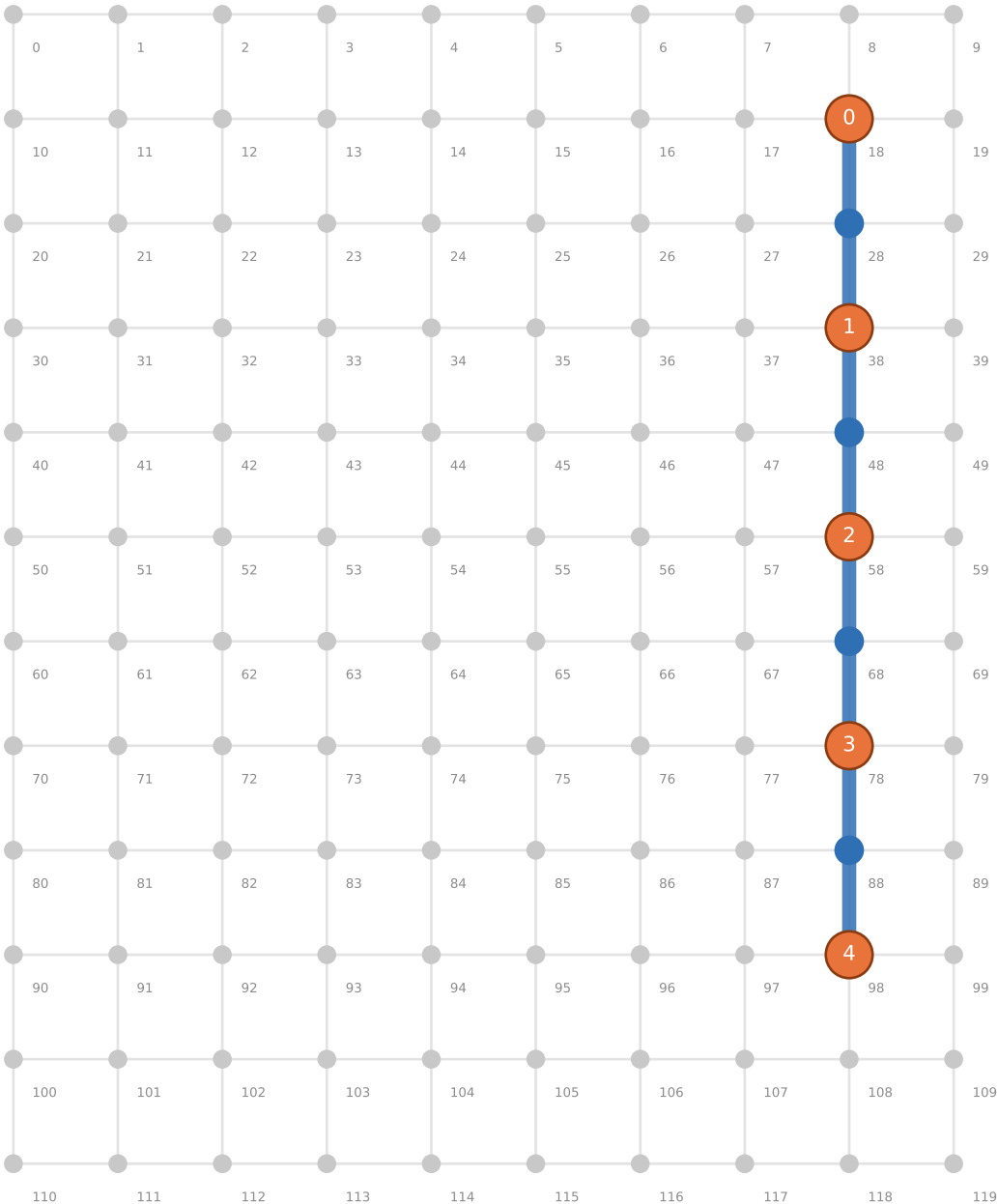
term 0: Z0 Z1 Z2 Z3 Z4
chain length 9, 4 ancillas, 0 |0> fillers



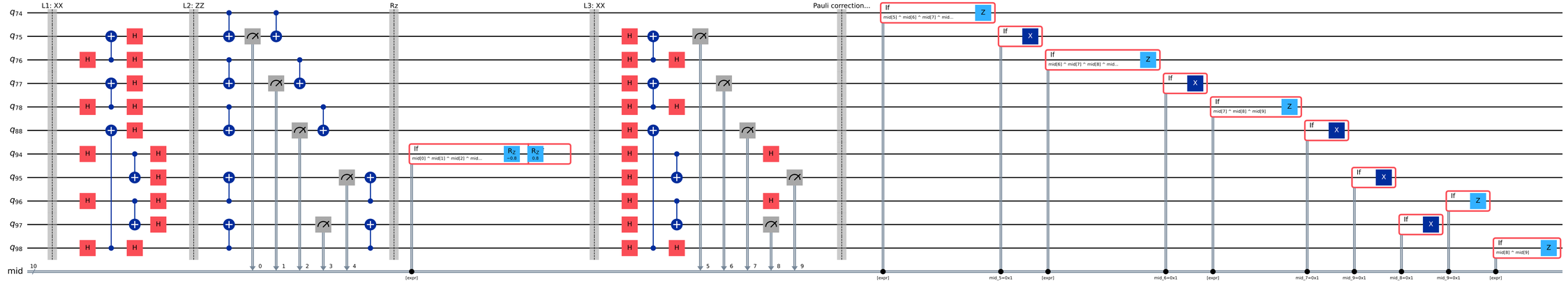
[Z5] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 12, 11, 10, 20] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 8, 2q gates 8 (+0 vs. untranspiled from routing)



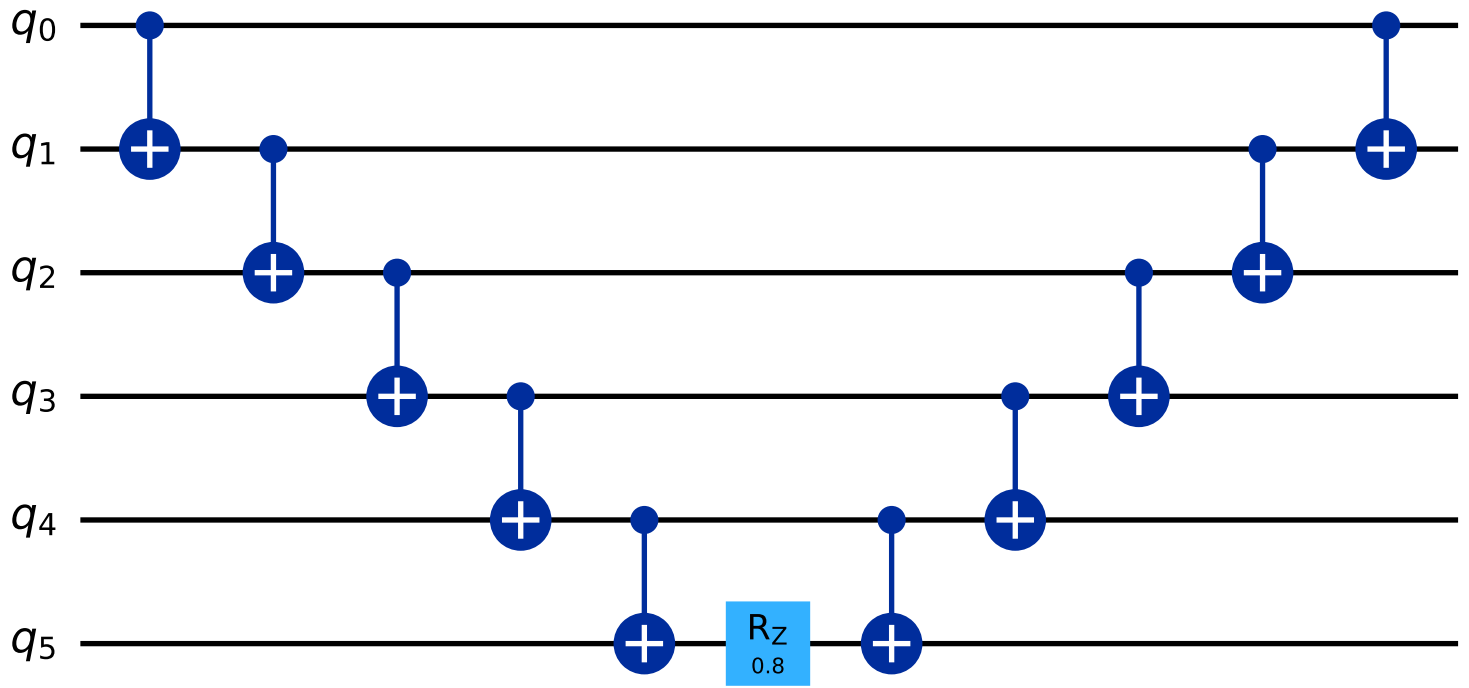
[Z5] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 16



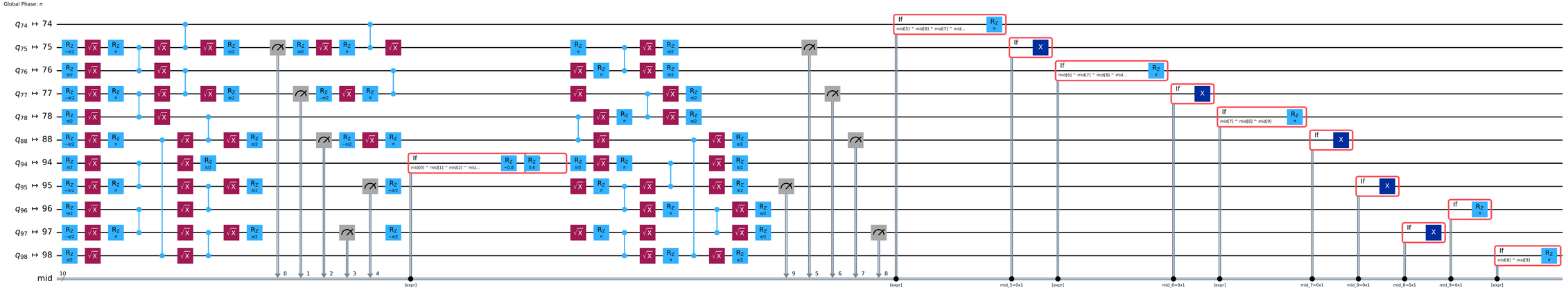
[Z6] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 4, 10 mid-circuit measurements, logical→physical [74, 76, 78, 98, 96, 94]



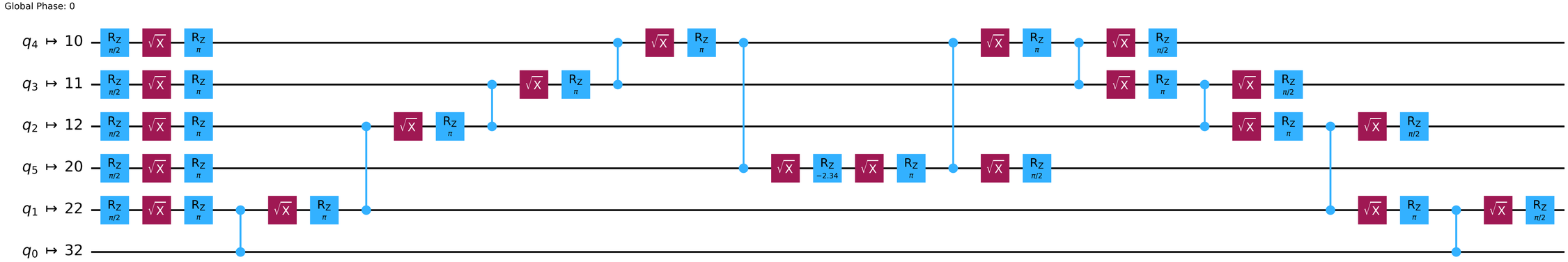
[Z6] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10



[Z6] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:50 rz:48 cz:20 if_else:11 measure:10)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 4, total depth 25

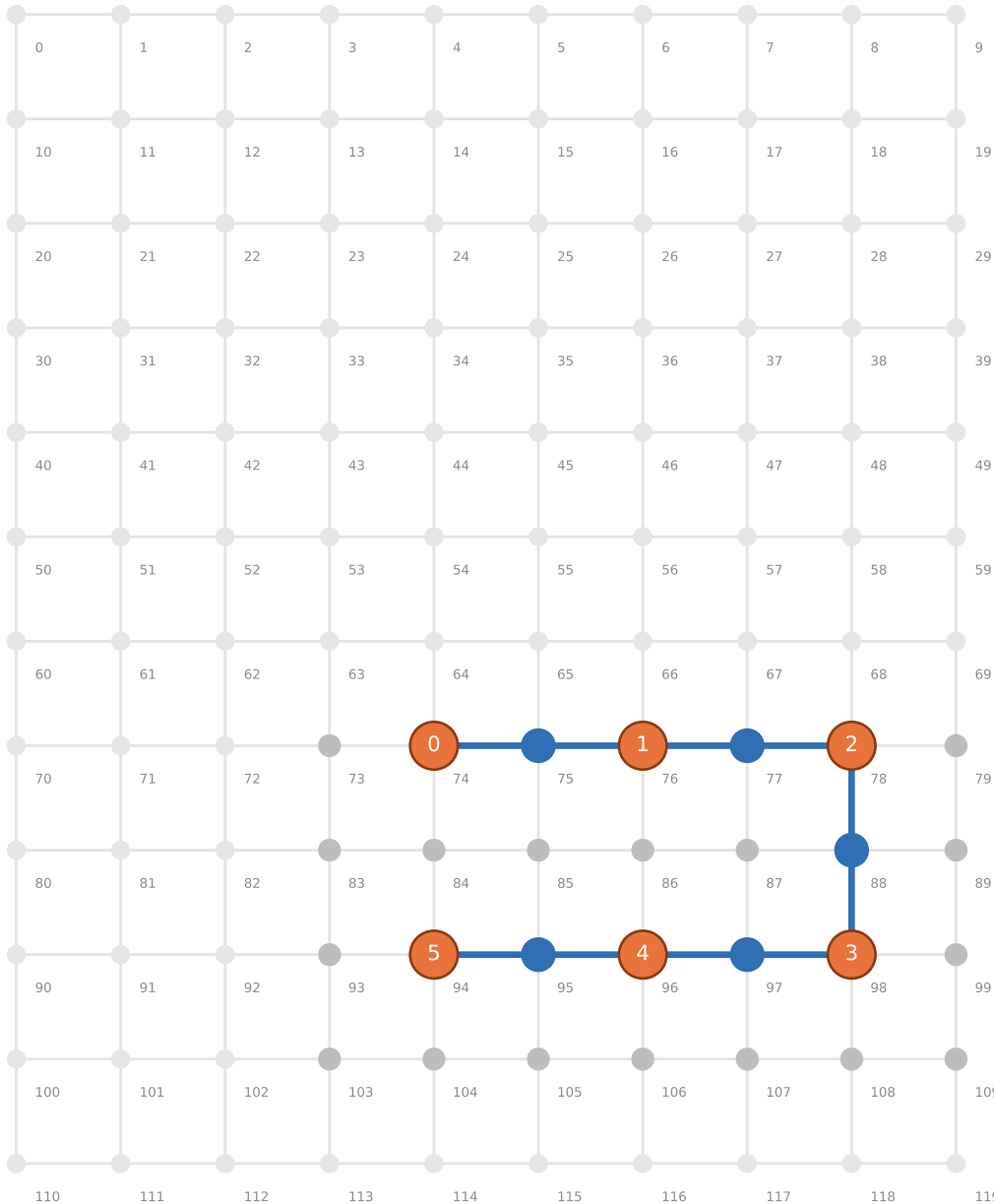


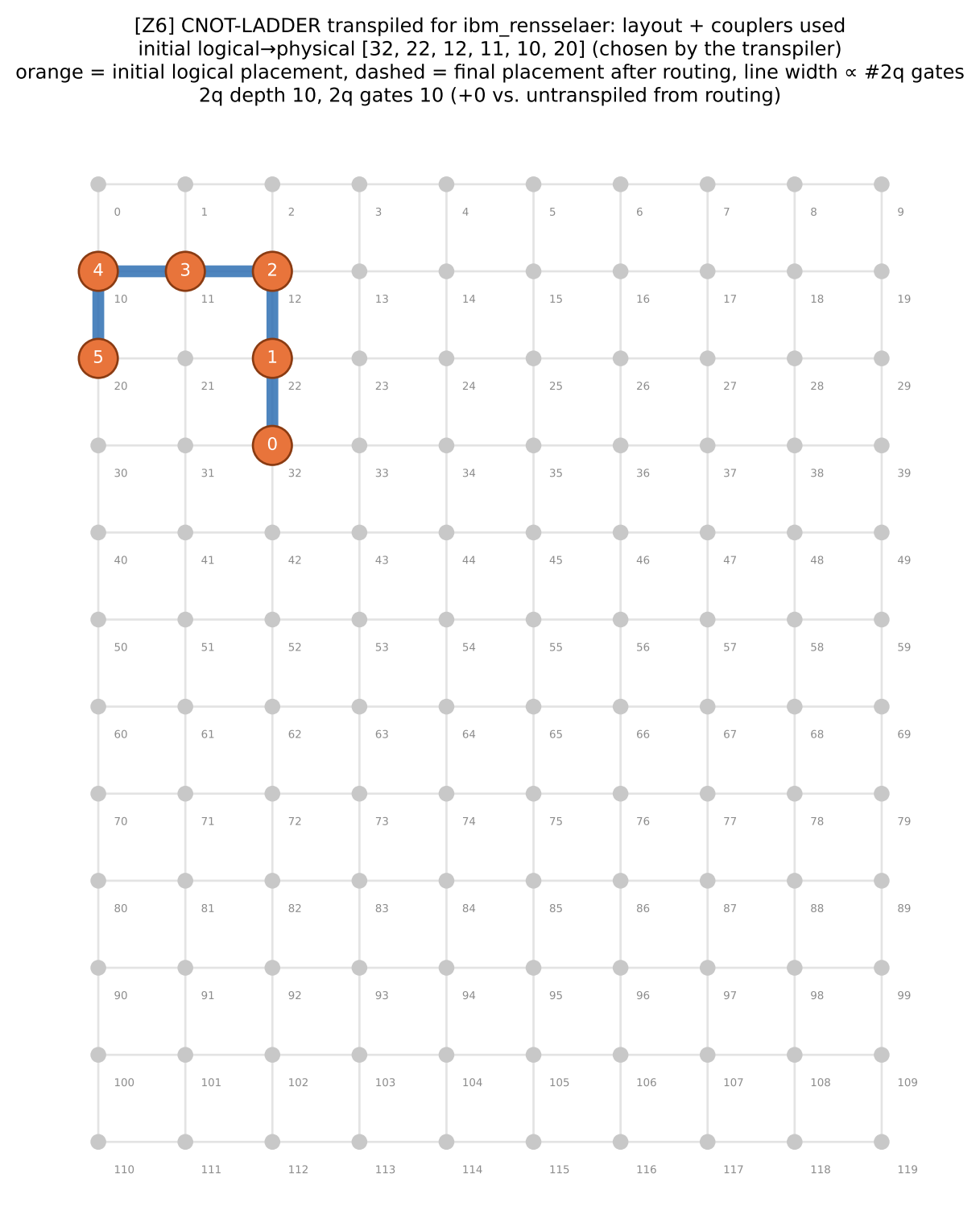
[Z6] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:25 sx:20 cz:10)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 (6 logical qubits)
2q depth 10, total depth 35



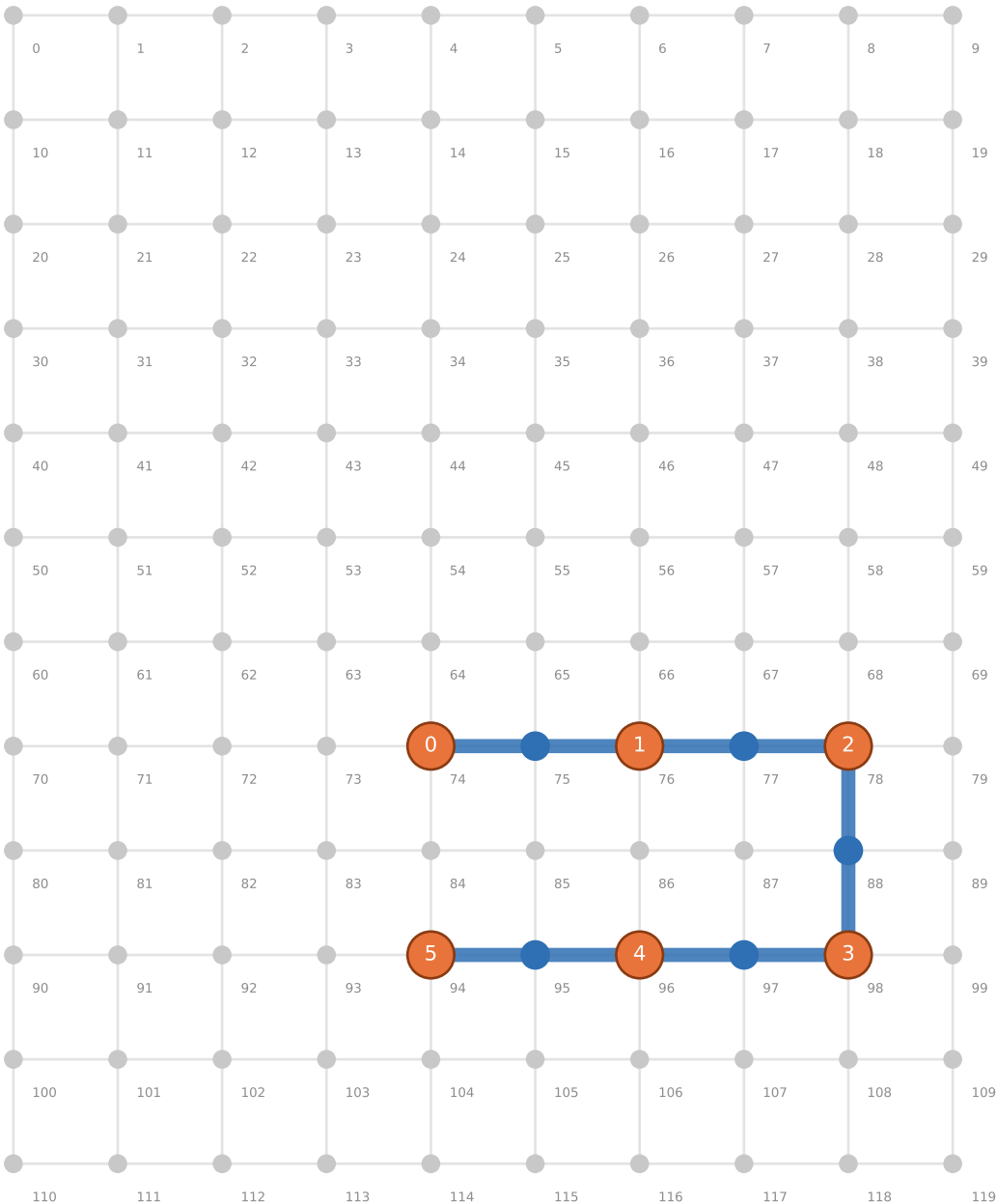
[Z6] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [74, 76, 78, 98, 96, 94]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5
chain length 11, 5 ancillas, 0 |0> fillers



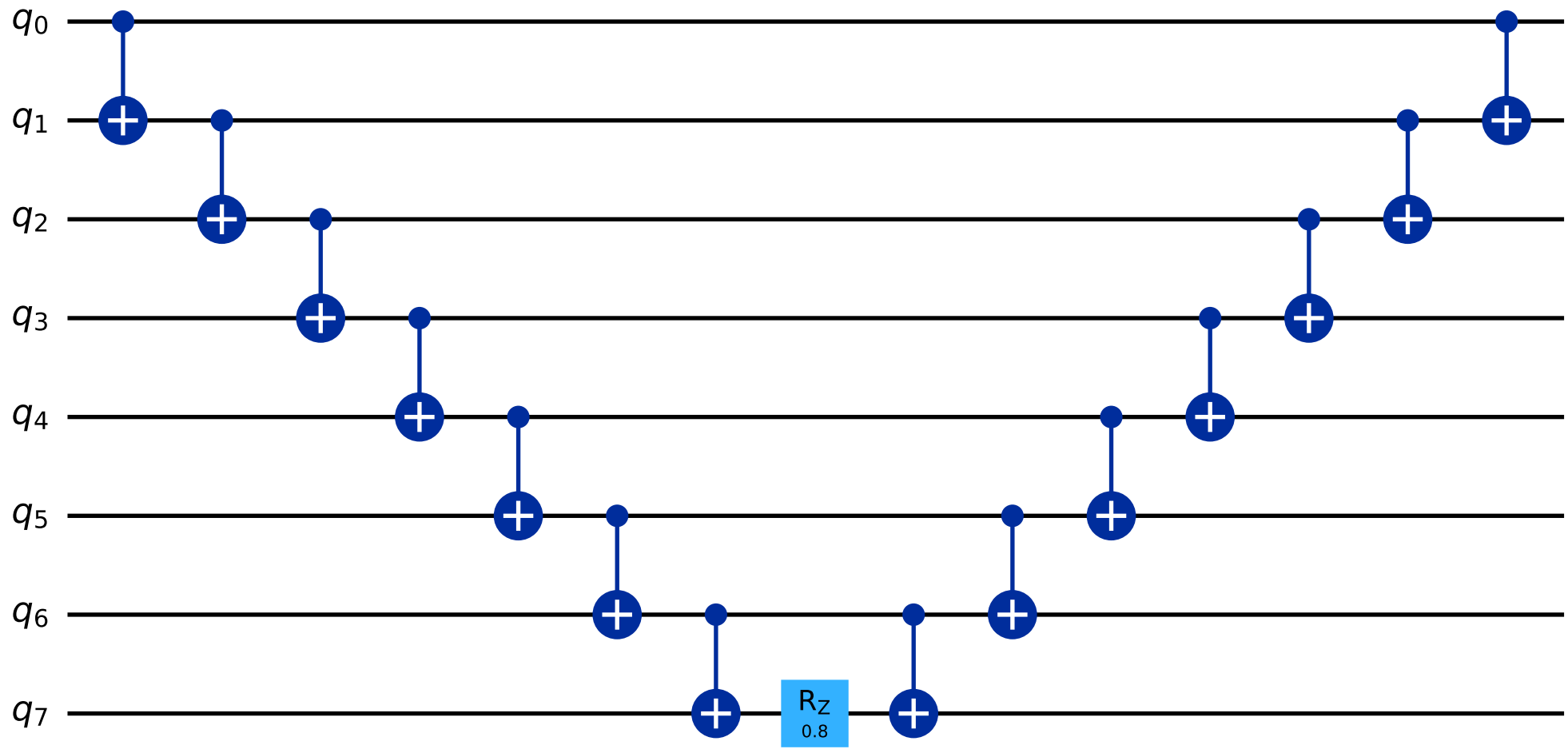

[Z6] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [32, 22, 12, 11, 10, 20] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 10, 2q gates 10 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
4
3
2
5
1
0

[Z6] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [74, 76, 78, 98, 96, 94]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 20

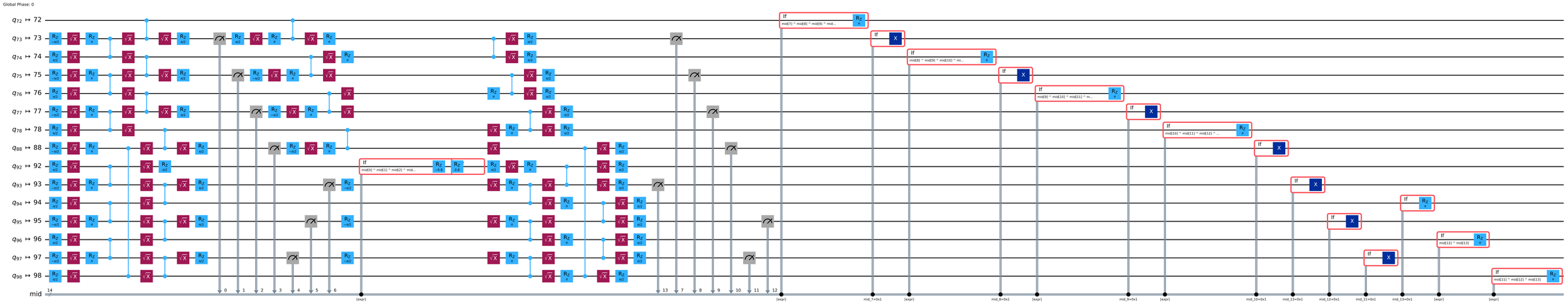


[Z8] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 4, 14 mid-circuit measurements, logical→physical [72, 74, 76, 78, 98, 96, 94, 92]

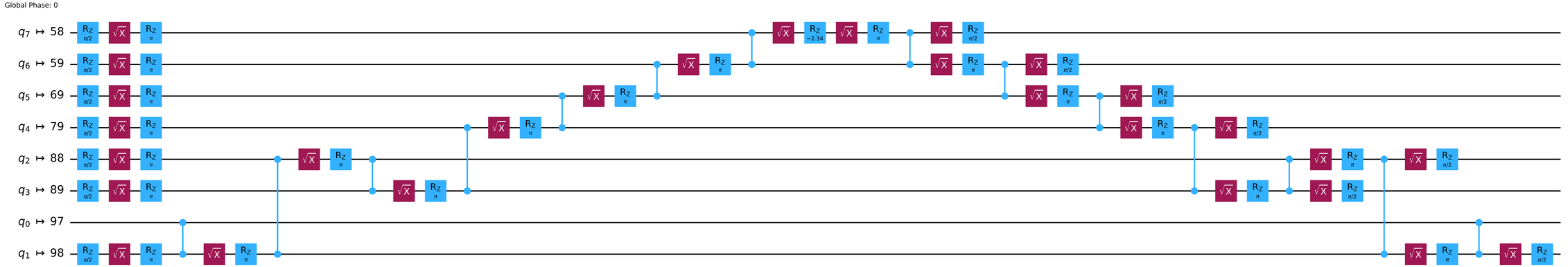
[Z8] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14



[Z8] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:70 rz:66 cz:28 if_else:15 measure:14)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 4, total depth 27

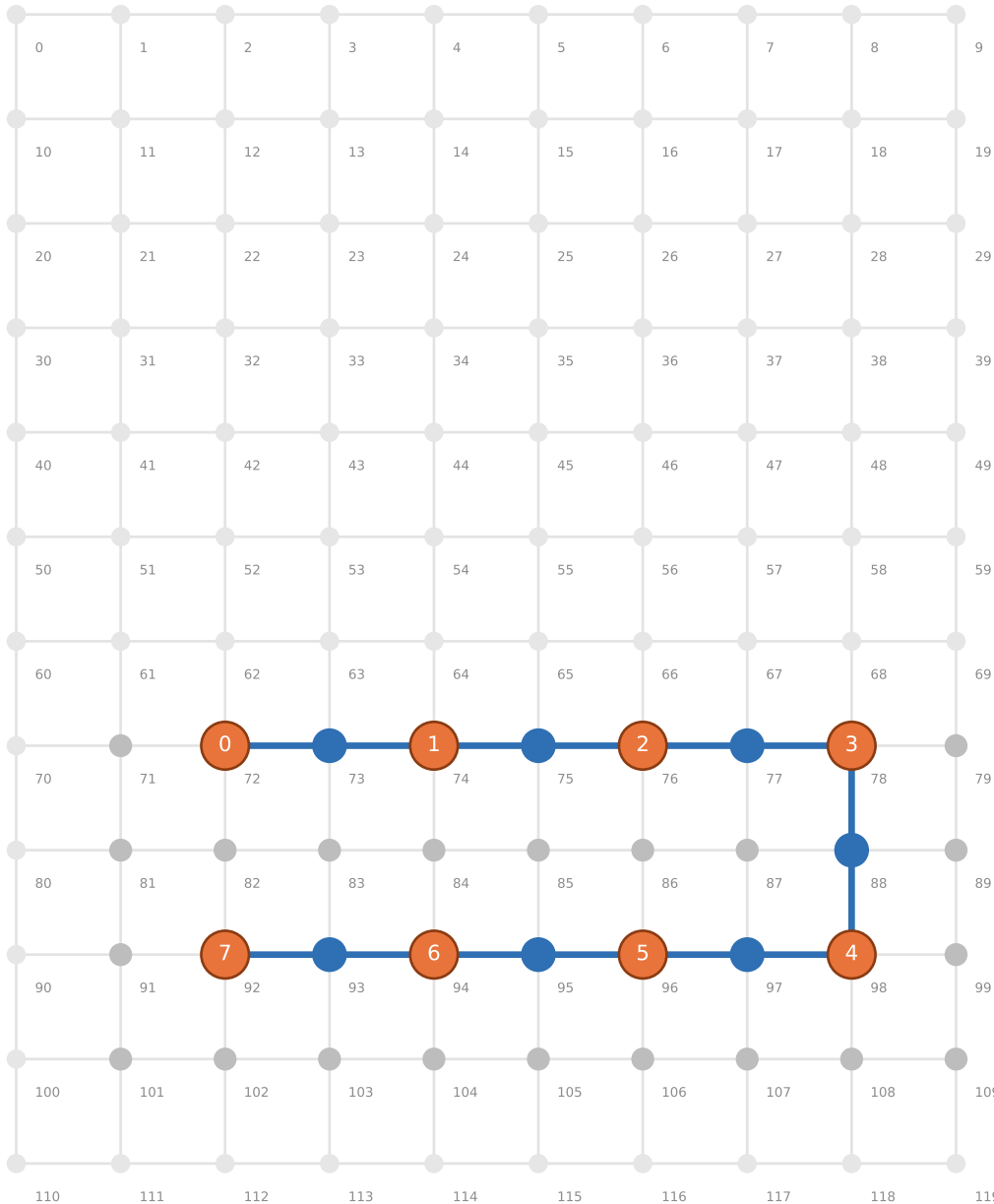


[Z8] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:35 sx:28 cz:14)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14, total depth 47

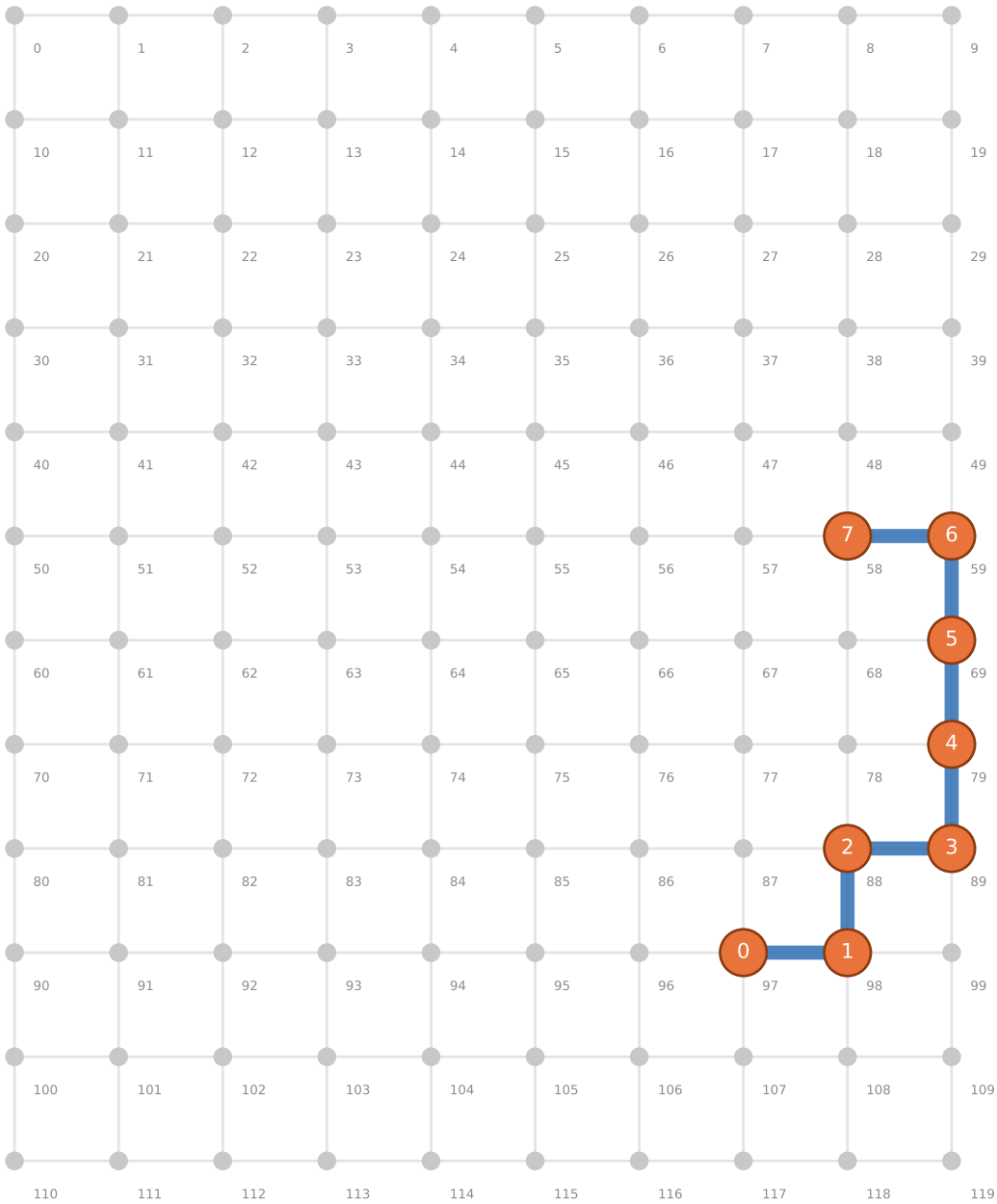


[Z8] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

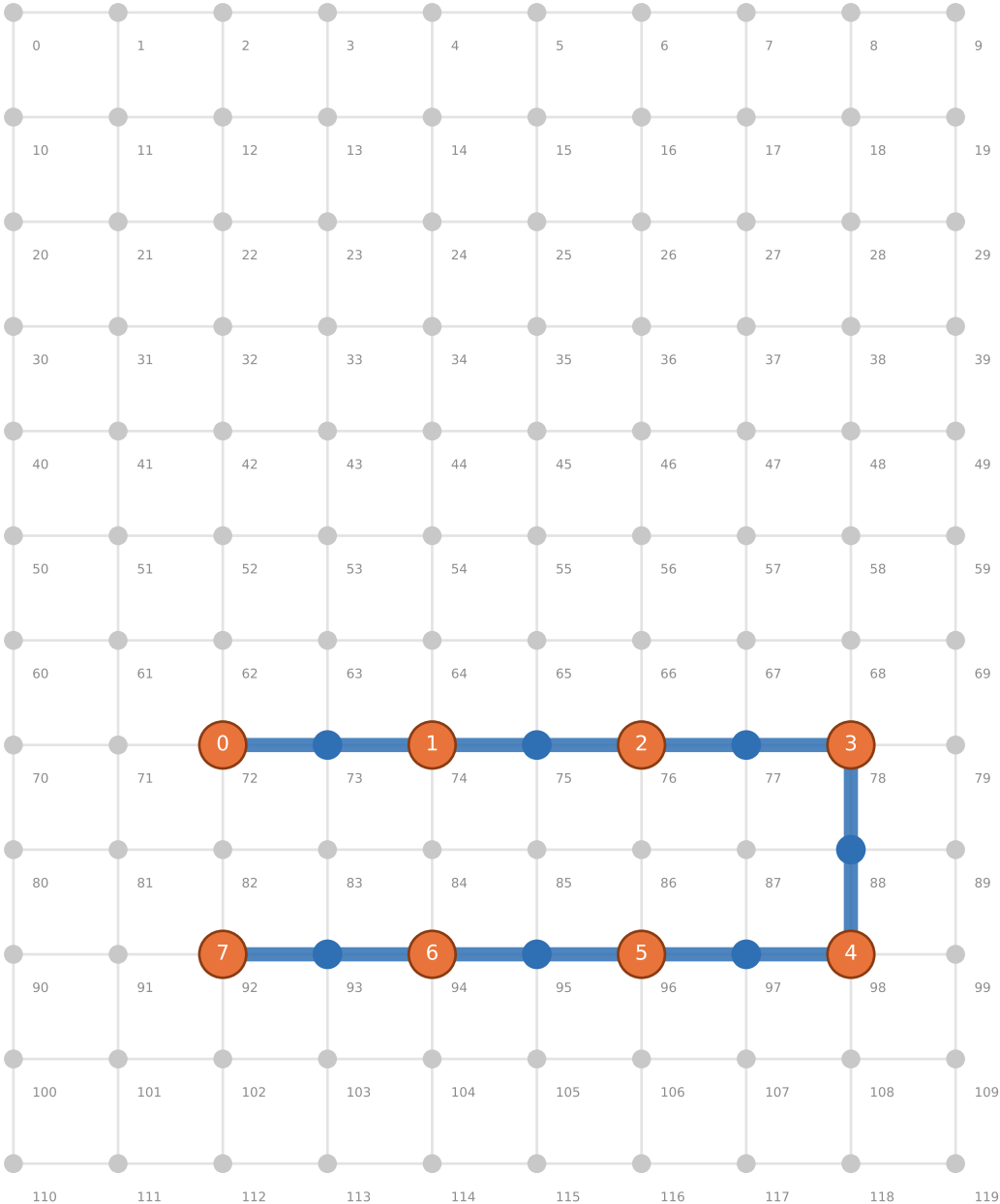
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7
chain length 15, 7 ancillas, 0 |0> fillers



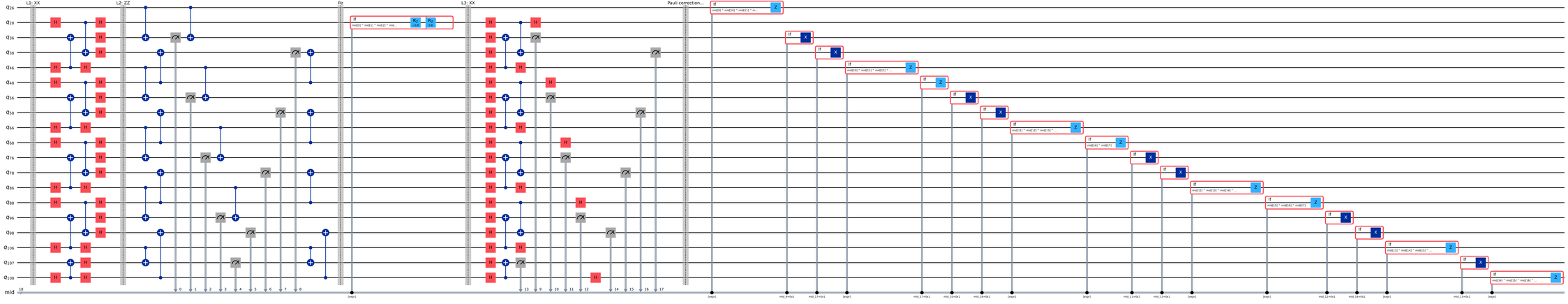
[Z8] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [97, 98, 88, 89, 79, 69, 59, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



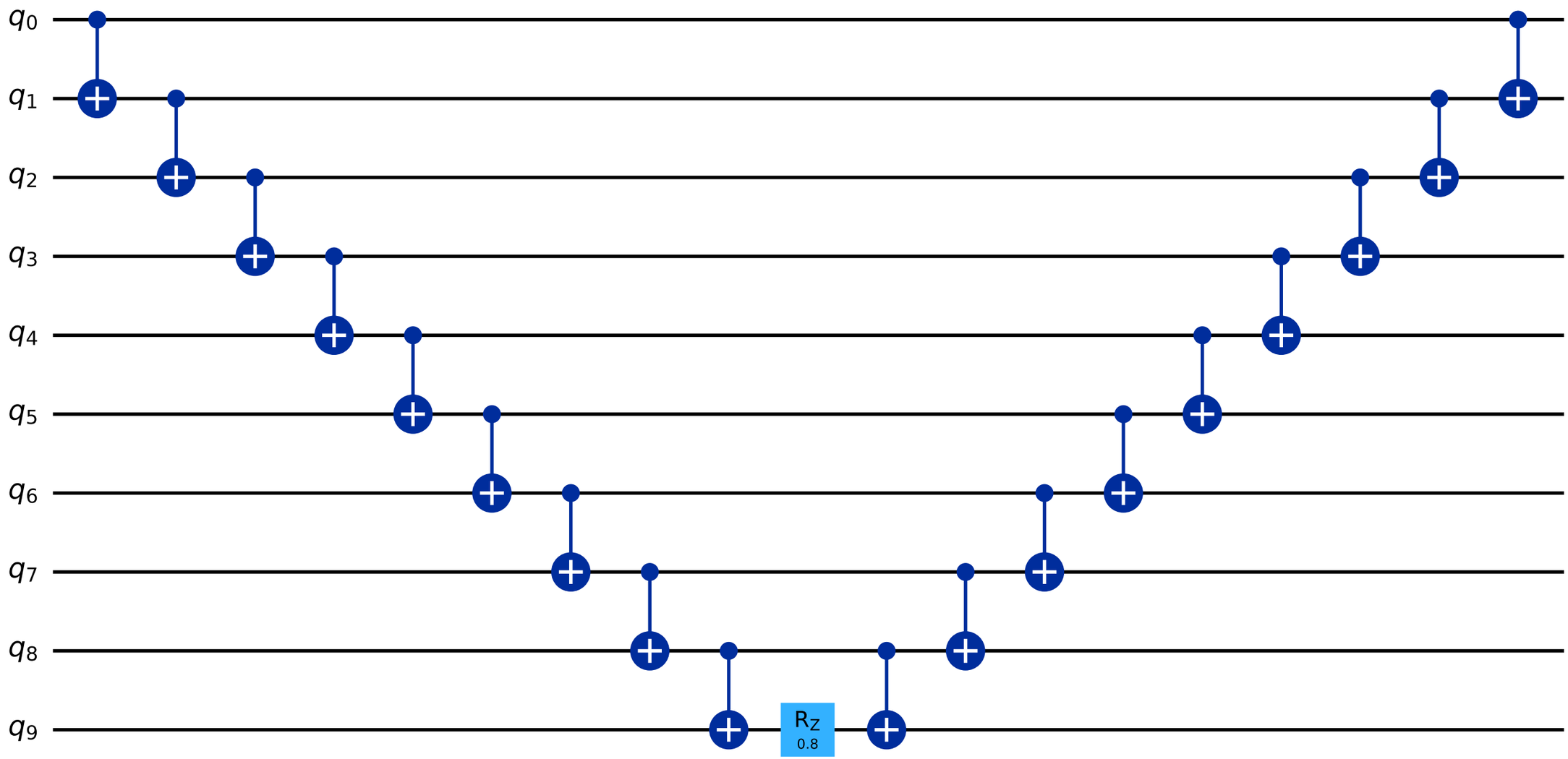
[Z8] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 28



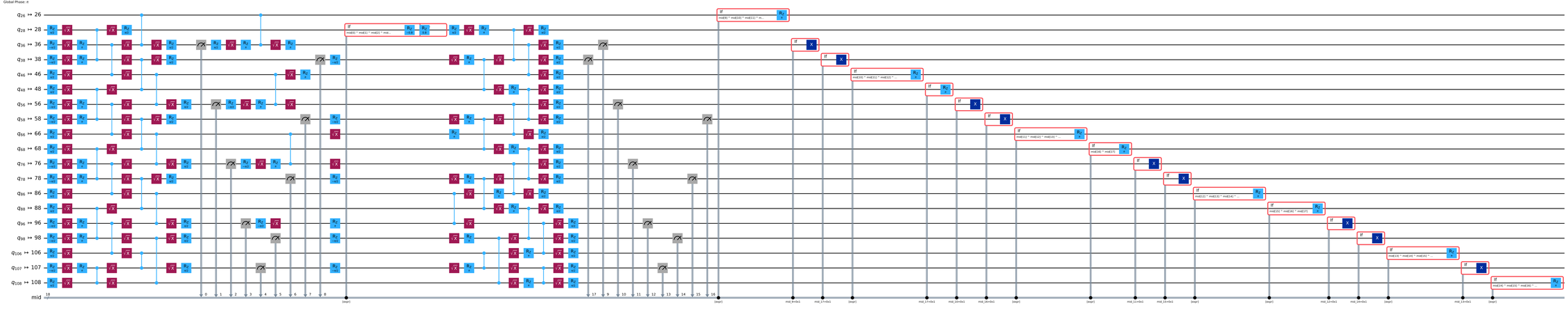
[Z10] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 (10 logical qubits)
2q depth 4, 18 mid-circuit measurements, logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]



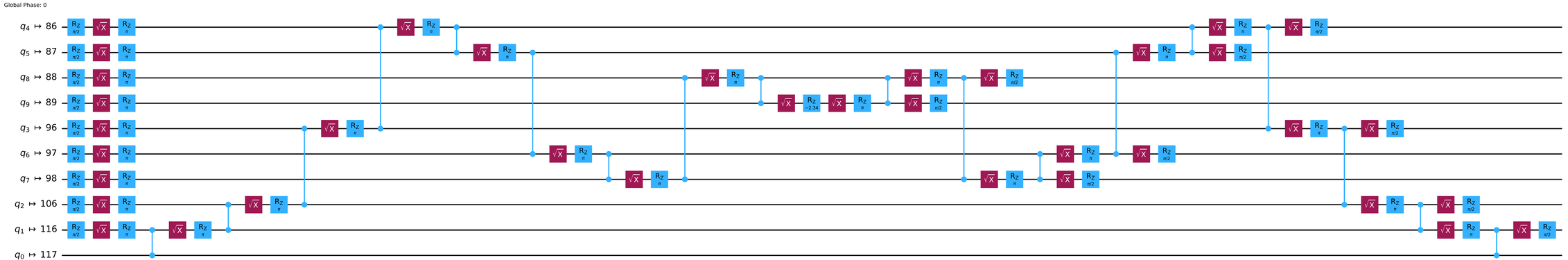
[Z10] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18



[Z10] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:90 rz:84 cz:36 if_else:19 measure:18)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 4, total depth 29

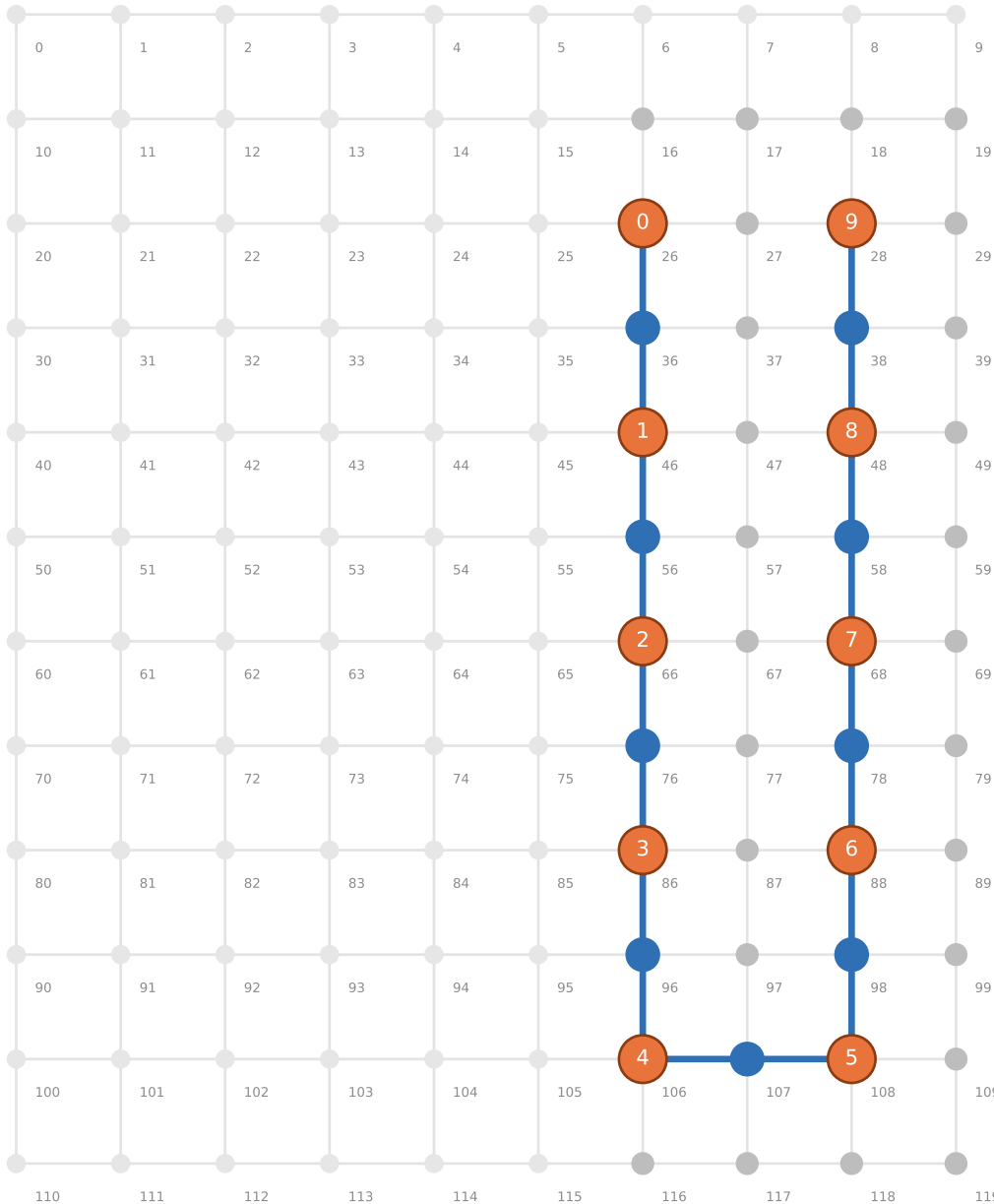


[Z10] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:45 sx:36 cz:18)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18, total depth 59

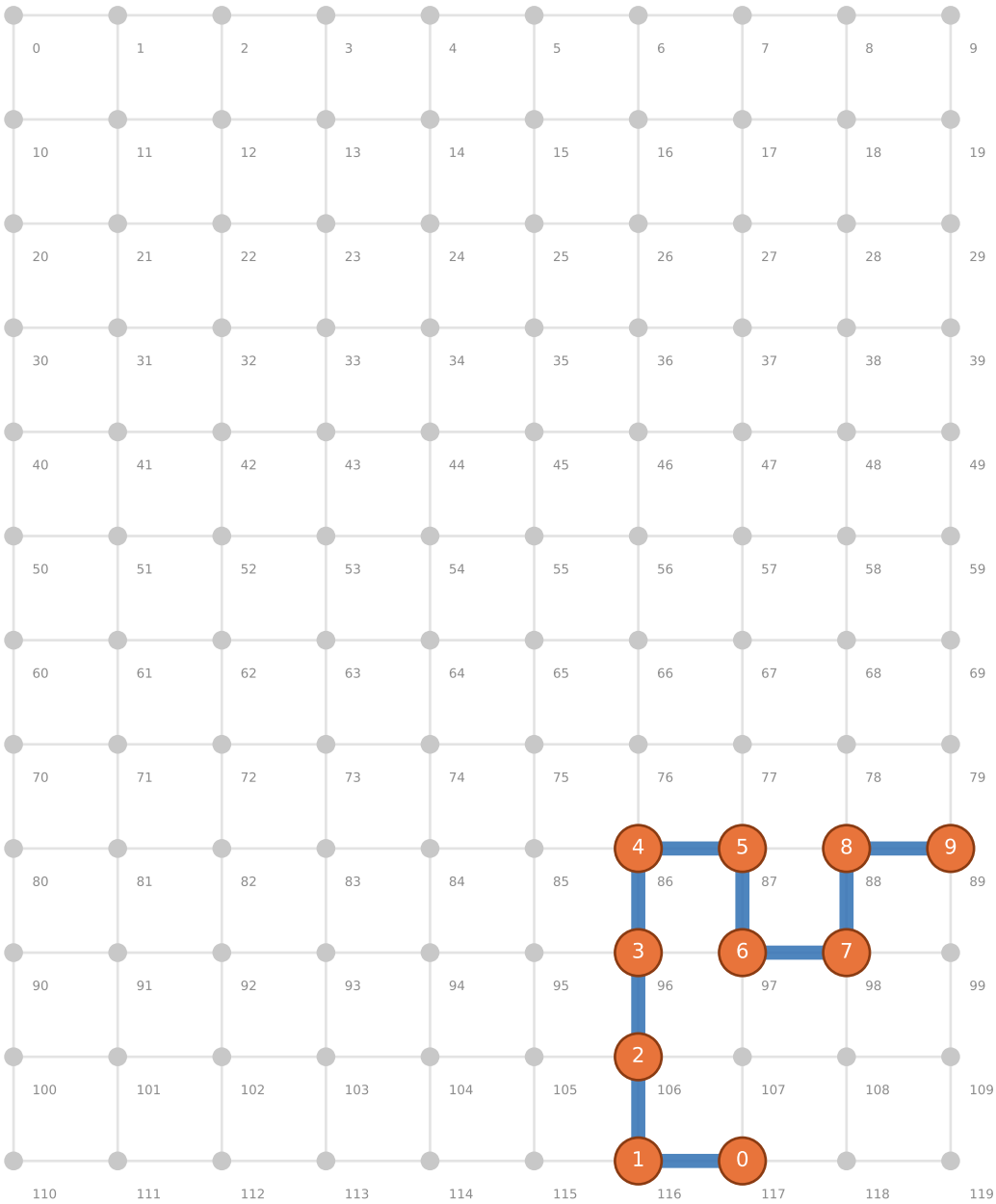


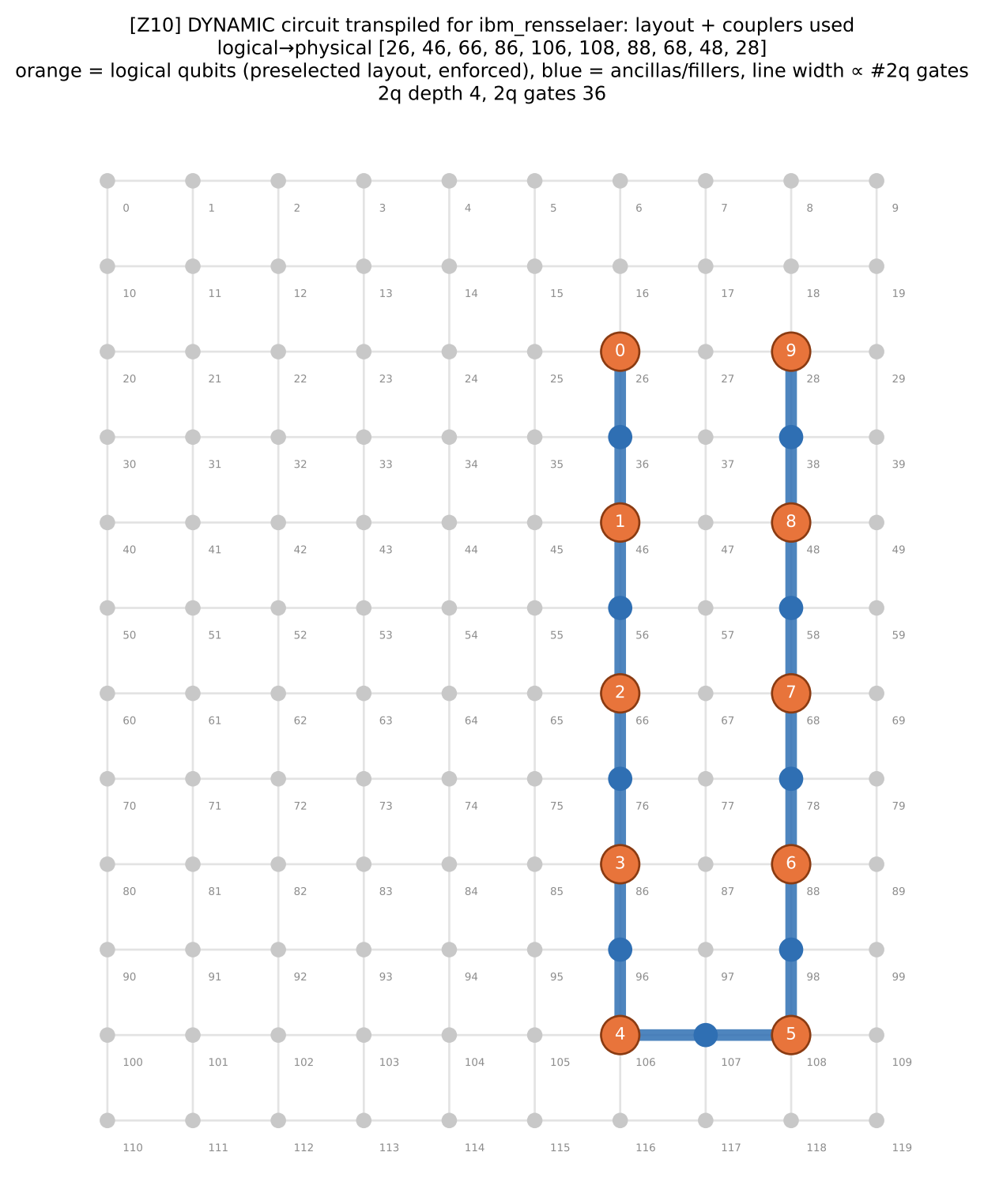
[Z10] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9
chain length 19, 9 ancillas, 0 |0> fillers

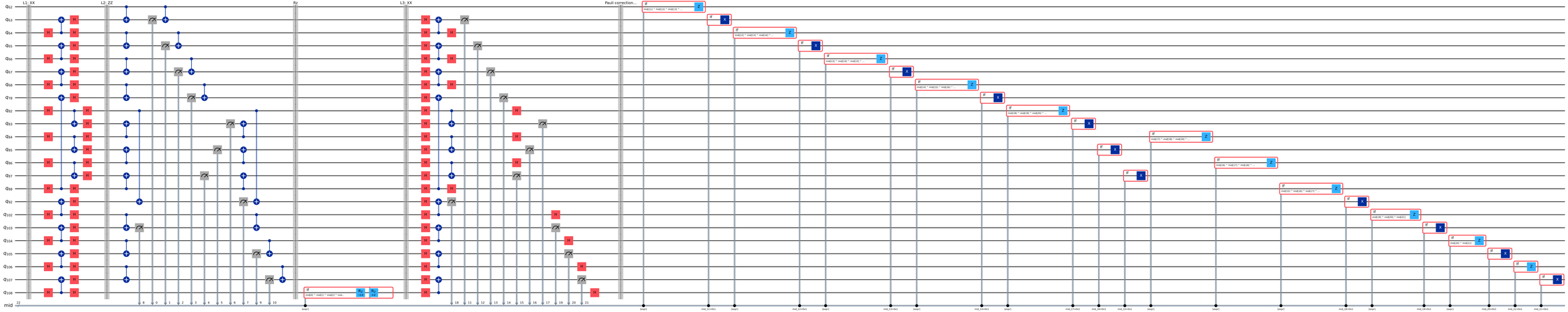


[Z10] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [117, 116, 106, 96, 86, 87, 97, 98, 88, 89] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 18, 2q gates 18 (+0 vs. untranspiled from routing)

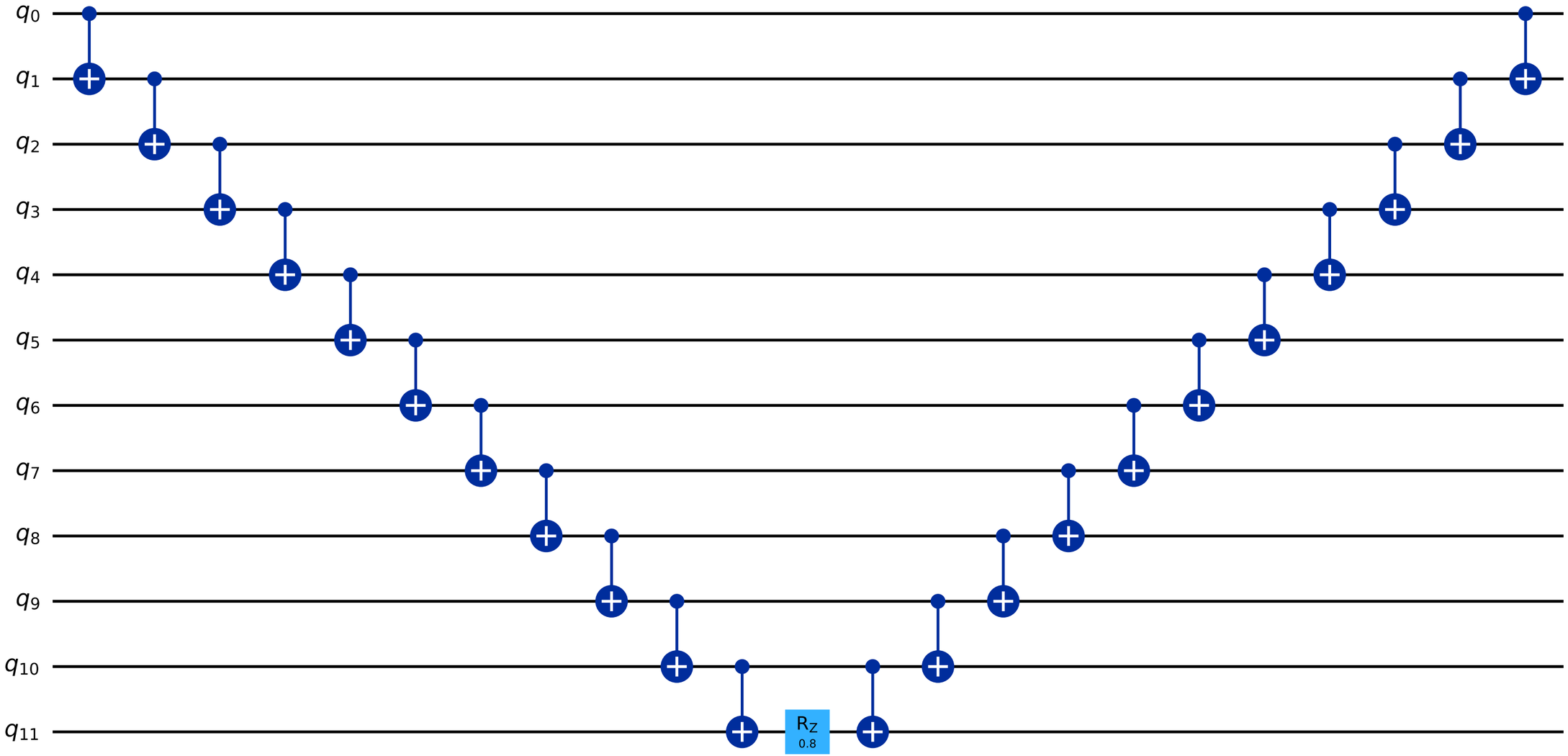



[Z10] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 36
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
3
4
5
6
7
8
9

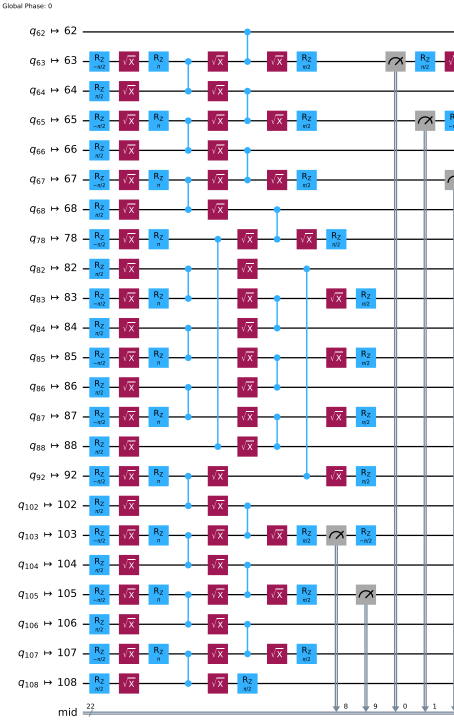
[Z12] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = ZQ Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 (12 logical qubits)
2q depth 4, 22 mid-circuit measurements, logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]

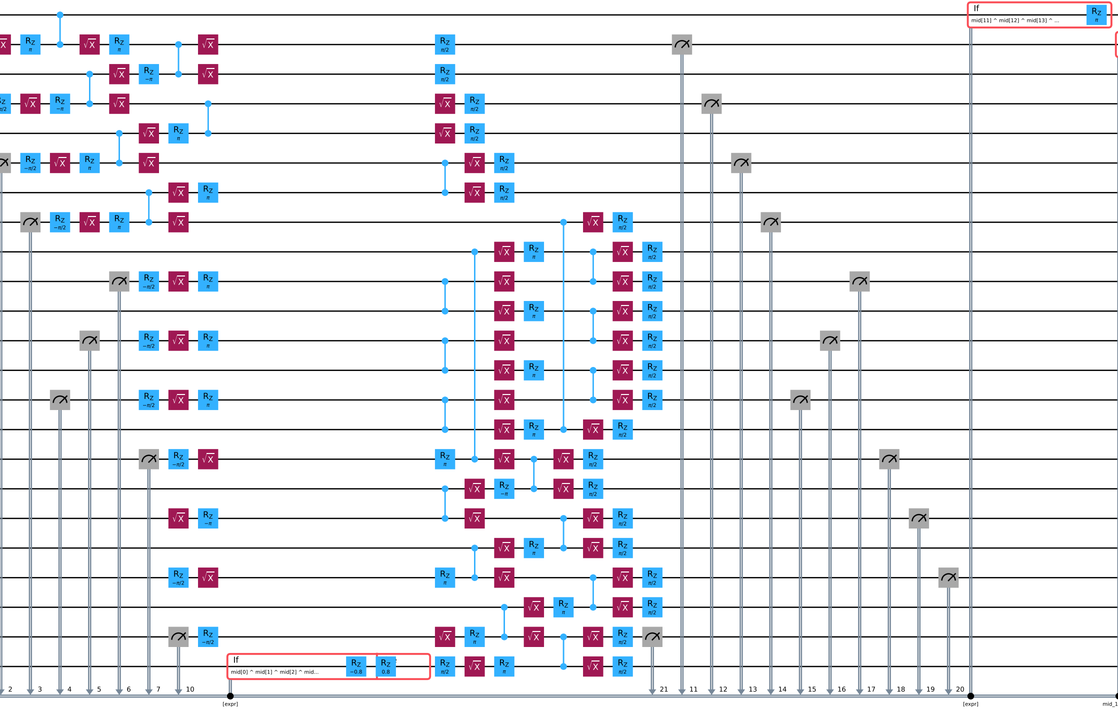


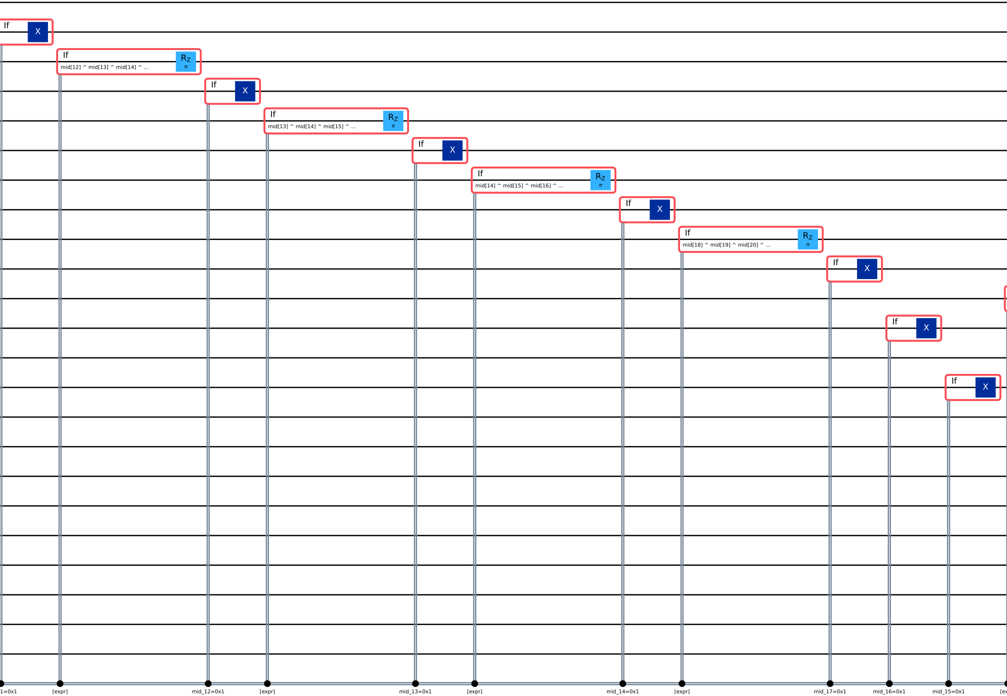
[Z12] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 22

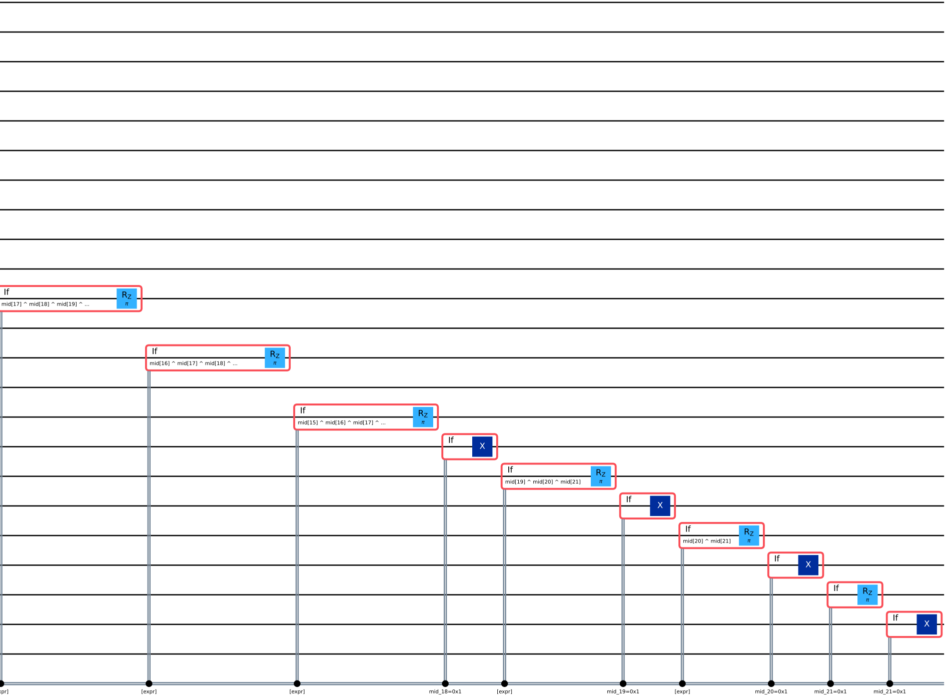


[Z12] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:110 rz:102 cz:44 if_else:23 measure:22)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 4, total depth 31

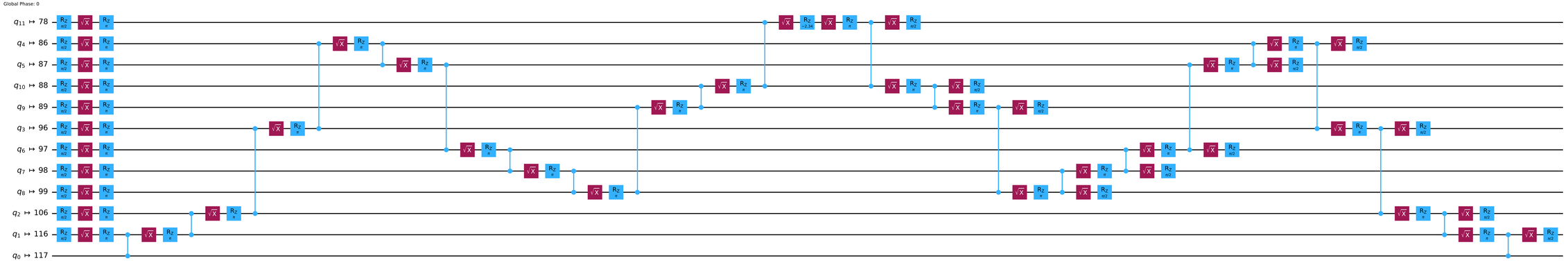

Global Phase: 0
$q_{62} \mapsto 62$
$q_{63} \mapsto 63$
$q_{64} \mapsto 64$
$q_{65} \mapsto 65$
$q_{66} \mapsto 66$
$q_{67} \mapsto 67$
$q_{68} \mapsto 68$
$q_{78} \mapsto 78$
$q_{82} \mapsto 82$
$q_{83} \mapsto 83$
$q_{84} \mapsto 84$
$q_{85} \mapsto 85$
$q_{86} \mapsto 86$
$q_{87} \mapsto 87$
$q_{88} \mapsto 88$
$q_{92} \mapsto 92$
$q_{102} \mapsto 102$
$q_{103} \mapsto 103$
$q_{104} \mapsto 104$
$q_{105} \mapsto 105$
$q_{106} \mapsto 106$
$q_{107} \mapsto 107$
$q_{108} \mapsto 108$
mid
22
R_Z
$-\pi/2$
$\pi/2$
π
$\sqrt{X}$
8
9
0
1


If
mid[11] ^ mid[12] ^ mid[13] ^ ...
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
RZ
π
π/2
−π/2
−π
−0.8
0.8
√X
[expr]
[expr]
2
3
4
5
6
7
10
21
11
12
13
14
15
16
17
18
19
20
mid_1


If
X
If
mid[12] ^ mid[13] ^ mid[14] ^ ...
R_Z
π
If
X
If
mid[13] ^ mid[14] ^ mid[15] ^ ...
R_Z
π
If
X
If
mid[14] ^ mid[15] ^ mid[16] ^ ...
R_Z
π
If
X
If
mid[18] ^ mid[19] ^ mid[20] ^ ...
R_Z
π
If
X
If
X
If
X
1=0x1
[expr]
mid_12=0x1
[expr]
mid_13=0x1
[expr]
mid_14=0x1
[expr]
mid_17=0x1
mid_16=0x1
mid_15=0x1
[ex

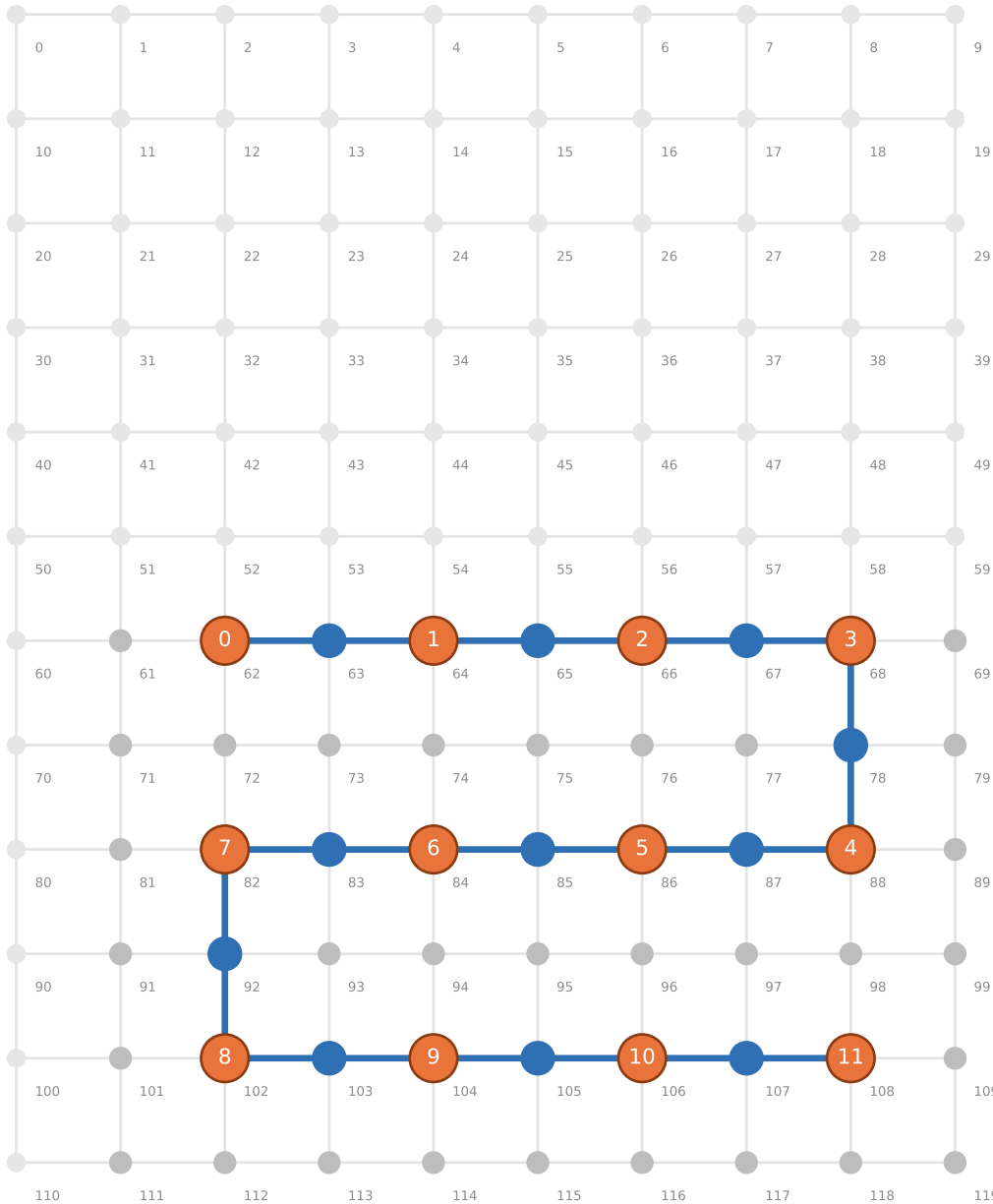

If
mid[17] ^ mid[18] ^ mid[19] ^ ...
R_Z
π
If
mid[16] ^ mid[17] ^ mid[18] ^ ...
R_Z
π
If
mid[15] ^ mid[16] ^ mid[17] ^ ...
R_Z
π
If
X
If
mid[19] ^ mid[20] ^ mid[21]
R_Z
π
If
X
If
mid[20] ^ mid[21]
R_Z
π
If
X
If
R_Z
π
If
X
xpr]
[expr]
[expr]
mid_18=0x1
[expr]
mid_19=0x1
[expr]
mid_20=0x1
mid_21=0x1
mid_21=0x1

[Z12] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:55 sx:44 cz:22)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 (12 logical qubits)
2q depth 22, total depth 71

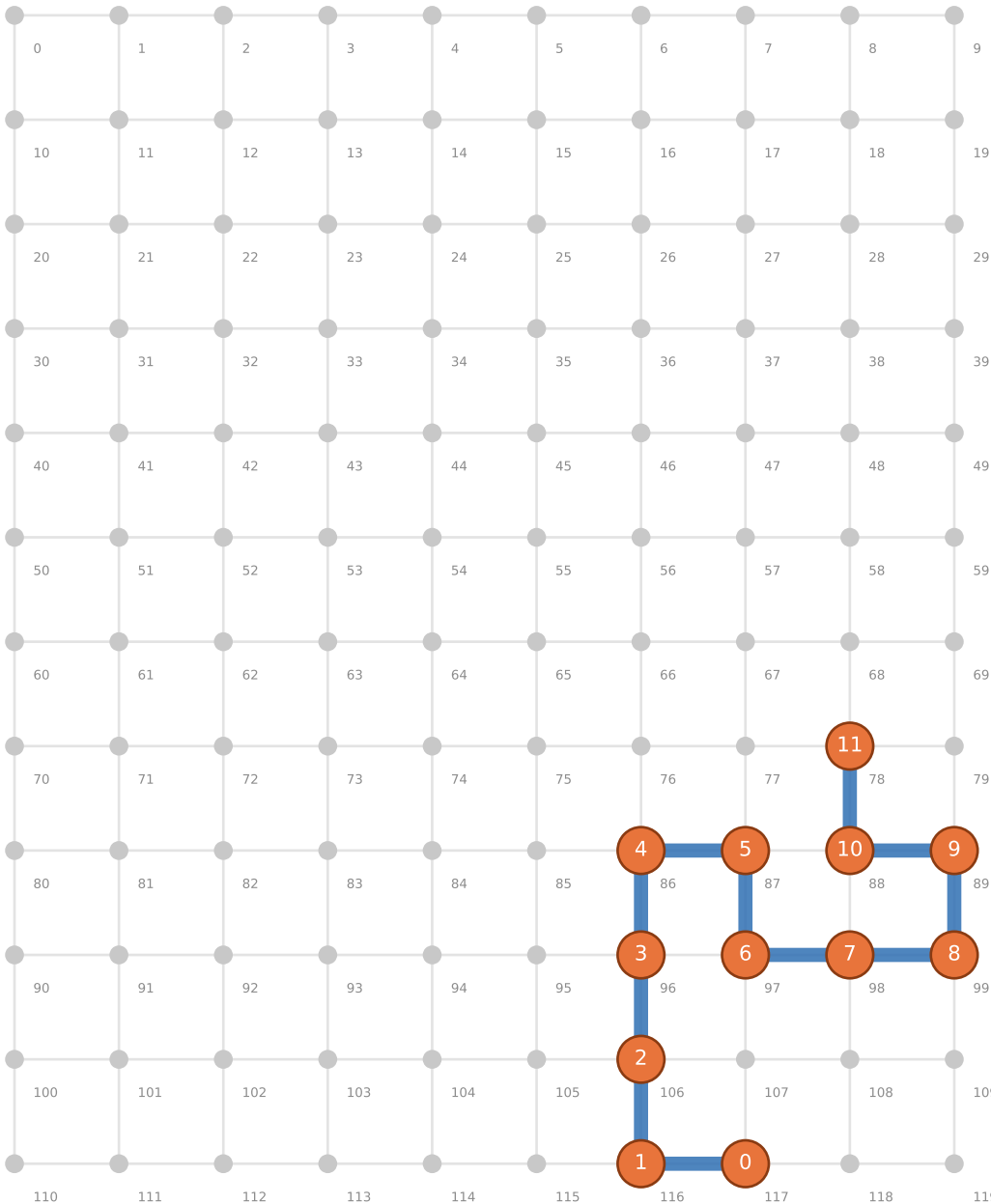


[Z12] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

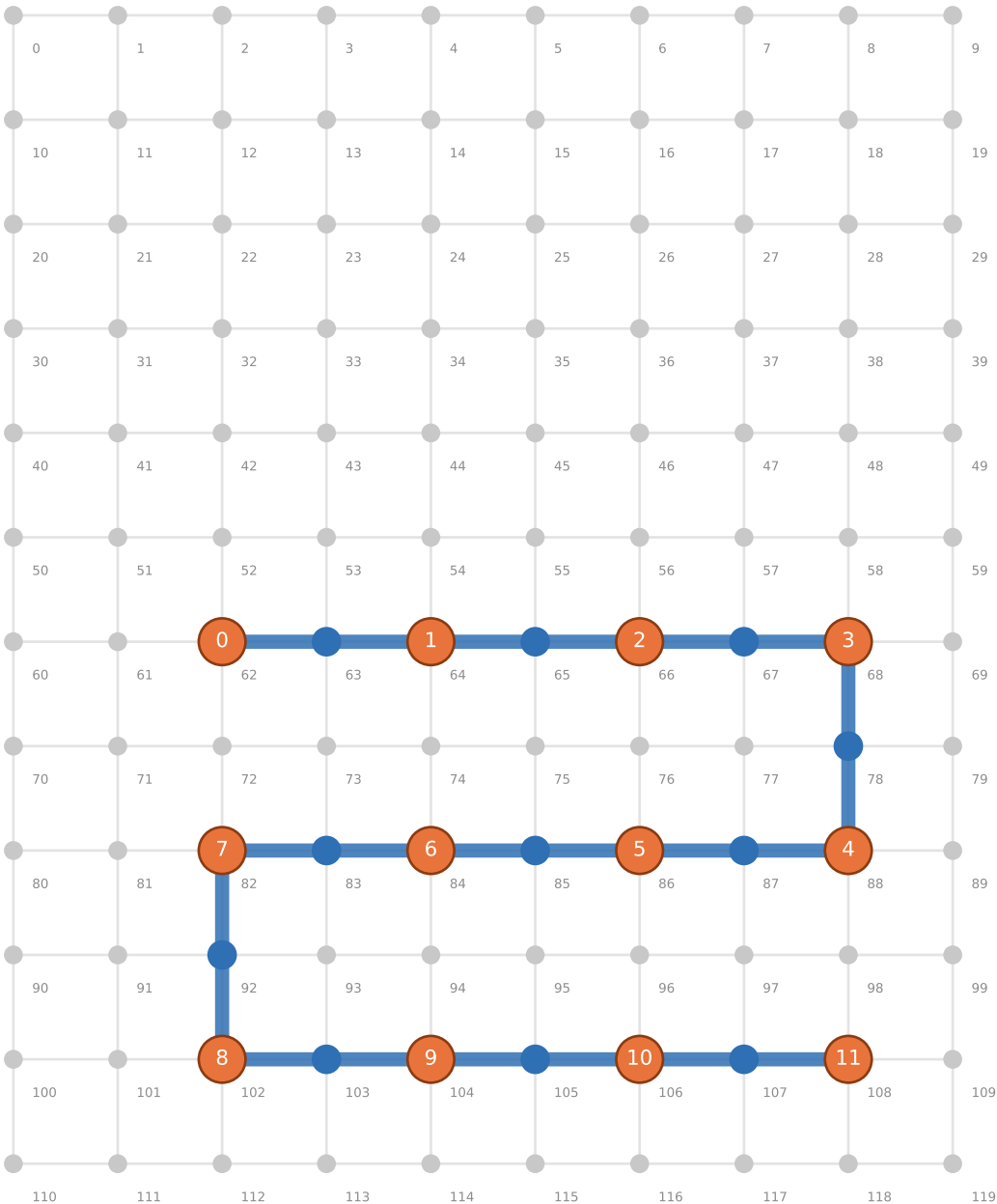
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11
chain length 23, 11 ancillas, 0 |0> fillers



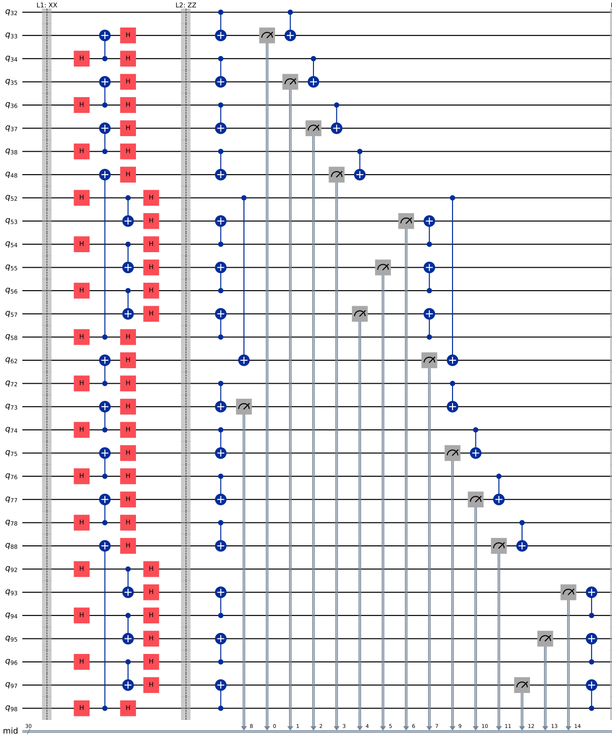
[Z12] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [117, 116, 106, 96, 86, 87, 97, 98, 99, 89, 88, 78] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 22, 2q gates 22 (+0 vs. untranspiled from routing)

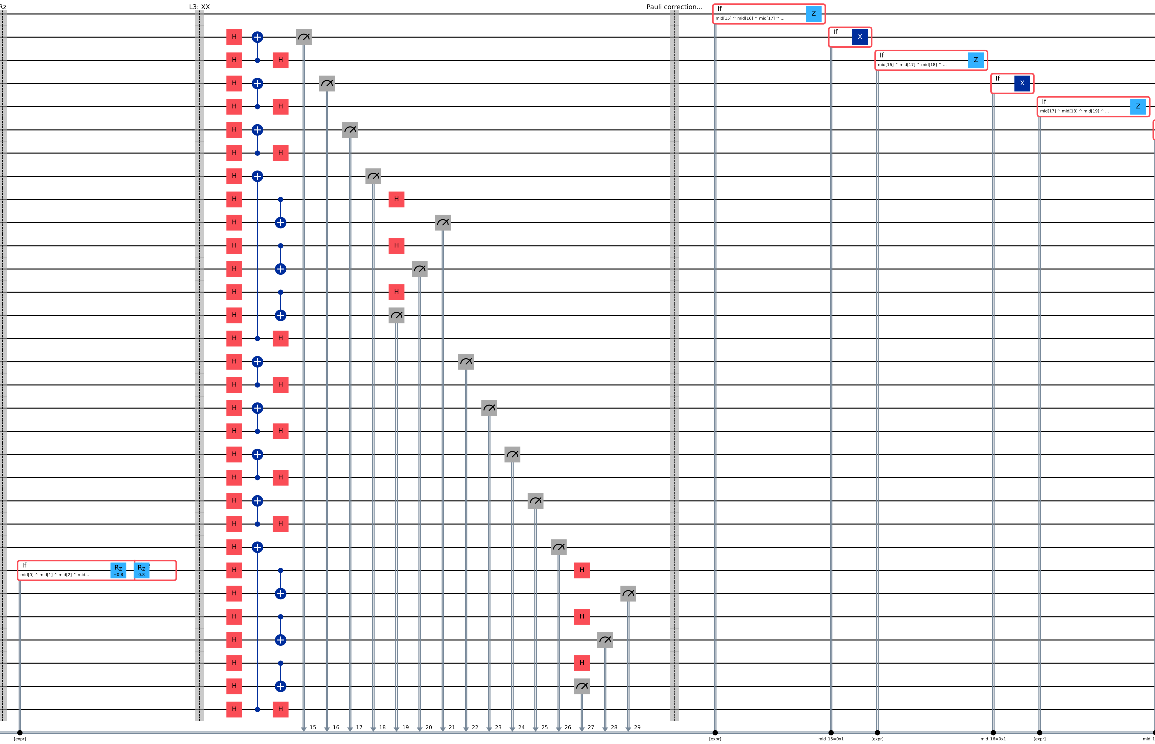


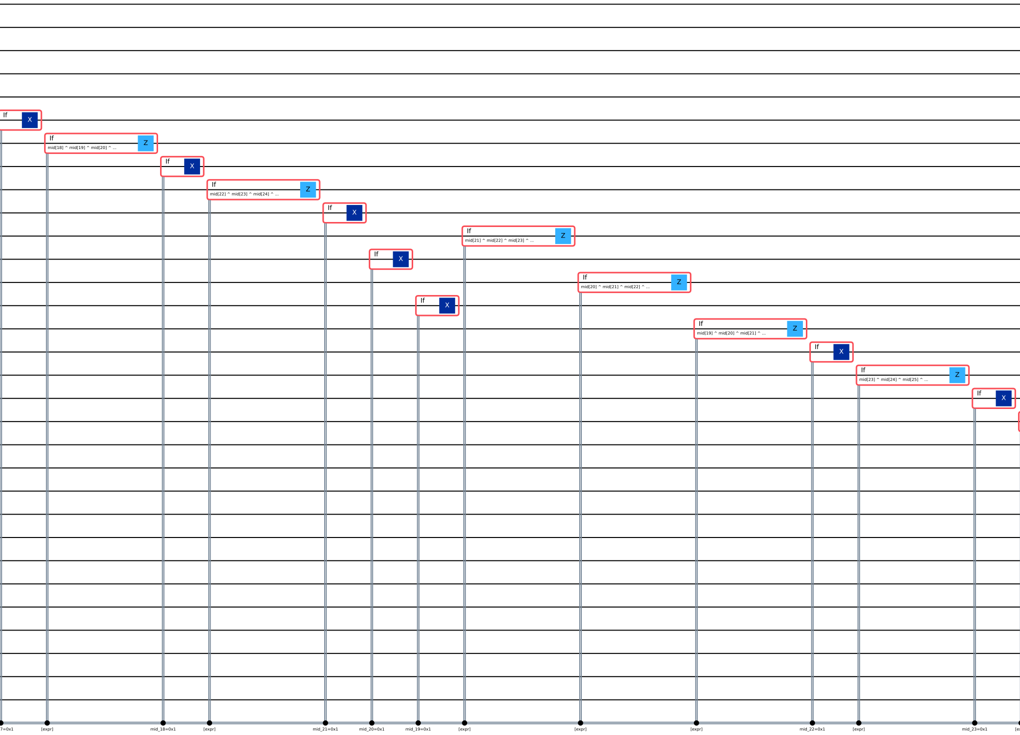
[Z12] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 44

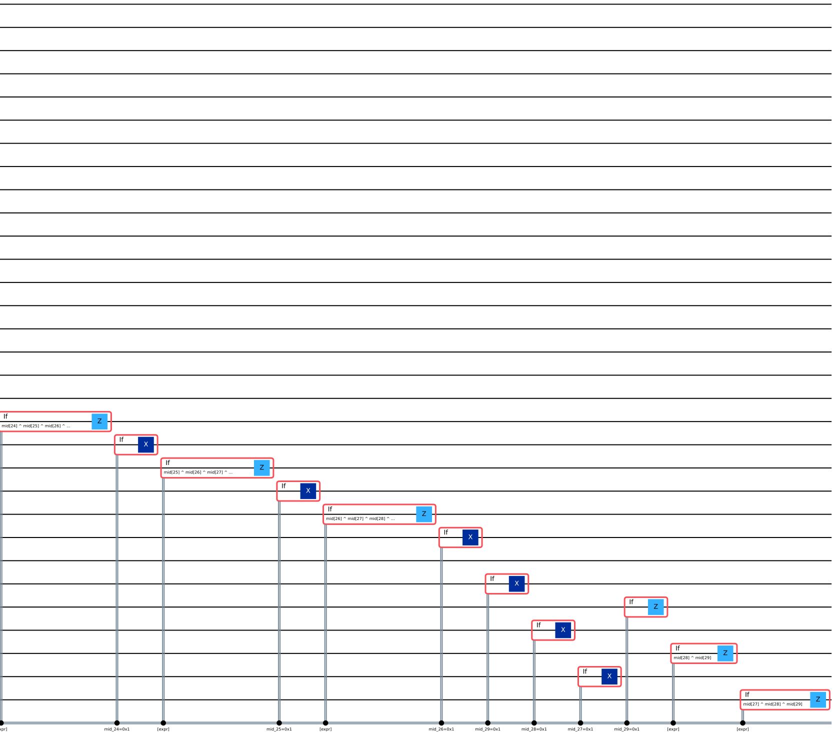


[Z16] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 4, 30 mid-circuit measurements, logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]

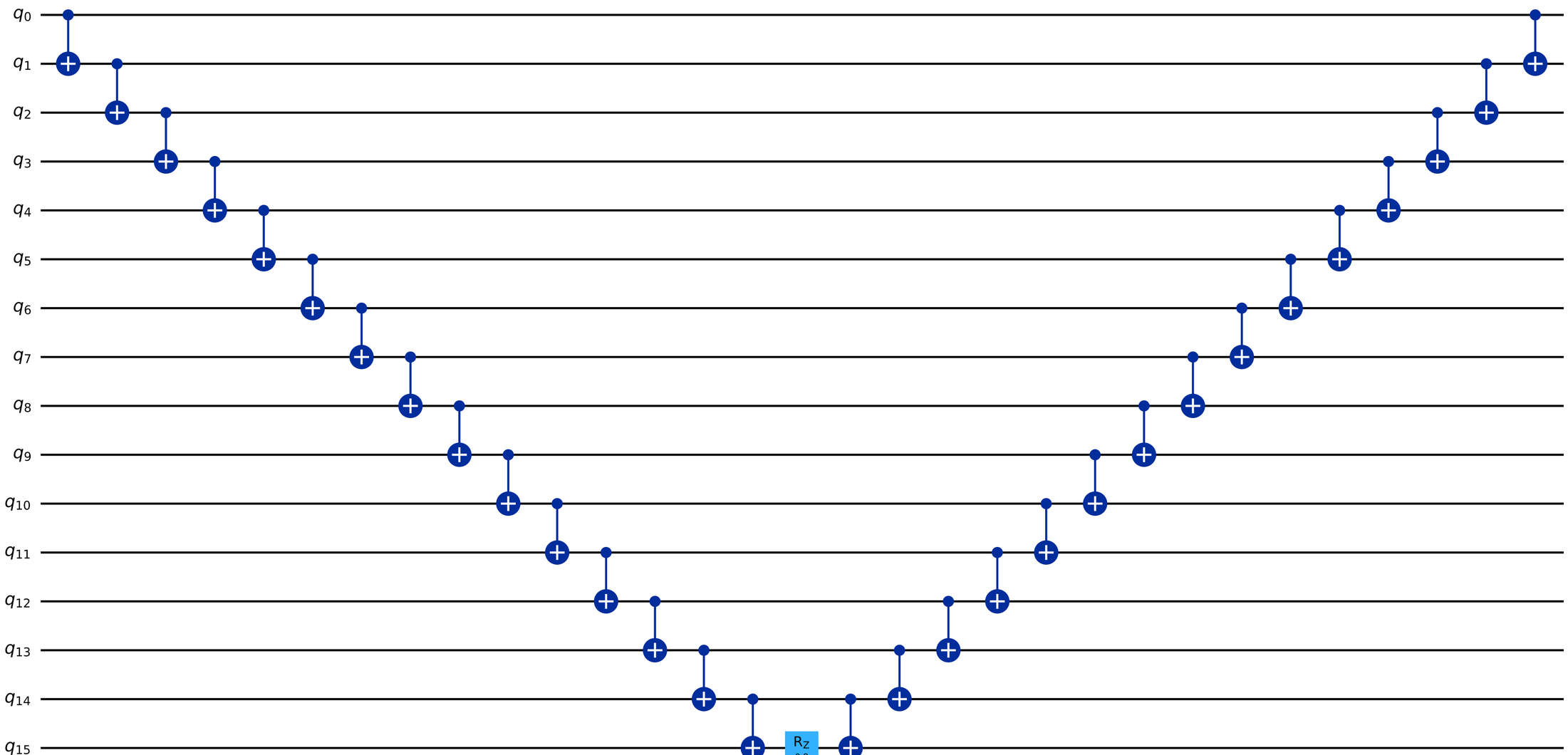

L1: XX
L2: ZZ
q_{32}
q_{33}
q_{34}
q_{35}
q_{36}
q_{37}
q_{38}
q_{48}
q_{52}
q_{53}
q_{54}
q_{55}
q_{56}
q_{57}
q_{58}
q_{62}
q_{72}
q_{73}
q_{74}
q_{75}
q_{76}
q_{77}
q_{78}
q_{88}
q_{92}
q_{93}
q_{94}
q_{95}
q_{96}
q_{97}
q_{98}
mid
30
H
8
0
1
2
3
4
5
6
7
9
10
11
12
13
14


Rz
L3: XX
Pauli correction...
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
RZ −0.8
RZ 0.8
[expr]
H
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
If
mid[15] ^ mid[16] ^ mid[17] ^ ...
Z
If
X
If
mid[16] ^ mid[17] ^ mid[18] ^ ...
Z
If
X
If
mid[17] ^ mid[18] ^ mid[19] ^ ...
Z
[expr]
mid_15=0x1
[expr]
mid_16=0x1
[expr]
mid_1

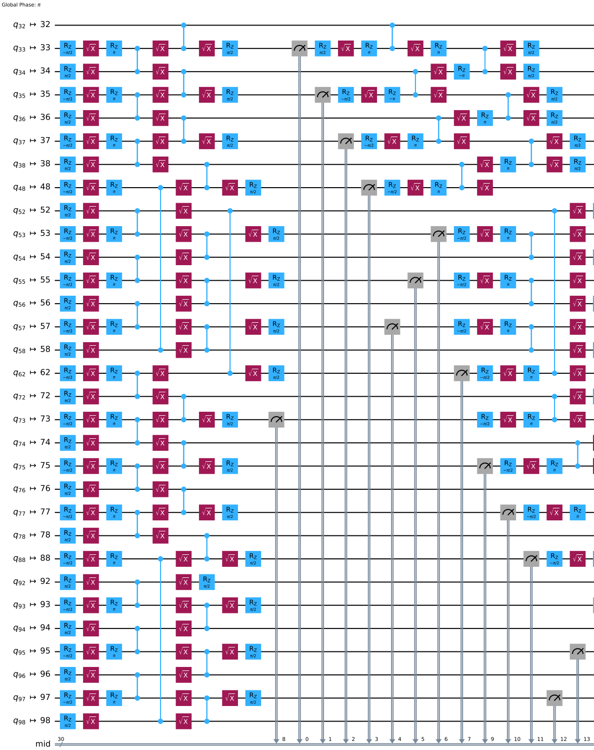

If
X
If
mid[18] ^ mid[19] ^ mid[20] ^ ...
Z
If
X
If
mid[22] ^ mid[23] ^ mid[24] ^ ...
Z
If
X
If
mid[21] ^ mid[22] ^ mid[23] ^ ...
Z
If
X
If
X
If
mid[20] ^ mid[21] ^ mid[22] ^ ...
Z
If
mid[19] ^ mid[20] ^ mid[21] ^ ...
Z
If
X
If
mid[23] ^ mid[24] ^ mid[25] ^ ...
Z
If
X
7=0x1
[expr]
mid_18=0x1
[expr]
mid_21=0x1
mid_20=0x1
mid_19=0x1
[expr]
[expr]
[expr]
mid_22=0x1
[expr]
mid_23=0x1

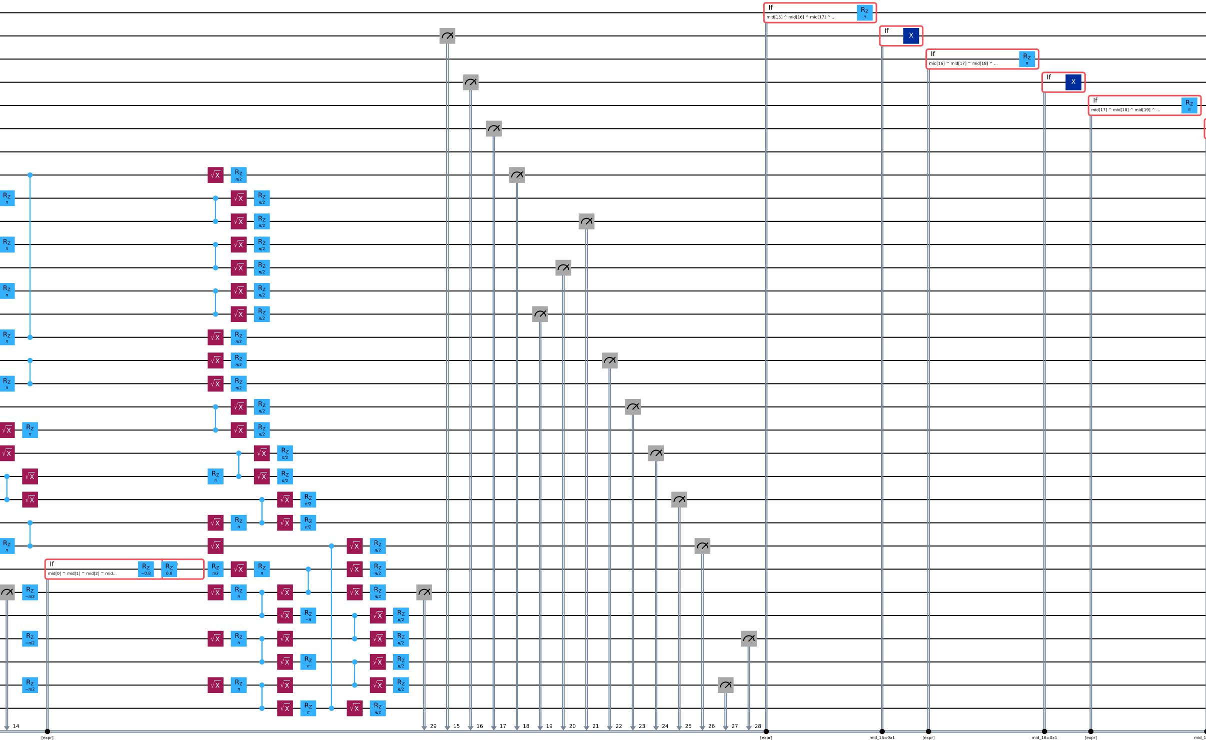

If
mid[24] ^ mid[25] ^ mid[26] ^ ...
Z
If
X
If
mid[25] ^ mid[26] ^ mid[27] ^ ...
Z
If
X
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
Z
If
X
If
X
If
X
If
X
If
Z
If
mid[28] ^ mid[29]
Z
If
mid[27] ^ mid[28] ^ mid[29]
Z
xpr]
mid_24=0x1
[expr]
mid_25=0x1
[expr]
mid_26=0x1
mid_29=0x1
mid_28=0x1
mid_27=0x1
mid_29=0x1
[expr]
[expr]

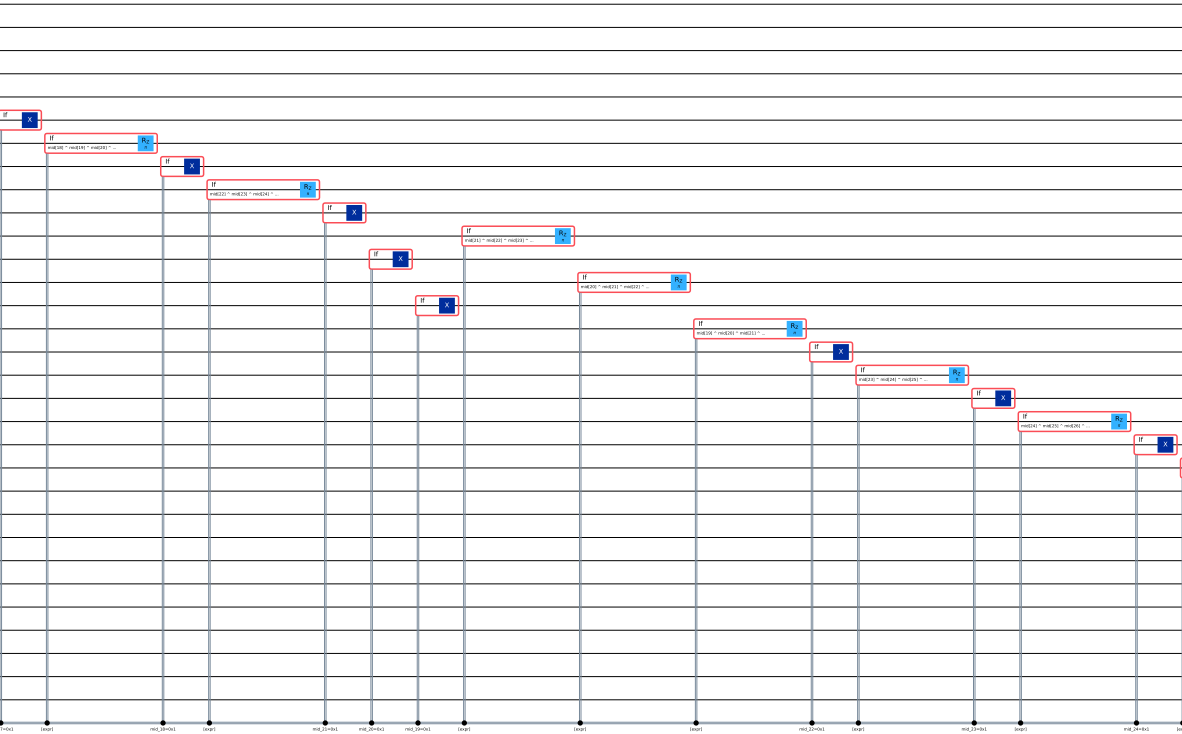
[Z16] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 30

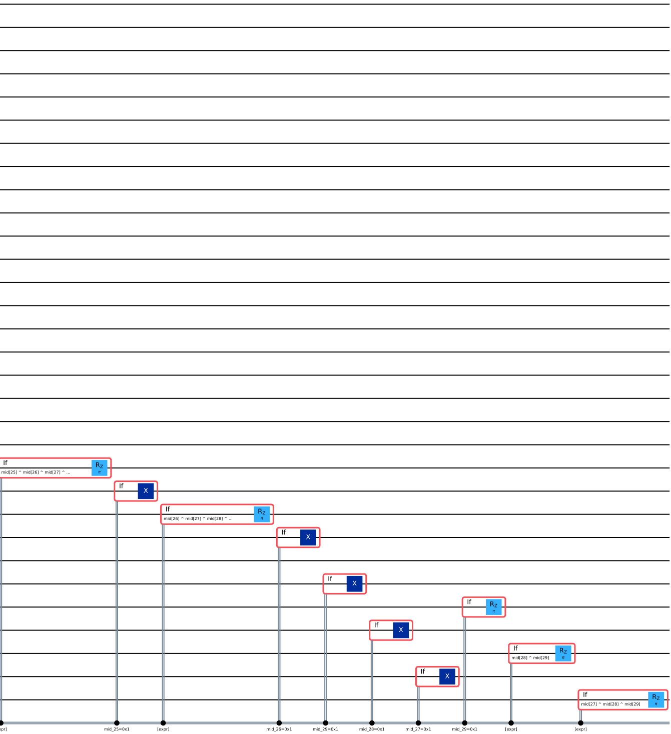


[Z16] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:150 rz:138 cz:60 if_else:31 measure:30)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 4, total depth 35

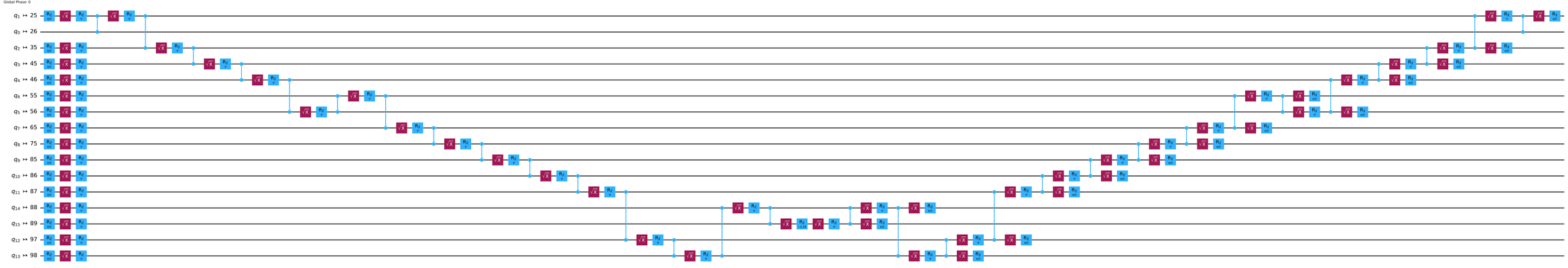

Global Phase: π
$q_{32} \mapsto 32$
$q_{33} \mapsto 33$
$q_{34} \mapsto 34$
$q_{35} \mapsto 35$
$q_{36} \mapsto 36$
$q_{37} \mapsto 37$
$q_{38} \mapsto 38$
$q_{48} \mapsto 48$
$q_{52} \mapsto 52$
$q_{53} \mapsto 53$
$q_{54} \mapsto 54$
$q_{55} \mapsto 55$
$q_{56} \mapsto 56$
$q_{57} \mapsto 57$
$q_{58} \mapsto 58$
$q_{62} \mapsto 62$
$q_{72} \mapsto 72$
$q_{73} \mapsto 73$
$q_{74} \mapsto 74$
$q_{75} \mapsto 75$
$q_{76} \mapsto 76$
$q_{77} \mapsto 77$
$q_{78} \mapsto 78$
$q_{88} \mapsto 88$
$q_{92} \mapsto 92$
$q_{93} \mapsto 93$
$q_{94} \mapsto 94$
$q_{95} \mapsto 95$
$q_{96} \mapsto 96$
$q_{97} \mapsto 97$
$q_{98} \mapsto 98$
mid
30
8
0
1
2
3
4
5
6
7
9
10
11
12
13


If
mid[15] ^ mid[16] ^ mid[17] ^ ...
Rz
π
If
X
If
mid[16] ^ mid[17] ^ mid[18] ^ ...
Rz
π
If
X
If
mid[17] ^ mid[18] ^ mid[19] ^ ...
Rz
π
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
Rz
−0.8
Rz
0.8
√X
Rz
π/2
−π/2
−π
14
[expr]
29
15
16
17
18
19
20
21
22
23
24
25
26
27
28
[expr]
mid_15=0x1
[expr]
mid_16=0x1
[expr]


If
X
If
mid[18] ^ mid[19] ^ mid[20] ^ ...
R_Z
π
If
X
If
mid[22] ^ mid[23] ^ mid[24] ^ ...
R_Z
π
If
X
If
mid[21] ^ mid[22] ^ mid[23] ^ ...
R_Z
π
If
X
If
mid[20] ^ mid[21] ^ mid[22] ^ ...
R_Z
π
If
X
If
mid[19] ^ mid[20] ^ mid[21] ^ ...
R_Z
π
If
X
If
mid[23] ^ mid[24] ^ mid[25] ^ ...
R_Z
π
If
X
If
mid[24] ^ mid[25] ^ mid[26] ^ ...
R_Z
π
If
X
7=0x1
[expr]
mid_18=0x1
[expr]
mid_21=0x1
mid_20=0x1
mid_19=0x1
[expr]
[expr]
[expr]
mid_22=0x1
[expr]
mid_23=0x1
[expr]
mid_24=0x1

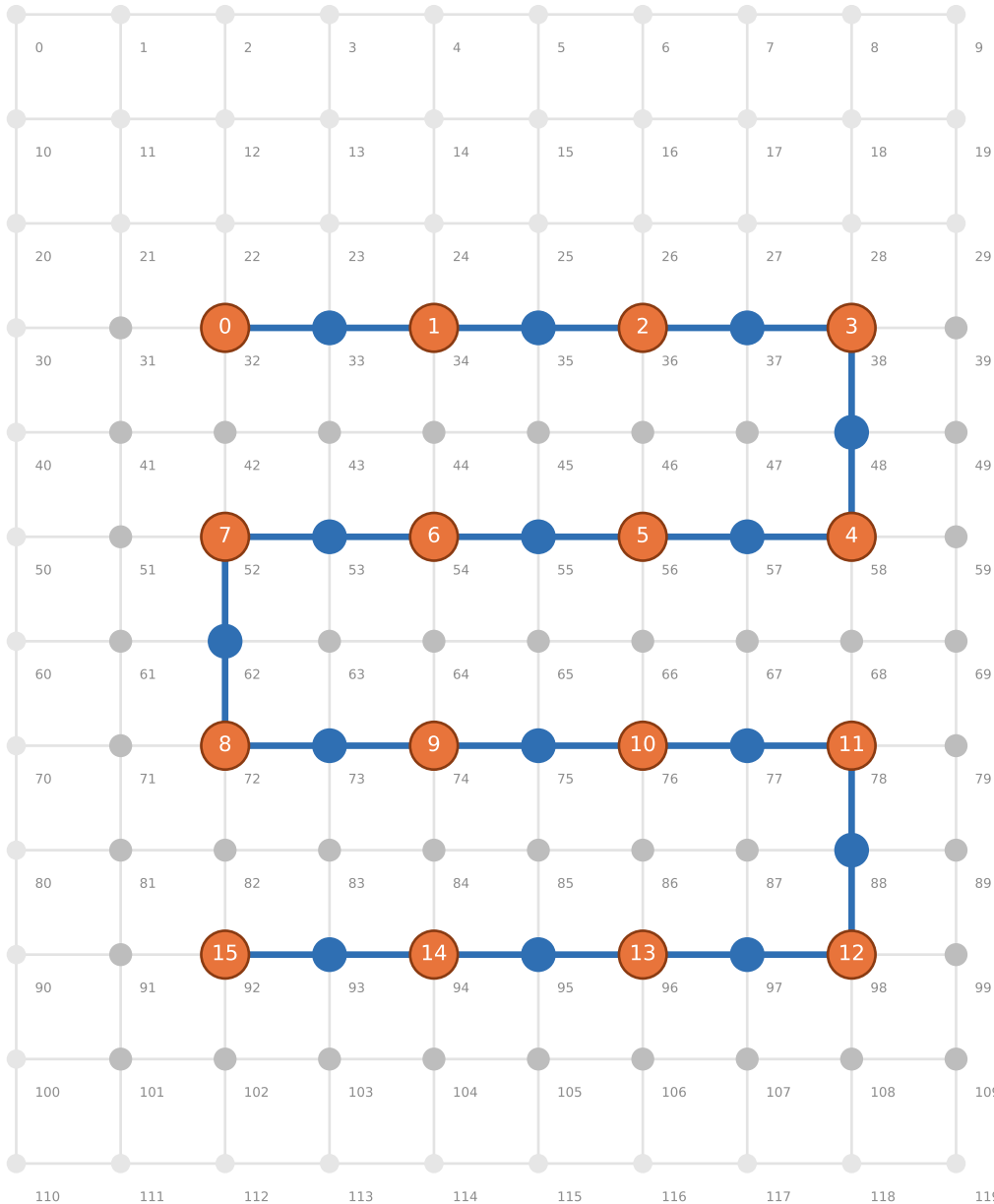

If
mid[25] ^ mid[26] ^ mid[27] ^ ...
R_Z
π
If
X
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
R_Z
π
If
X
If
X
If
X
If
X
If
R_Z
π
If
mid[28] ^ mid[29]
R_Z
π
If
mid[27] ^ mid[28] ^ mid[29]
R_Z
π
xpr]
mid_25=0x1
[expr]
mid_26=0x1
mid_29=0x1
mid_28=0x1
mid_27=0x1
mid_29=0x1
[expr]
[expr]

[Z16] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:75 sx:60 cz:30)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 (16 logical qubits)
2q depth 30, total depth 95

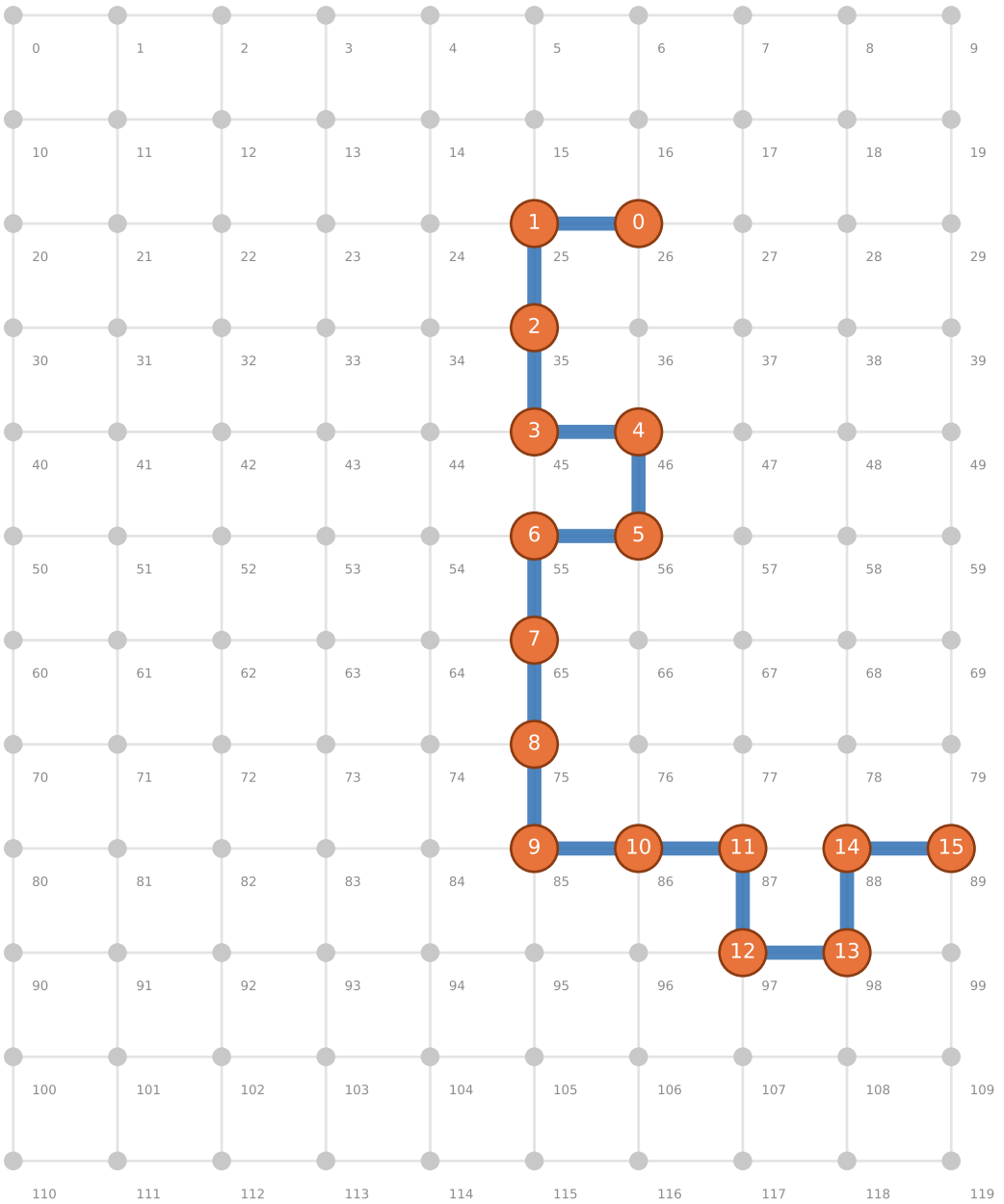


[Z16] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

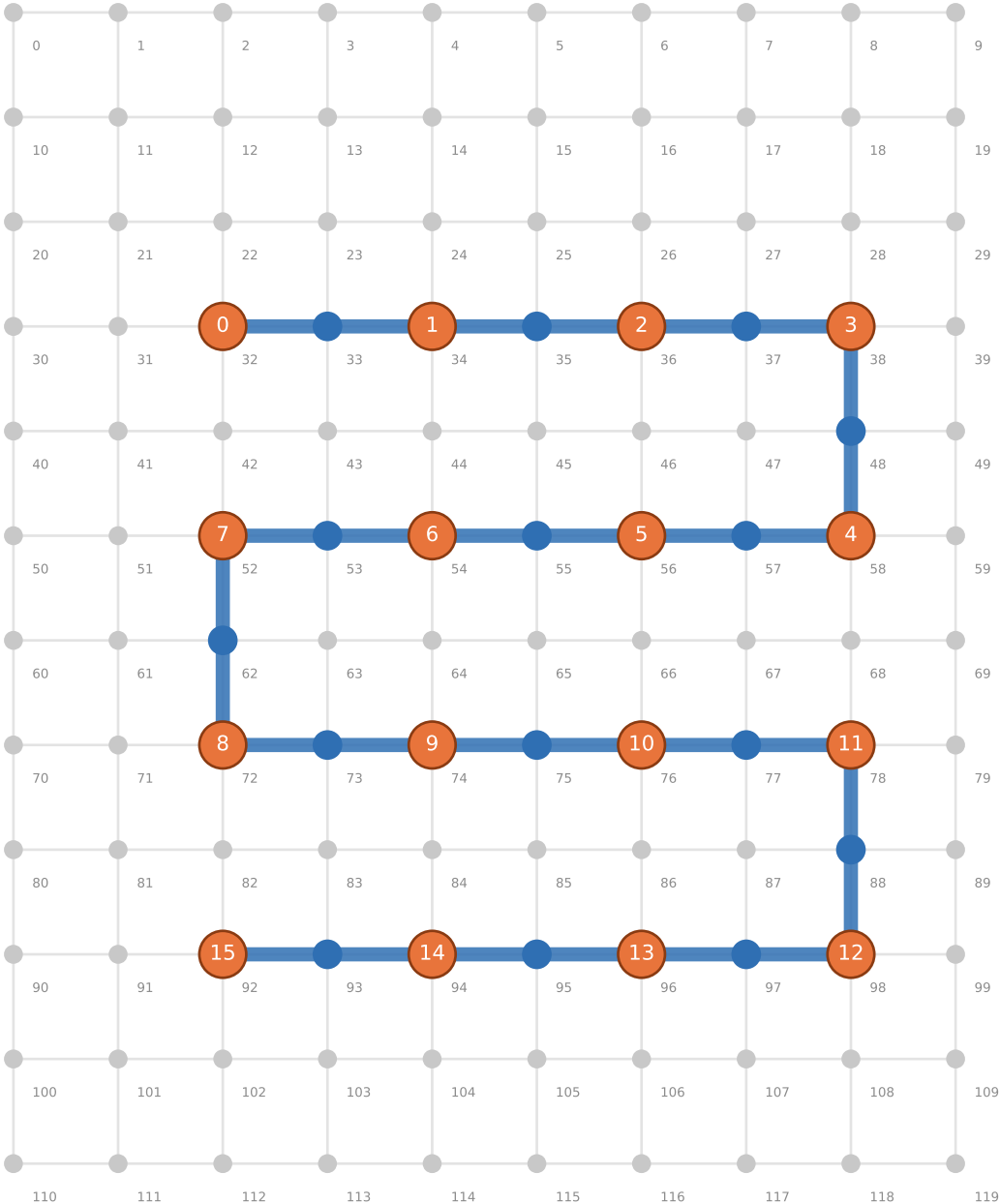
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15
chain length 31, 15 ancillas, 0 |0> fillers



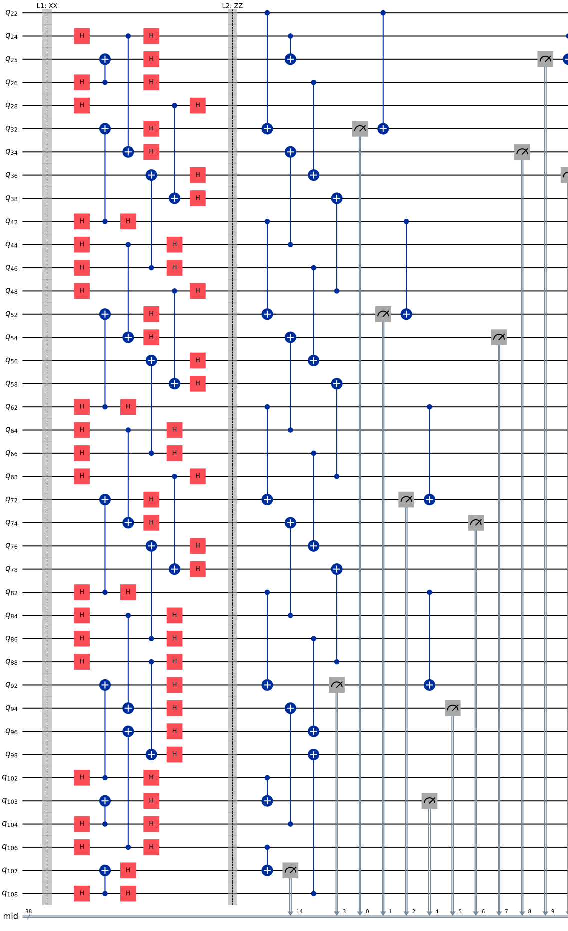
[Z16] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [26, 25, 35, 45, 46, 56, 55, 65, 75, 85, 86, 87, 97, 98, 88, 89] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 30, 2q gates 30 (+0 vs. untranspiled from routing)

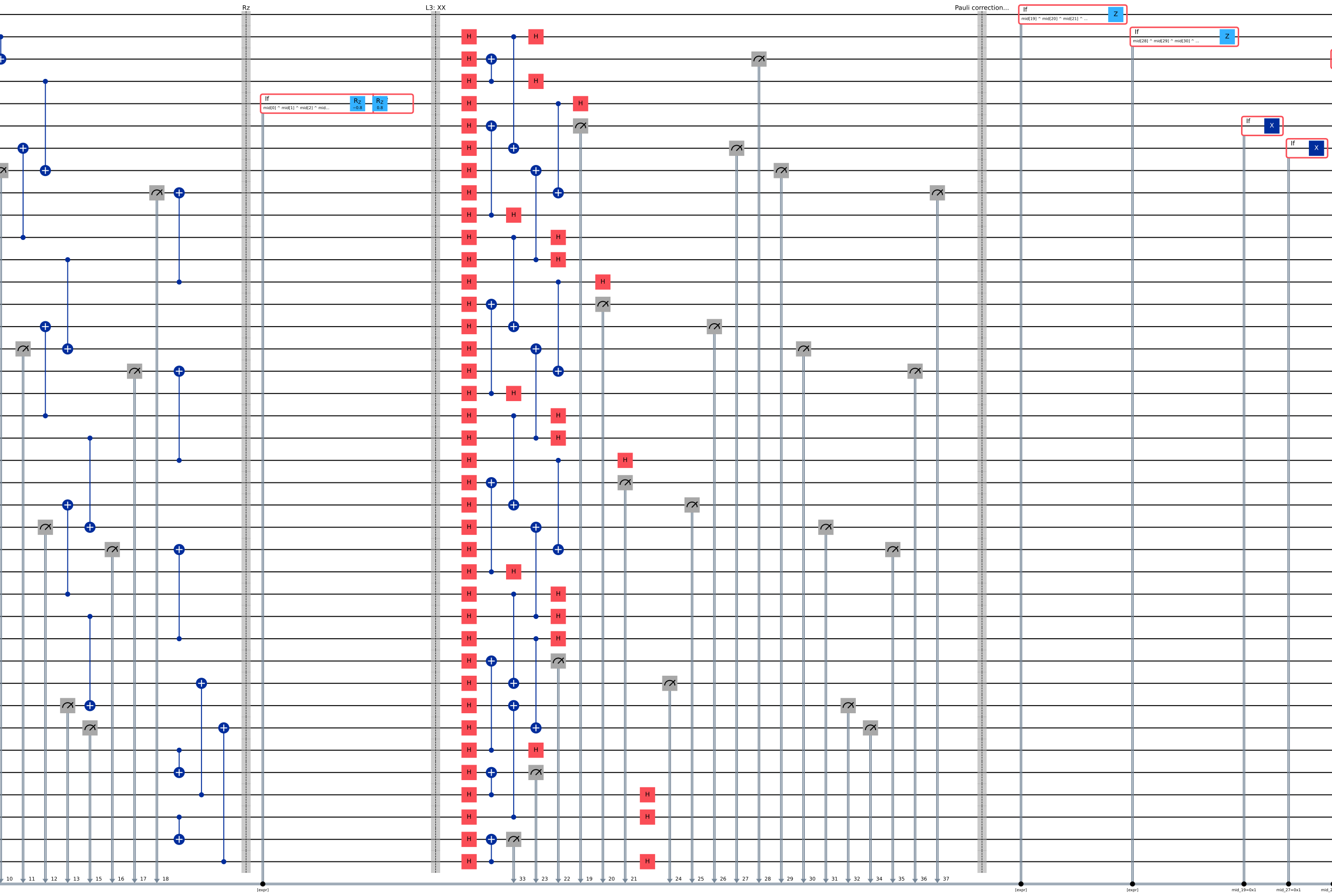


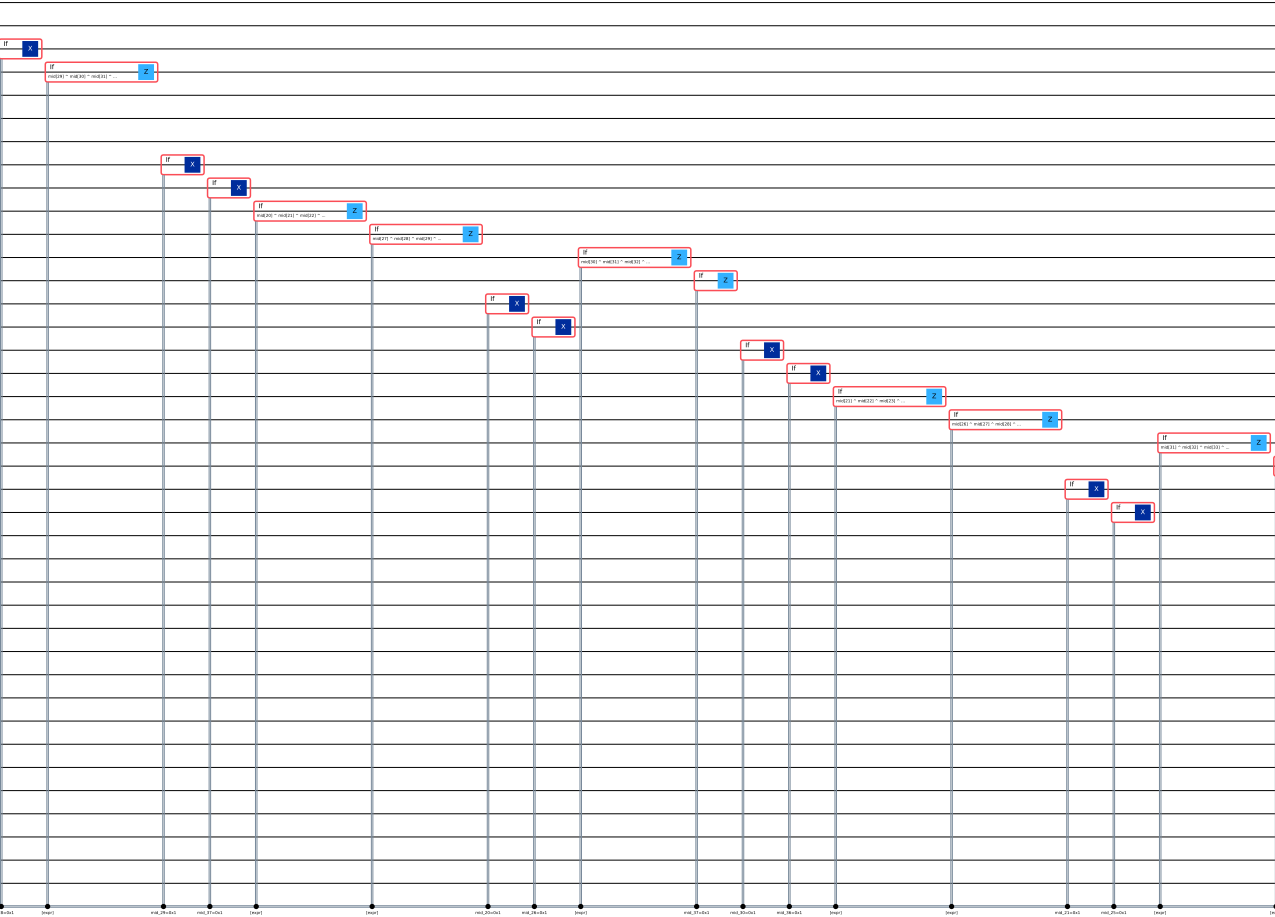
[Z16] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 60

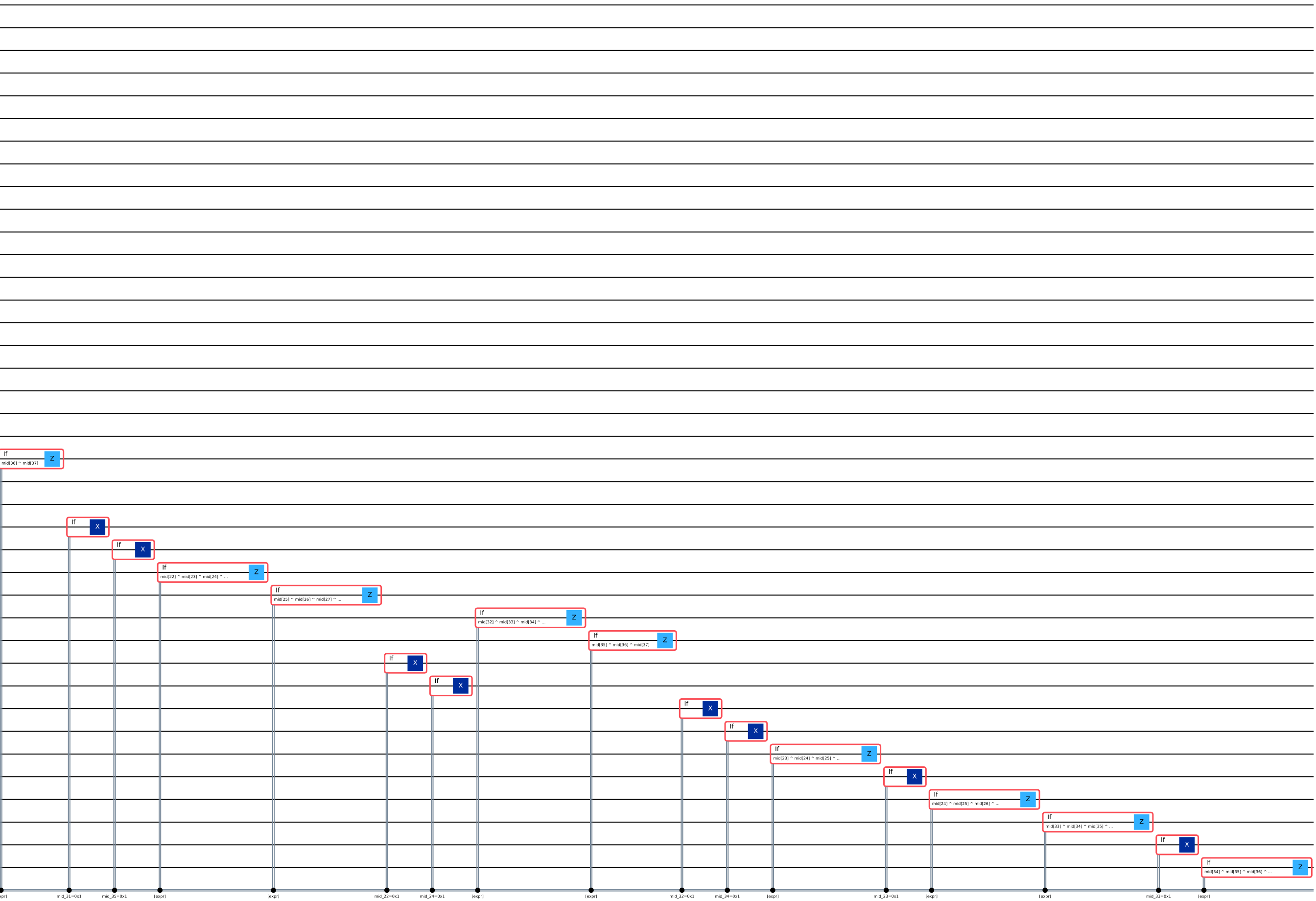


[Z20] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 4, 38 mid-circuit measurements, logical→physical [22, 42, 62, 82, 102, 104, 84, 64, 44, 24, 26, 46, 66, 86, 106, 108, 88, 68, 48, 28]

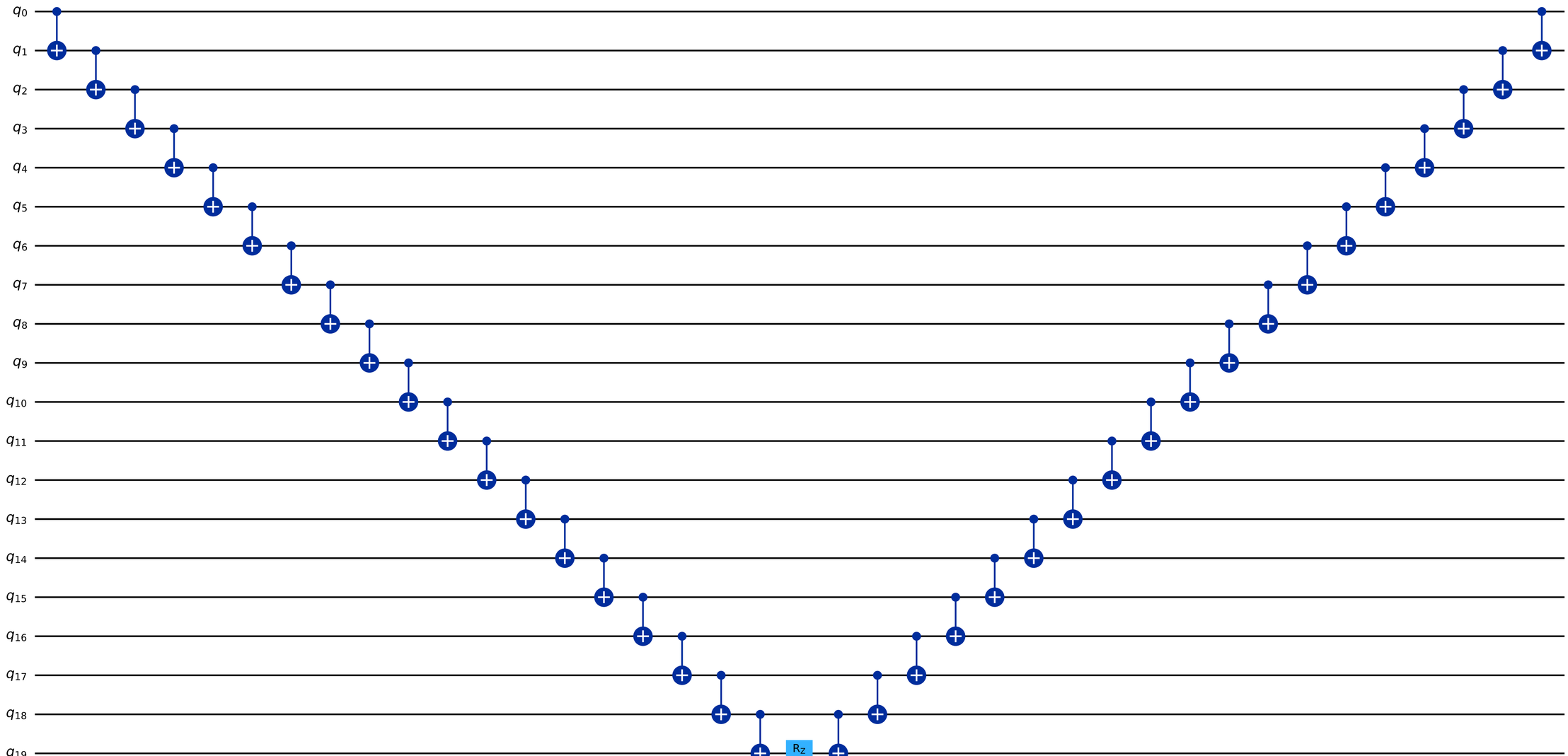

L1: XX
L2: ZZ
q_{22}
q_{24}
q_{25}
q_{26}
q_{28}
q_{32}
q_{34}
q_{36}
q_{38}
q_{42}
q_{44}
q_{46}
q_{48}
q_{52}
q_{54}
q_{56}
q_{58}
q_{62}
q_{64}
q_{66}
q_{68}
q_{72}
q_{74}
q_{76}
q_{78}
q_{82}
q_{84}
q_{86}
q_{88}
q_{92}
q_{94}
q_{96}
q_{98}
q_{102}
q_{103}
q_{104}
q_{106}
q_{107}
q_{108}
mid
38
H
14
3
0
1
2
4
5
6
7
8
9


Rz
L3: XX
Pauli correction...
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
R_Z
−0.8
R_Z
0.8
If
mid[19] ^ mid[20] ^ mid[21] ^ ...
Z
If
mid[28] ^ mid[29] ^ mid[30] ^ ...
Z
If
X
If
X
H
10
11
12
13
15
16
17
18
[expr]
33
23
22
19
20
21
24
25
26
27
28
29
30
31
32
34
35
36
37
[expr]
[expr]
mid_19=0x1
mid_27=0x1

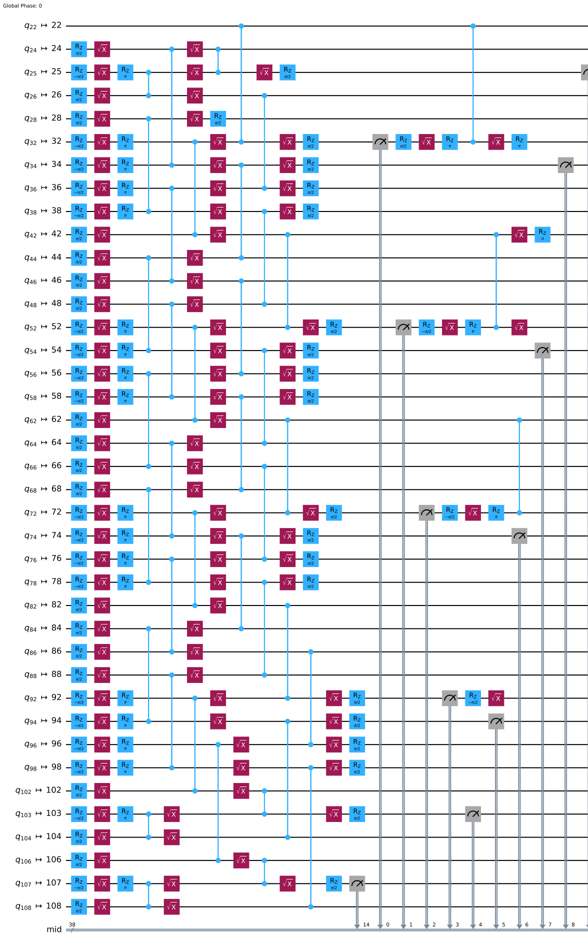

If
X
If
mid[29] ^ mid[30] ^ mid[31] ^ ...
Z
If
X
If
X
If
mid[20] ^ mid[21] ^ mid[22] ^ ...
Z
If
mid[27] ^ mid[28] ^ mid[29] ^ ...
Z
If
X
If
X
If
mid[30] ^ mid[31] ^ mid[32] ^ ...
Z
If
Z
If
X
If
X
If
mid[21] ^ mid[22] ^ mid[23] ^ ...
Z
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
Z
If
X
If
X
If
mid[31] ^ mid[32] ^ mid[33] ^ ...
Z
8=0x1
[expr]
mid_29=0x1
mid_37=0x1
[expr]
[expr]
mid_20=0x1
mid_26=0x1
[expr]
mid_37=0x1
mid_30=0x1
mid_36=0x1
[expr]
[expr]
mid_21=0x1
mid_25=0x1
[expr]
[ex

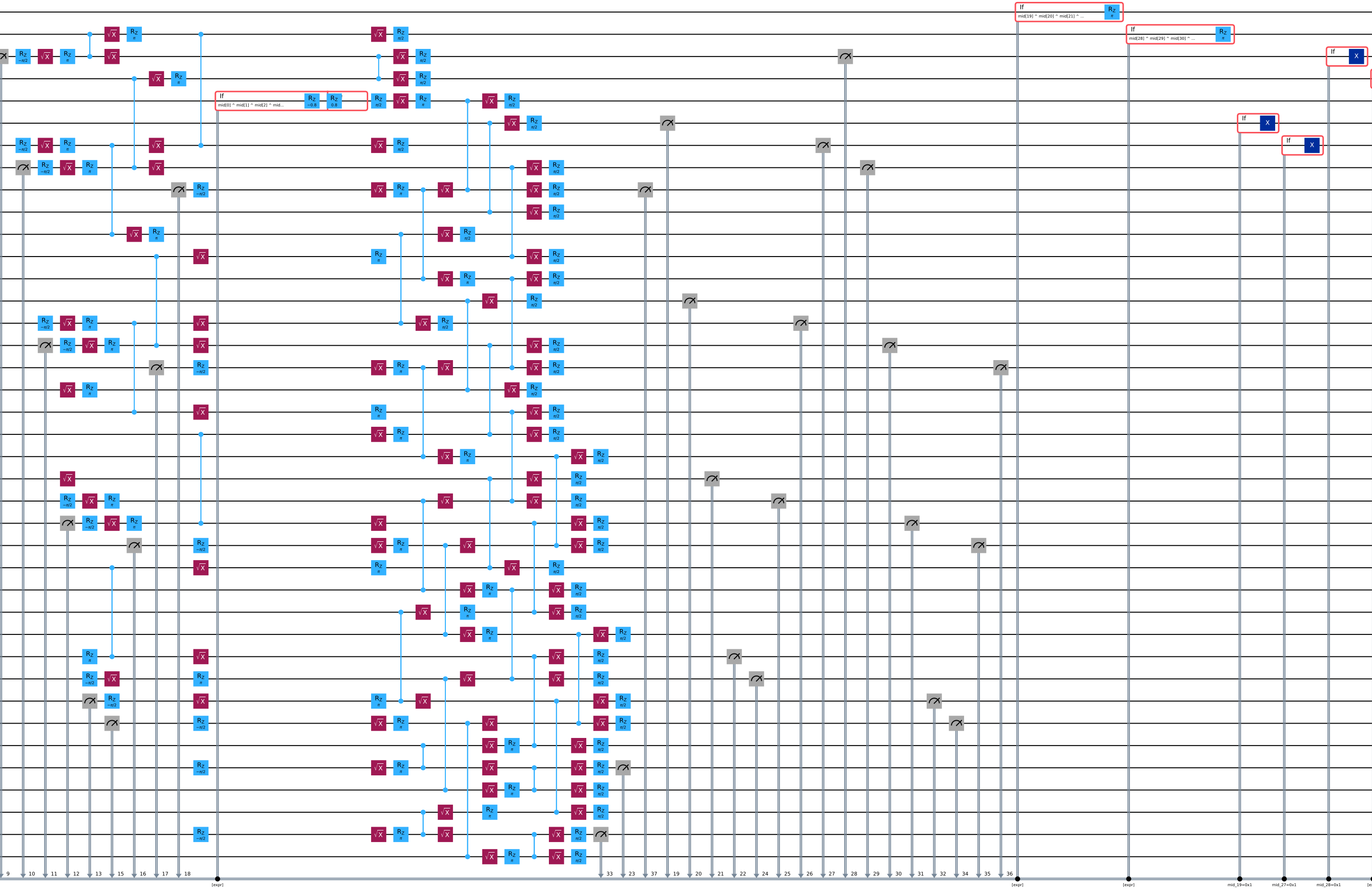

If
Z
mid[36] ^ mid[37]
xpr]
If
X
mid_31=0x1
If
X
mid_35=0x1
If
Z
mid[22] ^ mid[23] ^ mid[24] ^ ...
[expr]
If
Z
mid[25] ^ mid[26] ^ mid[27] ^ ...
[expr]
If
X
mid_22=0x1
If
X
mid_24=0x1
If
Z
mid[32] ^ mid[33] ^ mid[34] ^ ...
[expr]
If
Z
mid[35] ^ mid[36] ^ mid[37]
[expr]
If
X
mid_32=0x1
If
X
mid_34=0x1
If
Z
mid[23] ^ mid[24] ^ mid[25] ^ ...
[expr]
If
X
mid_23=0x1
If
Z
mid[24] ^ mid[25] ^ mid[26] ^ ...
[expr]
If
Z
mid[33] ^ mid[34] ^ mid[35] ^ ...
[expr]
If
X
mid_33=0x1
If
Z
mid[34] ^ mid[35] ^ mid[36] ^ ...
[expr]

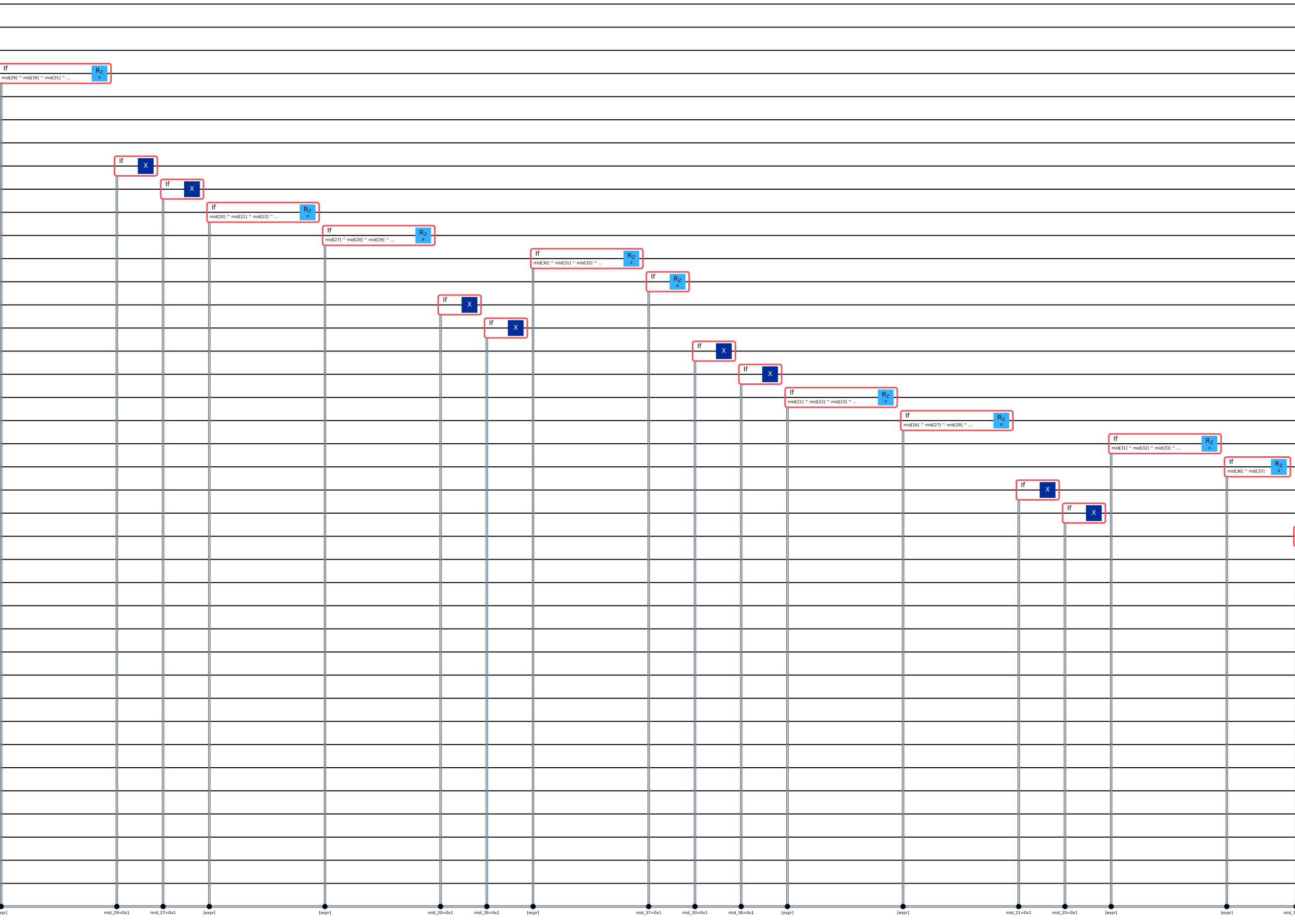
[Z20] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 38

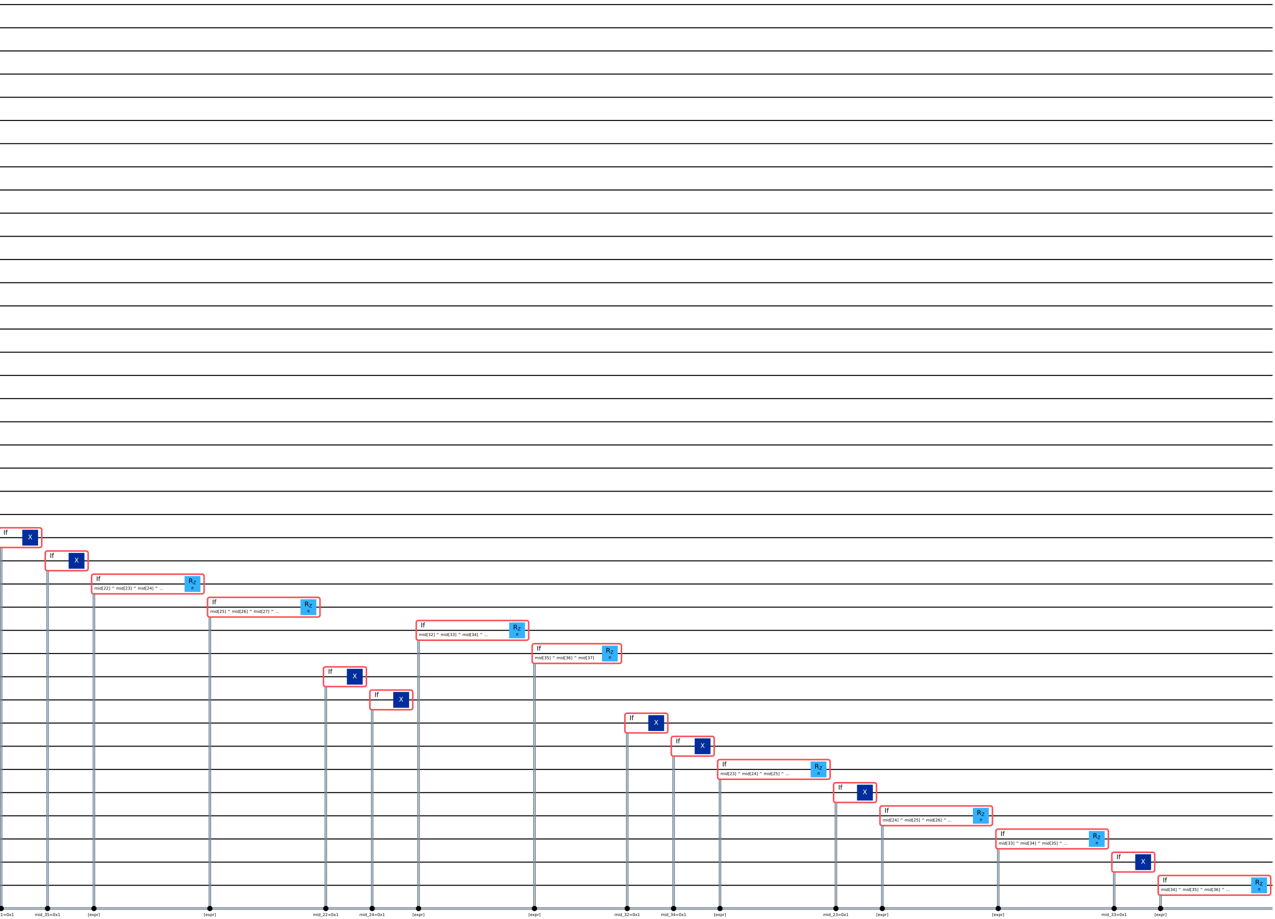


[Z20] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:190 rz:174 cz:76 if_else:39 measure:38)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 4, total depth 39

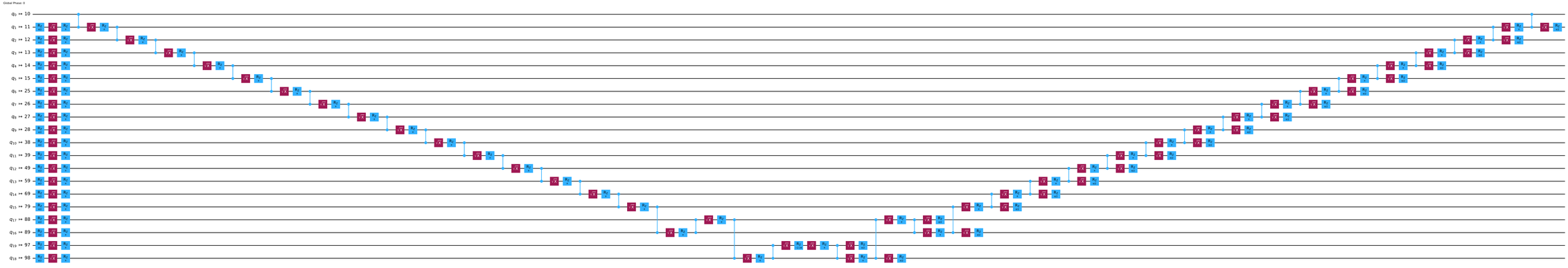

Global Phase: 0
$q_{22} \mapsto 22$
$q_{24} \mapsto 24$
$q_{25} \mapsto 25$
$q_{26} \mapsto 26$
$q_{28} \mapsto 28$
$q_{32} \mapsto 32$
$q_{34} \mapsto 34$
$q_{36} \mapsto 36$
$q_{38} \mapsto 38$
$q_{42} \mapsto 42$
$q_{44} \mapsto 44$
$q_{46} \mapsto 46$
$q_{48} \mapsto 48$
$q_{52} \mapsto 52$
$q_{54} \mapsto 54$
$q_{56} \mapsto 56$
$q_{58} \mapsto 58$
$q_{62} \mapsto 62$
$q_{64} \mapsto 64$
$q_{66} \mapsto 66$
$q_{68} \mapsto 68$
$q_{72} \mapsto 72$
$q_{74} \mapsto 74$
$q_{76} \mapsto 76$
$q_{78} \mapsto 78$
$q_{82} \mapsto 82$
$q_{84} \mapsto 84$
$q_{86} \mapsto 86$
$q_{88} \mapsto 88$
$q_{92} \mapsto 92$
$q_{94} \mapsto 94$
$q_{96} \mapsto 96$
$q_{98} \mapsto 98$
$q_{102} \mapsto 102$
$q_{103} \mapsto 103$
$q_{104} \mapsto 104$
$q_{106} \mapsto 106$
$q_{107} \mapsto 107$
$q_{108} \mapsto 108$
mid
38
R_Z
$\pi/2$
$-\pi/2$
π
$\sqrt{X}$
14
0
1
2
3
4
5
6
7
8


If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
If
mid[19] ^ mid[20] ^ mid[21] ^ ...
If
mid[28] ^ mid[29] ^ mid[30] ^ ...
If
If
If
R_Z
√X
X
−π/2
π
π/2
−0.8
0.8
9
10
11
12
13
15
16
17
18
[expr]
33
23
37
19
20
21
22
24
25
26
27
28
29
30
31
32
34
35
36
[expr]
[expr]
mid_19=0x1
mid_27=0x1
mid_28=0x1


If
mid[29] ^ mid[30] ^ mid[31] ^ ...
R_Z
π
If
X
If
X
If
mid[20] ^ mid[21] ^ mid[22] ^ ...
R_Z
π
If
mid[27] ^ mid[28] ^ mid[29] ^ ...
R_Z
π
If
mid[30] ^ mid[31] ^ mid[32] ^ ...
R_Z
π
If
R_Z
π
If
X
If
X
If
X
If
X
If
mid[21] ^ mid[22] ^ mid[23] ^ ...
R_Z
π
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
R_Z
π
If
mid[31] ^ mid[32] ^ mid[33] ^ ...
R_Z
π
If
mid[36] ^ mid[37]
R_Z
π
If
X
If
X
xpr]
mid_29=0x1
mid_37=0x1
[expr]
[expr]
mid_20=0x1
mid_26=0x1
[expr]
mid_37=0x1
mid_30=0x1
mid_36=0x1
[expr]
[expr]
mid_21=0x1
mid_25=0x1
[expr]
[expr]
mid_3

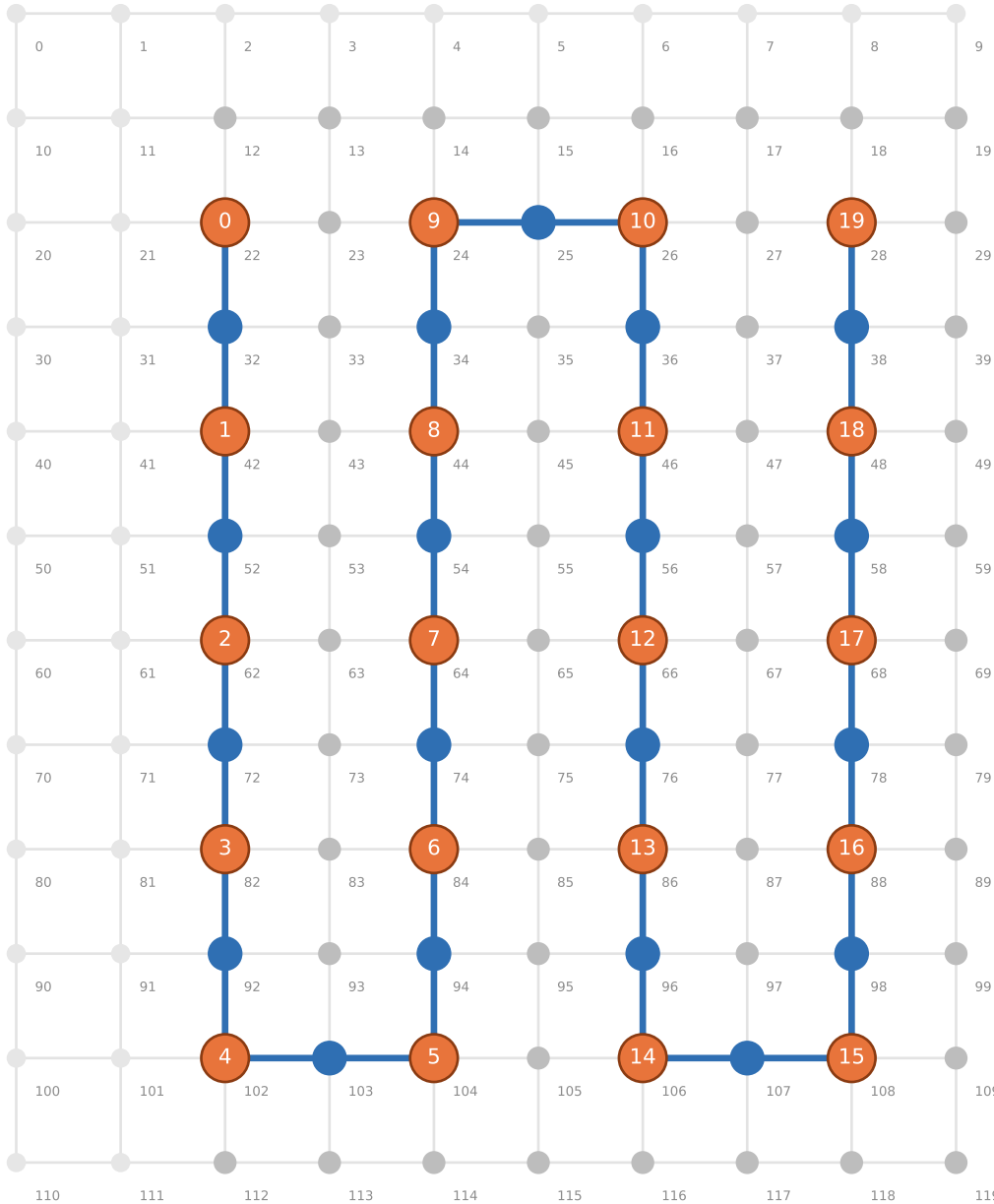

If
X
1=0x1
If
X
mid_35=0x1
If
mid[22] ^ mid[23] ^ mid[24] ^ ...
R_Z
π
[expr]
If
mid[25] ^ mid[26] ^ mid[27] ^ ...
R_Z
π
[expr]
If
X
mid_22=0x1
If
X
mid_24=0x1
If
mid[32] ^ mid[33] ^ mid[34] ^ ...
R_Z
π
[expr]
If
mid[35] ^ mid[36] ^ mid[37]
R_Z
π
[expr]
If
X
mid_32=0x1
If
X
mid_34=0x1
If
mid[23] ^ mid[24] ^ mid[25] ^ ...
R_Z
π
[expr]
If
X
mid_23=0x1
If
mid[24] ^ mid[25] ^ mid[26] ^ ...
R_Z
π
[expr]
If
mid[33] ^ mid[34] ^ mid[35] ^ ...
R_Z
π
[expr]
If
X
mid_33=0x1
If
mid[34] ^ mid[35] ^ mid[36] ^ ...
R_Z
π
[expr]

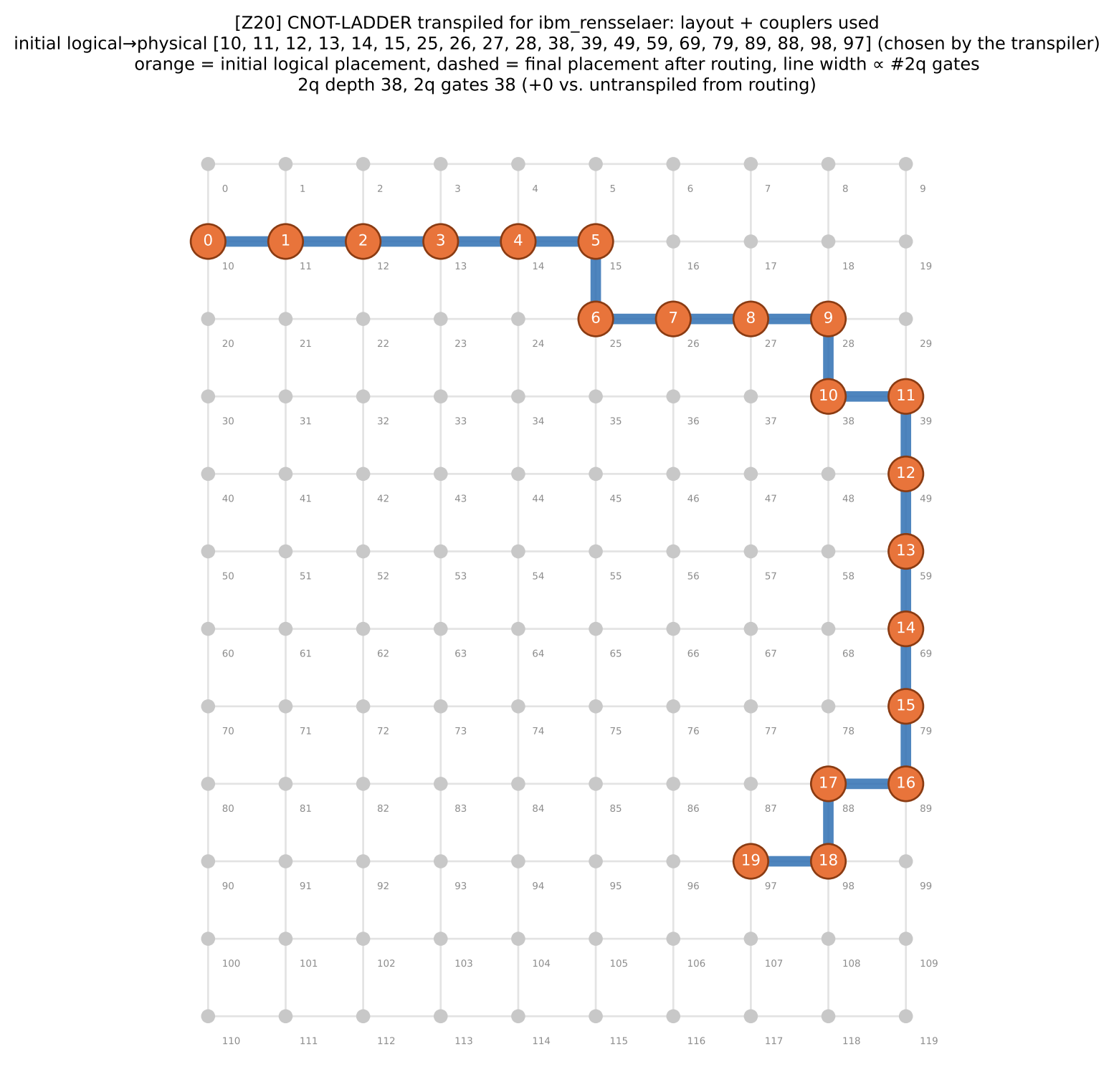
[Z20] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:95 sx:76 cz:38)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 (20 logical qubits)
2q depth 38, total depth 119



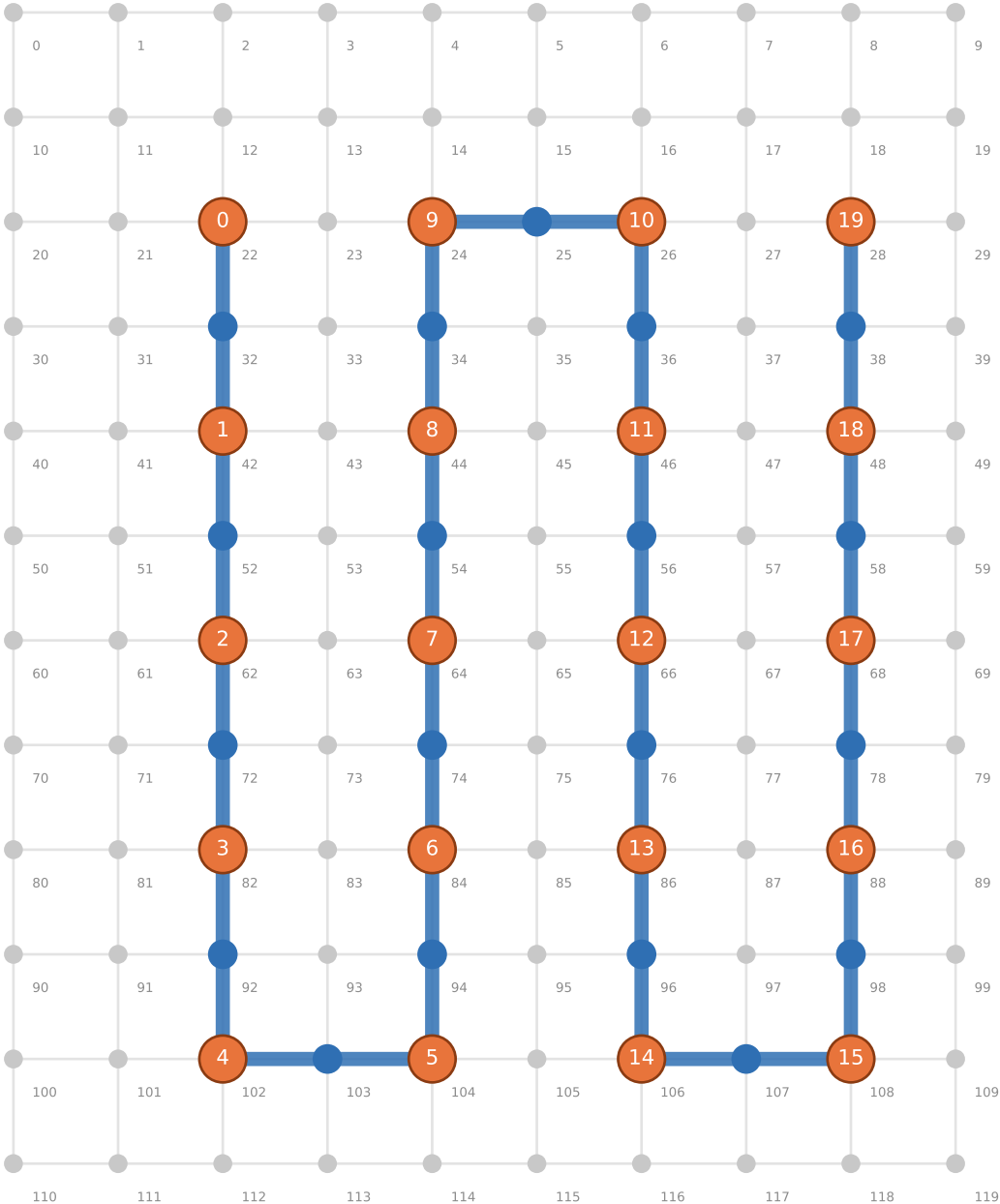
[Z20] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [22, 42, 62, 82, 102, 104, 84, 64, 44, 24, 26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19
chain length 39, 19 ancillas, 0 |0> fillers



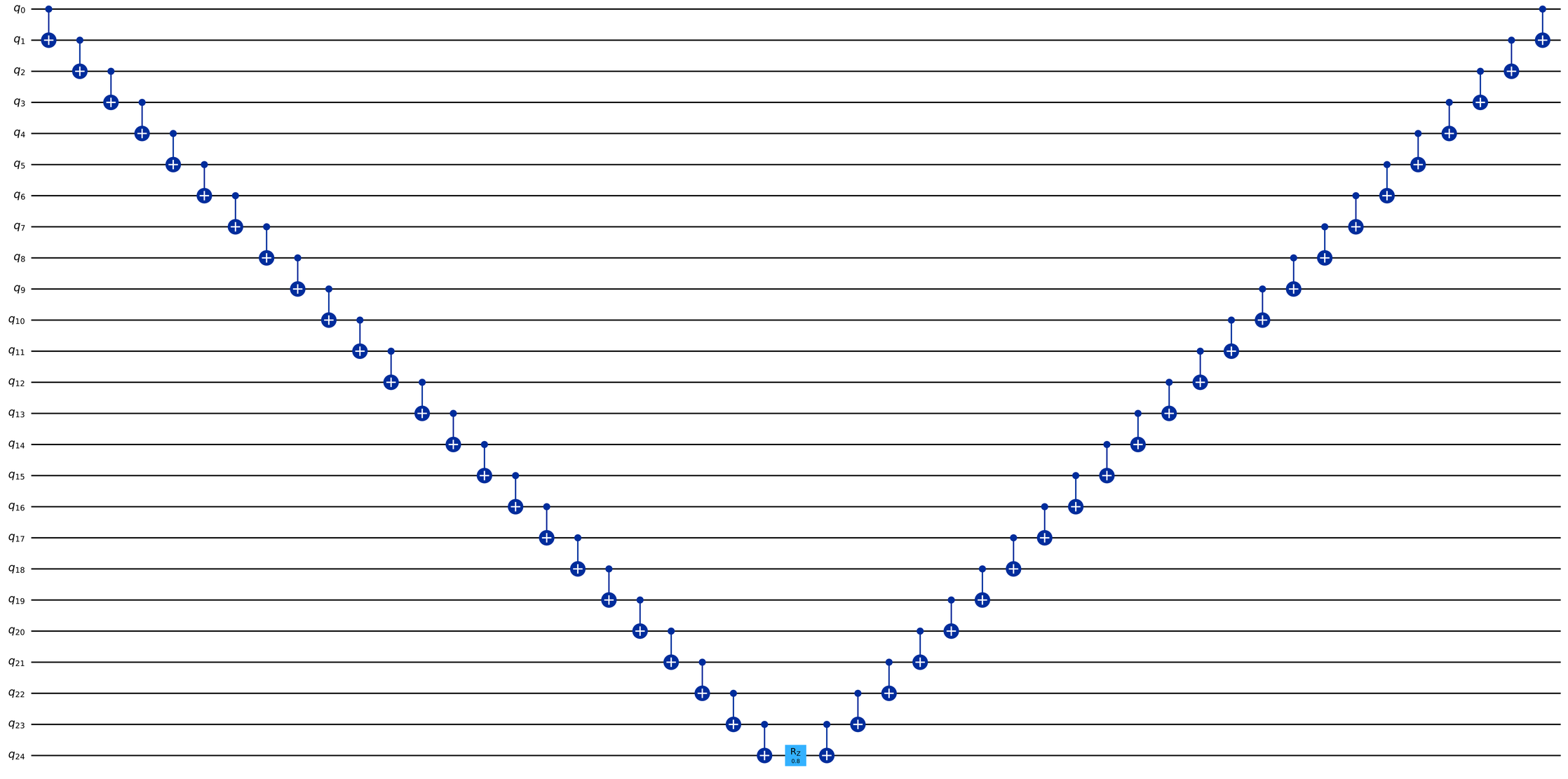

[Z20] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [10, 11, 12, 13, 14, 15, 25, 26, 27, 28, 38, 39, 49, 59, 69, 79, 89, 88, 98, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 38, 2q gates 38 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

[Z20] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [22, 42, 62, 82, 102, 104, 84, 64, 44, 24, 26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 76

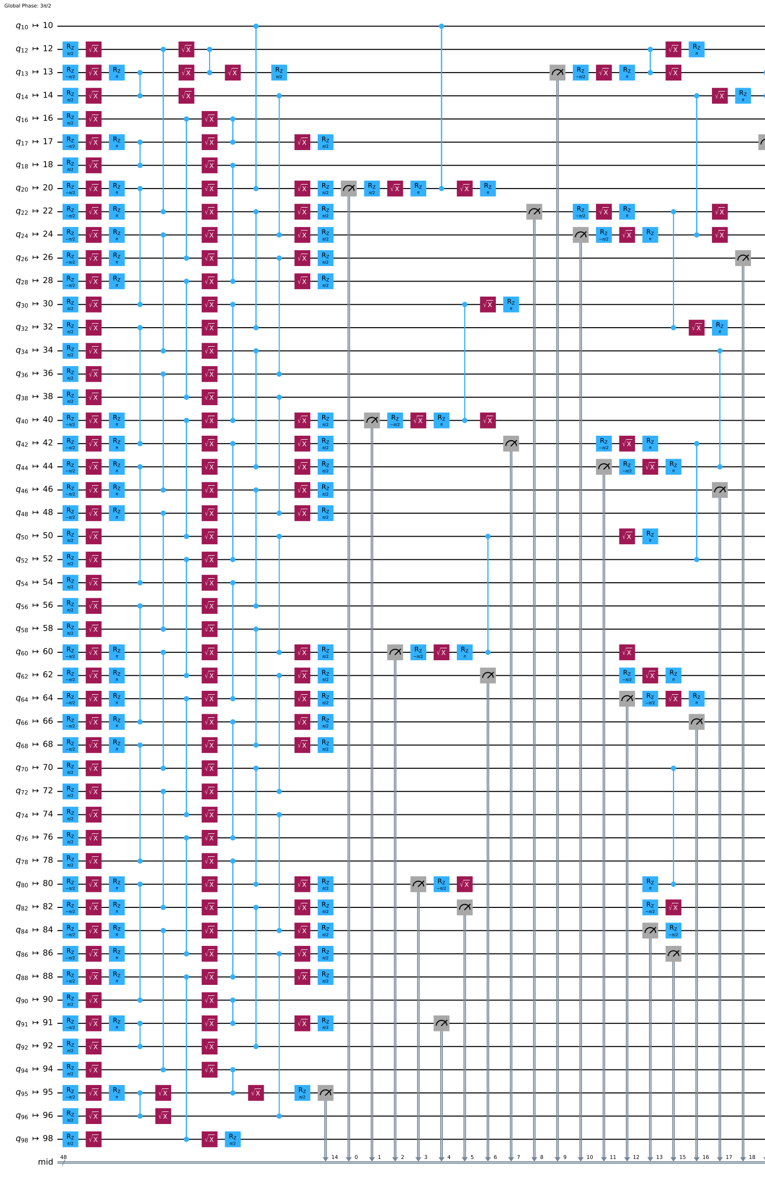


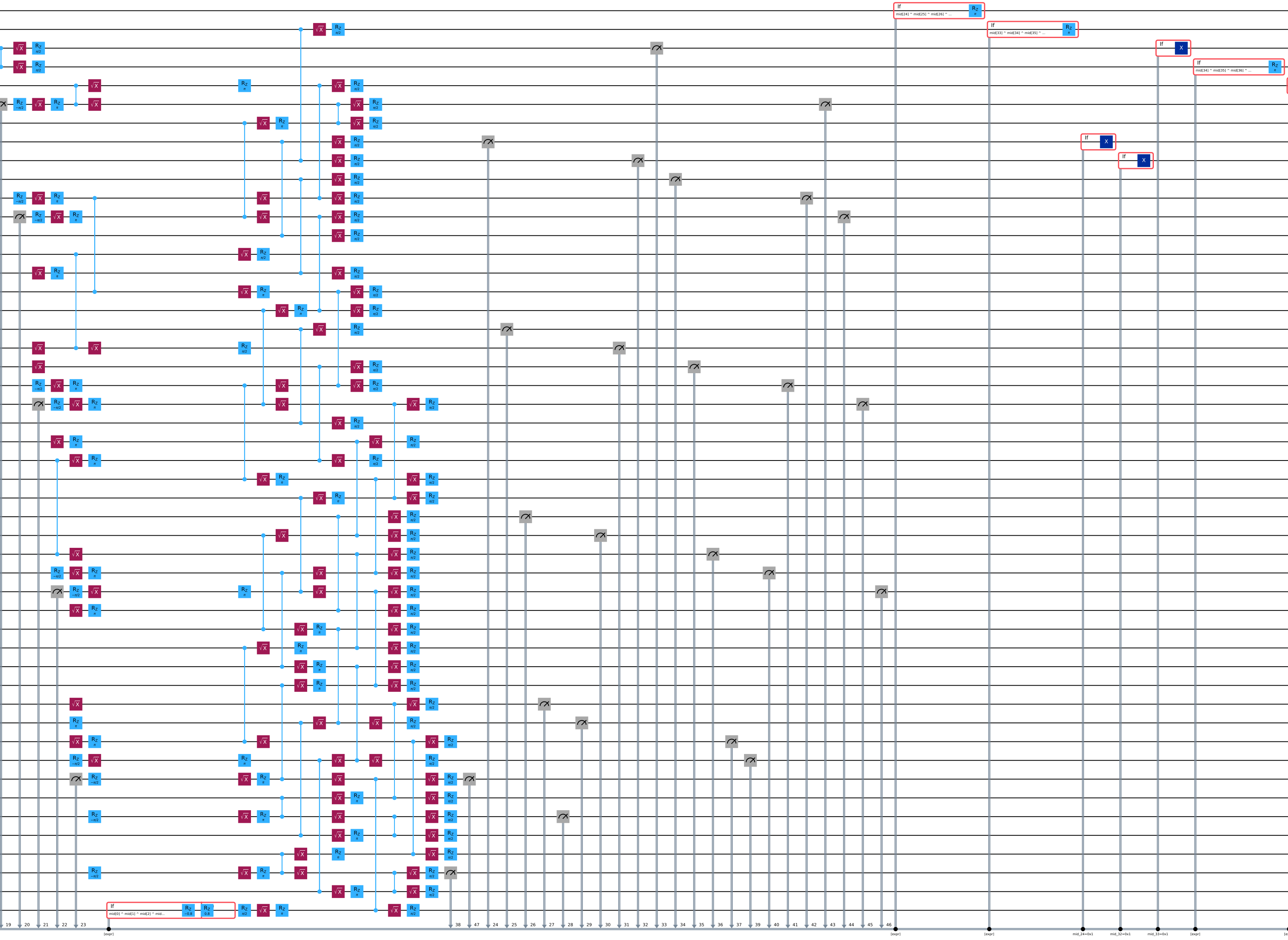
LCL_DP
LCL_DT
Xor
LCL_DH
Final Inspection

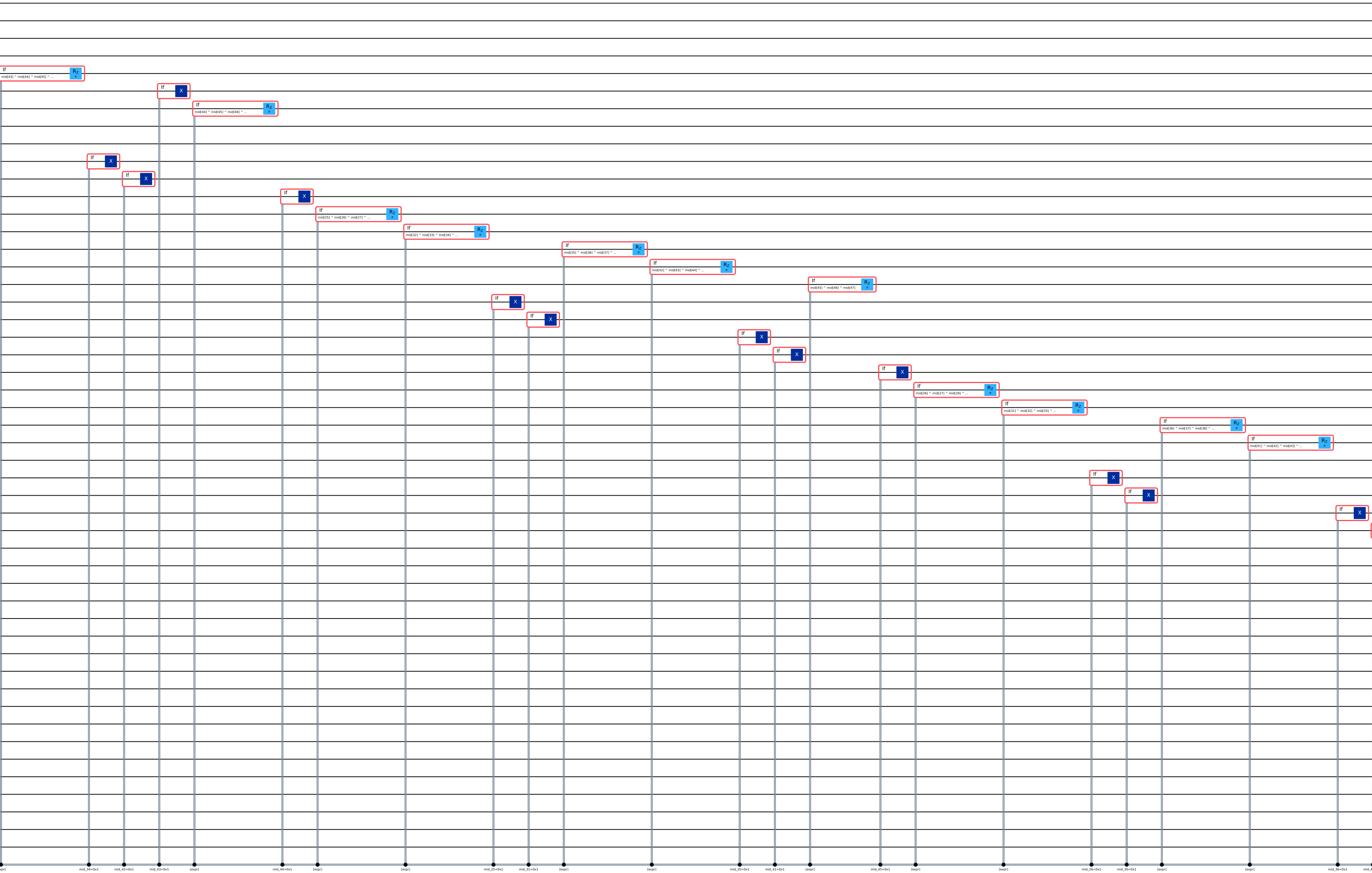
[Z25] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48

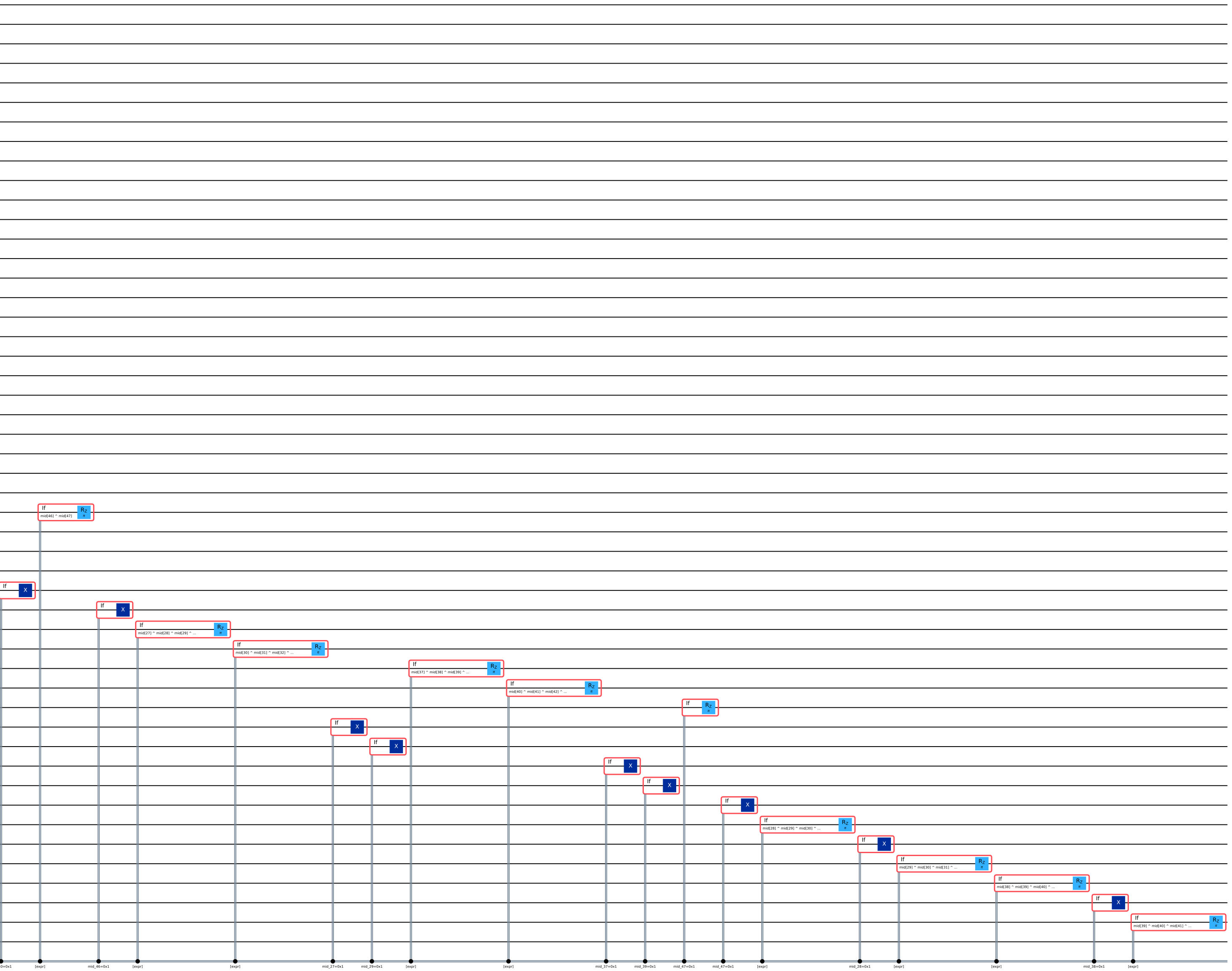


[Z25] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:240 rz:219 cz:96 if_else:49 measure:48)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 4, total depth 47

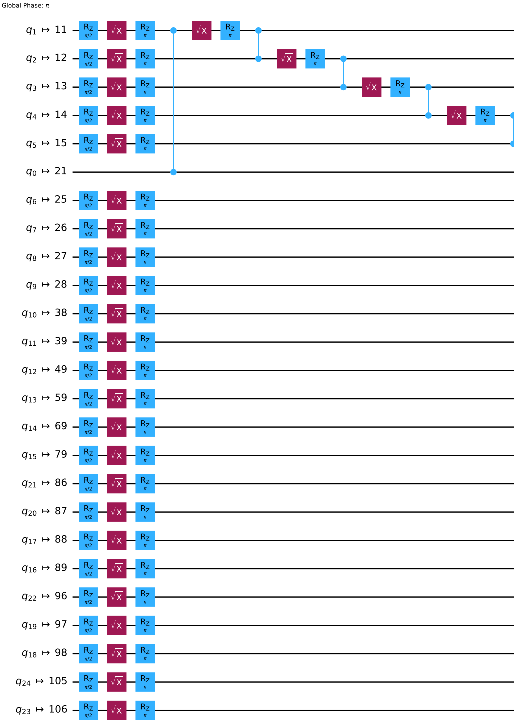

Global Phase: 3π/2
q_{10} ↦ 10
q_{12} ↦ 12
q_{13} ↦ 13
q_{14} ↦ 14
q_{16} ↦ 16
q_{17} ↦ 17
q_{18} ↦ 18
q_{20} ↦ 20
q_{22} ↦ 22
q_{24} ↦ 24
q_{26} ↦ 26
q_{28} ↦ 28
q_{30} ↦ 30
q_{32} ↦ 32
q_{34} ↦ 34
q_{36} ↦ 36
q_{38} ↦ 38
q_{40} ↦ 40
q_{42} ↦ 42
q_{44} ↦ 44
q_{46} ↦ 46
q_{48} ↦ 48
q_{50} ↦ 50
q_{52} ↦ 52
q_{54} ↦ 54
q_{56} ↦ 56
q_{58} ↦ 58
q_{60} ↦ 60
q_{62} ↦ 62
q_{64} ↦ 64
q_{66} ↦ 66
q_{68} ↦ 68
q_{70} ↦ 70
q_{72} ↦ 72
q_{74} ↦ 74
q_{76} ↦ 76
q_{78} ↦ 78
q_{80} ↦ 80
q_{82} ↦ 82
q_{84} ↦ 84
q_{86} ↦ 86
q_{88} ↦ 88
q_{90} ↦ 90
q_{91} ↦ 91
q_{92} ↦ 92
q_{94} ↦ 94
q_{95} ↦ 95
q_{96} ↦ 96
q_{98} ↦ 98
mid
48
14 0 1 2 3 4 5 6 7 8 9 10 11 12 13 15 16 17 18

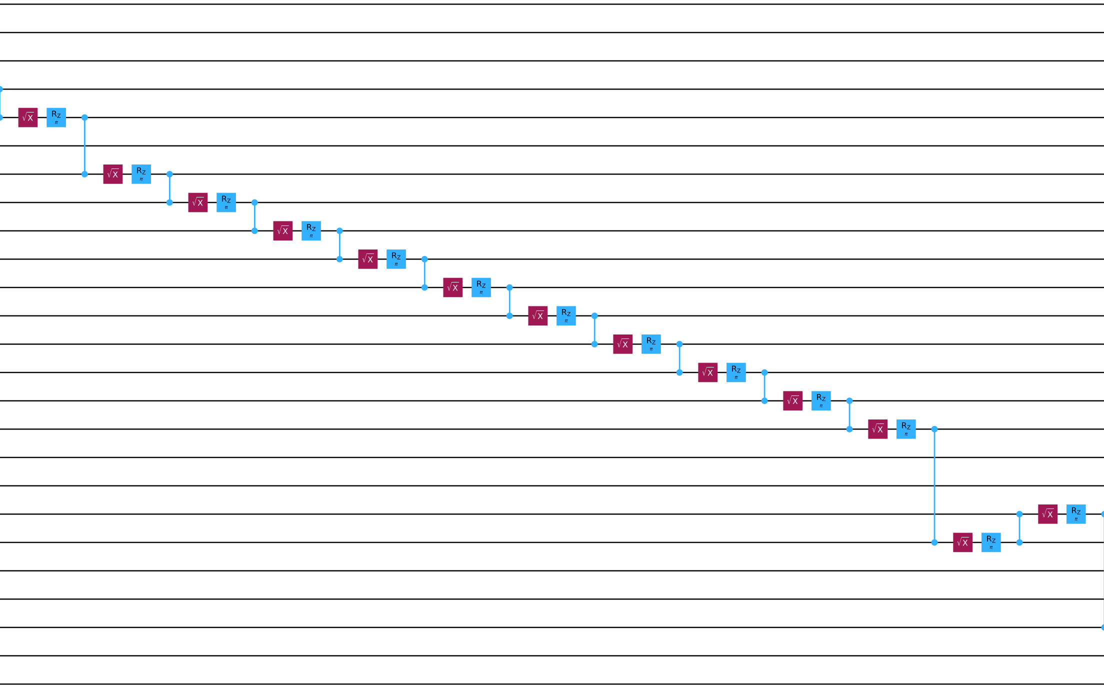

If
mid[24] ^ mid[25] ^ mid[26] ^ ...
If
mid[33] ^ mid[34] ^ mid[35] ^ ...
If
If
mid[34] ^ mid[35] ^ mid[36] ^ ...
If
If
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
19
20
21
22
23
[expr]
38
47
24
25
26
27
28
29
30
31
32
33
34
35
36
37
39
40
41
42
43
44
45
46
[expr]
[expr]
mid_24=0x1
mid_32=0x1
mid_33=0x1
[expr]

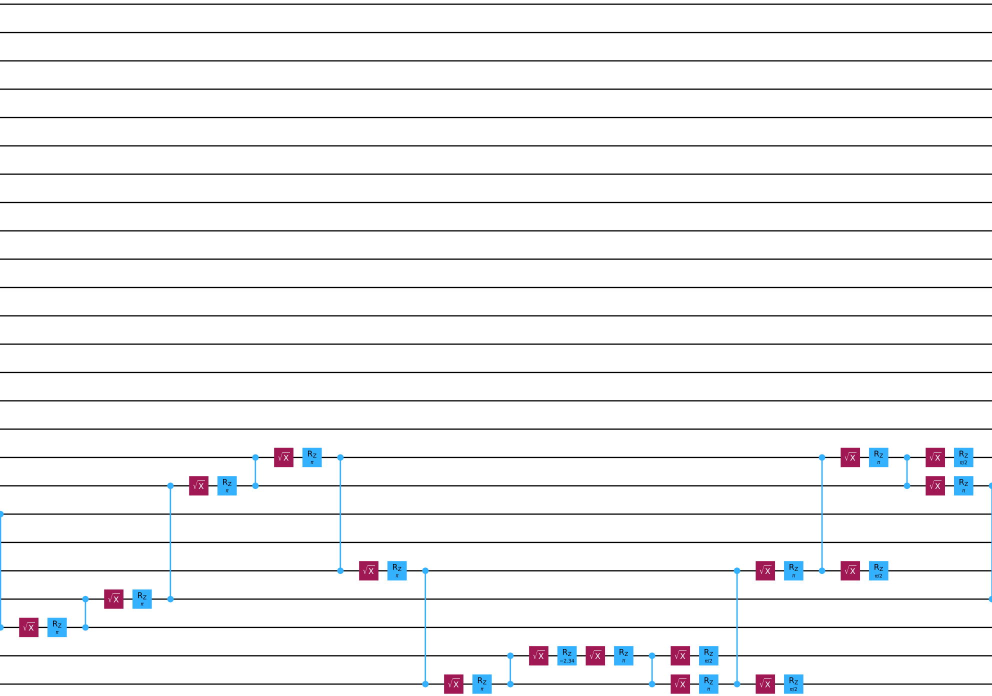

If
R_Z
π
mid[43] ^ mid[44] ^ mid[45] ^ ...
If
X
If
X
If
X
If
R_Z
π
mid[44] ^ mid[45] ^ mid[46] ^ ...
If
X
If
R_Z
π
mid[25] ^ mid[26] ^ mid[27] ^ ...
If
R_Z
π
mid[32] ^ mid[33] ^ mid[34] ^ ...
If
X
If
X
If
R_Z
π
mid[35] ^ mid[36] ^ mid[37] ^ ...
If
R_Z
π
mid[42] ^ mid[43] ^ mid[44] ^ ...
If
X
If
X
If
R_Z
π
mid[45] ^ mid[46] ^ mid[47]
If
X
If
R_Z
π
mid[26] ^ mid[27] ^ mid[28] ^ ...
If
R_Z
π
mid[31] ^ mid[32] ^ mid[33] ^ ...
If
X
If
X
If
R_Z
π
mid[36] ^ mid[37] ^ mid[38] ^ ...
If
R_Z
π
mid[41] ^ mid[42] ^ mid[43] ^ ...
If
X
expr]
mid_34=0x1
mid_42=0x1
mid_43=0x1
[expr]
mid_44=0x1
[expr]
[expr]
mid_25=0x1
mid_31=0x1
[expr]
[expr]
mid_35=0x1
mid_41=0x1
[expr]
mid_45=0x1
[expr]
[expr]
mid_26=0x1
mid_30=0x1
[expr]
[expr]
mid_36=0x1
mid_4

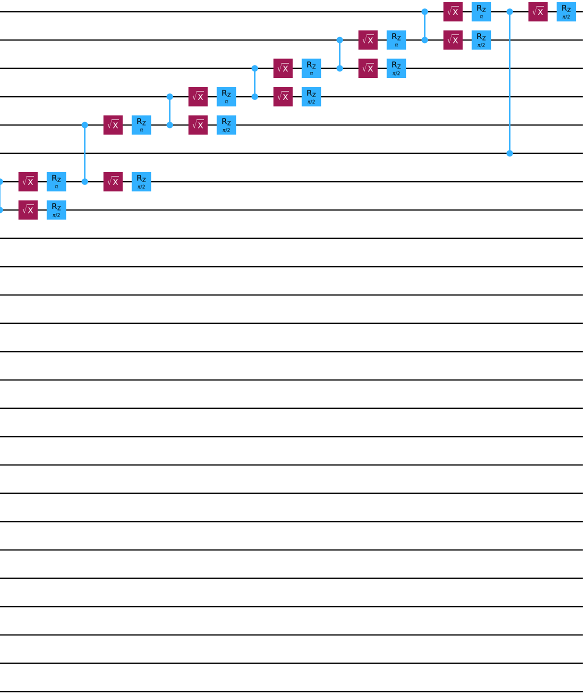

If
X
If
mid[46] ^ mid[47]
R_Z
π
If
X
If
mid[27] ^ mid[28] ^ mid[29] ^ ...
R_Z
π
If
mid[30] ^ mid[31] ^ mid[32] ^ ...
R_Z
π
If
X
If
X
If
mid[37] ^ mid[38] ^ mid[39] ^ ...
R_Z
π
If
mid[40] ^ mid[41] ^ mid[42] ^ ...
R_Z
π
If
X
If
X
If
R_Z
π
If
X
If
mid[28] ^ mid[29] ^ mid[30] ^ ...
R_Z
π
If
X
If
mid[29] ^ mid[30] ^ mid[31] ^ ...
R_Z
π
If
mid[38] ^ mid[39] ^ mid[40] ^ ...
R_Z
π
If
X
If
mid[39] ^ mid[40] ^ mid[41] ^ ...
R_Z
π
0=0x1
[expr]
mid_46=0x1
[expr]
[expr]
mid_27=0x1
mid_29=0x1
[expr]
[expr]
mid_37=0x1
mid_39=0x1
mid_47=0x1
mid_47=0x1
[expr]
mid_28=0x1
[expr]
[expr]
mid_38=0x1
[expr]

[Z25] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:120 sx:96 cz:48)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48, total depth 149


Global Phase: π
q₁ ↦ 11
q₂ ↦ 12
q₃ ↦ 13
q₄ ↦ 14
q₅ ↦ 15
q₀ ↦ 21
q₆ ↦ 25
q₇ ↦ 26
q₈ ↦ 27
q₉ ↦ 28
q₁₀ ↦ 38
q₁₁ ↦ 39
q₁₂ ↦ 49
q₁₃ ↦ 59
q₁₄ ↦ 69
q₁₅ ↦ 79
q₂₁ ↦ 86
q₂₀ ↦ 87
q₁₇ ↦ 88
q₁₆ ↦ 89
q₂₂ ↦ 96
q₁₉ ↦ 97
q₁₈ ↦ 98
q₂₄ ↦ 105
q₂₃ ↦ 106
Rz
π/2
√X
π

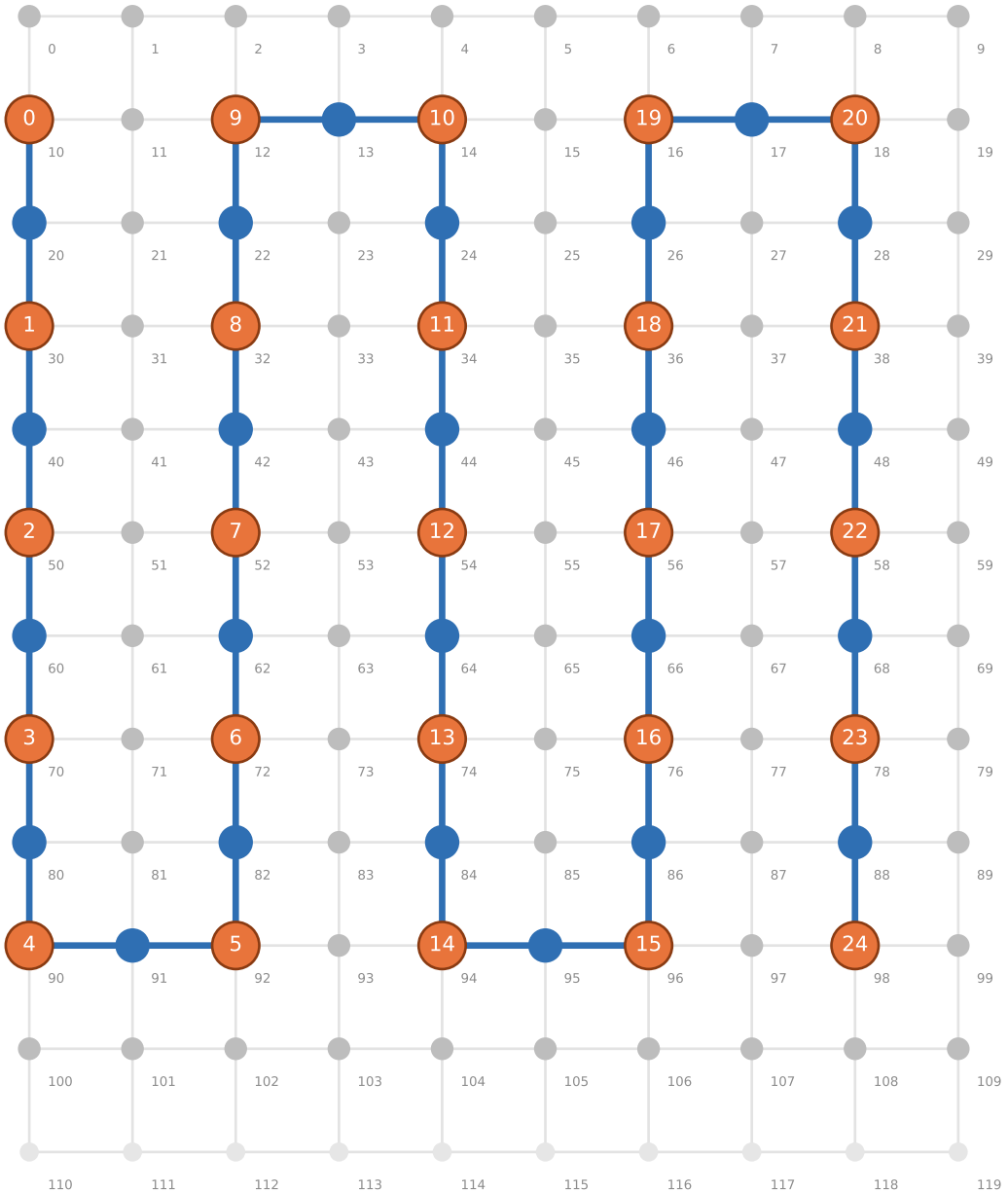



√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ −2.34
√X
RZ π
√X
RZ π/2
√X
RZ π
√X
RZ π
√X
RZ π/2
√X
RZ π
√X
RZ π/2
√X
RZ π/2
√X
RZ π

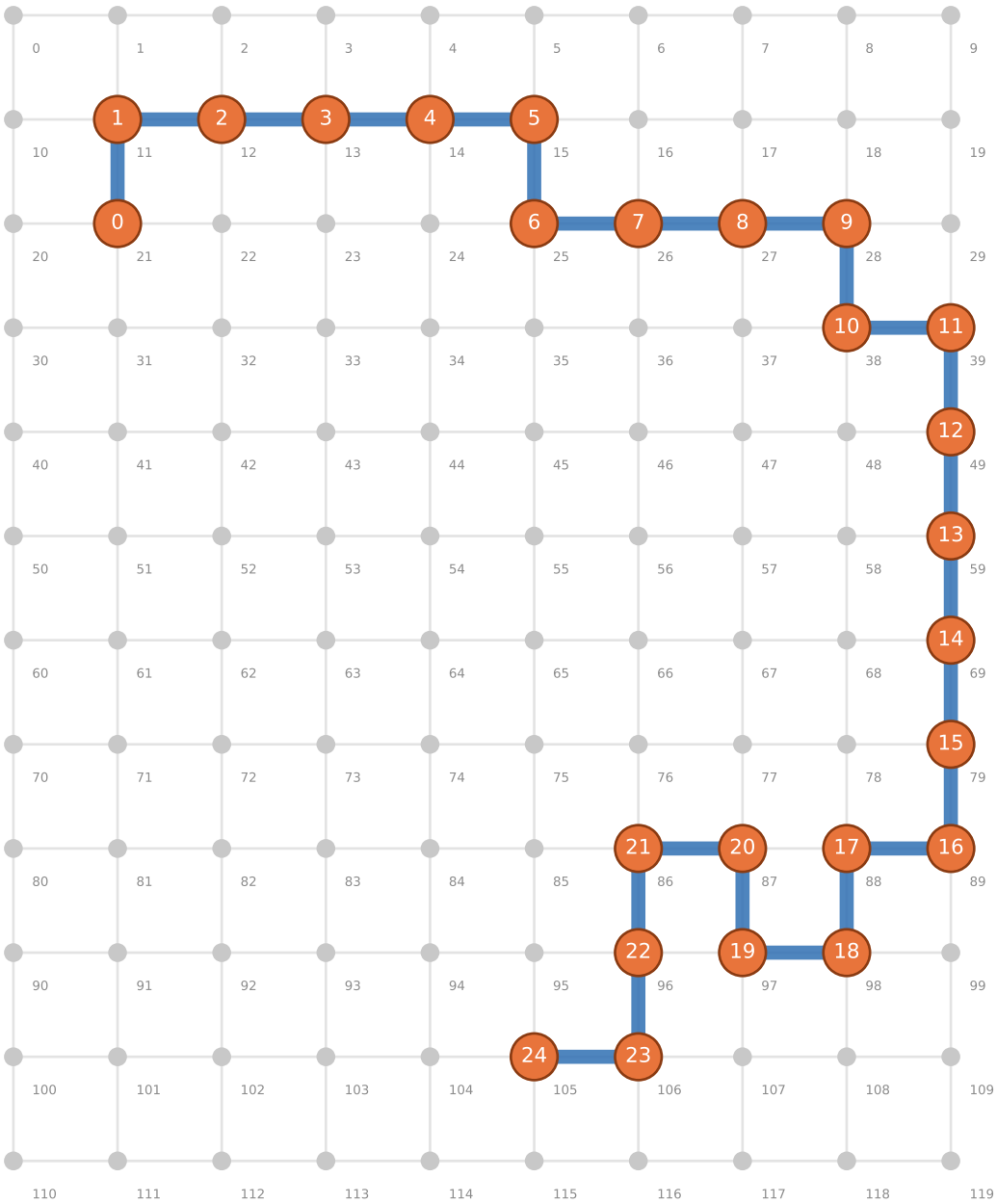


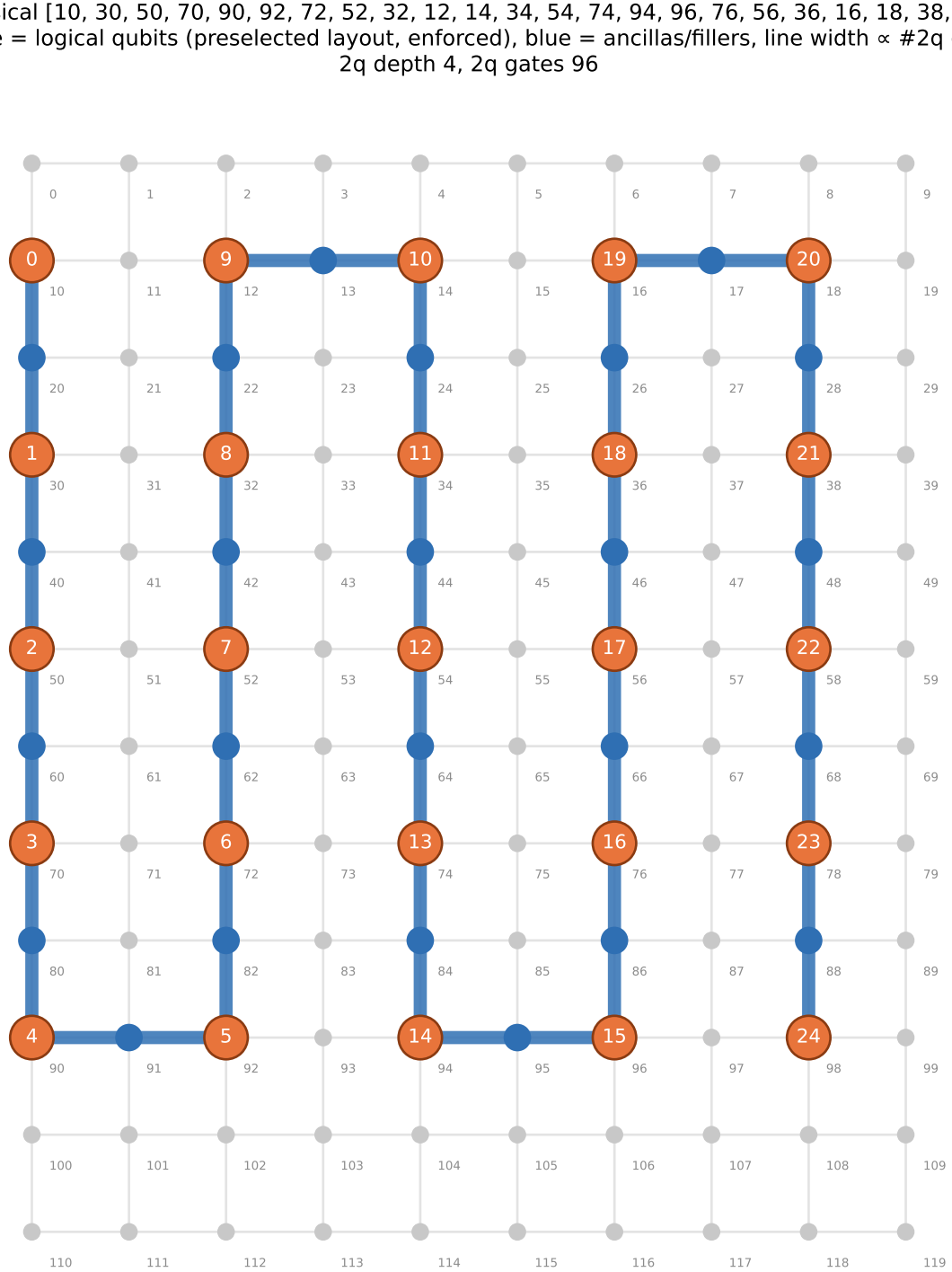
[Z25] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [10, 30, 50, 70, 90, 92, 72, 52, 32, 12, 14, 34, 54, 74, 94, 96, 76, 56, 36, 16, 18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24
chain length 49, 24 ancillas, 0 |0> fillers

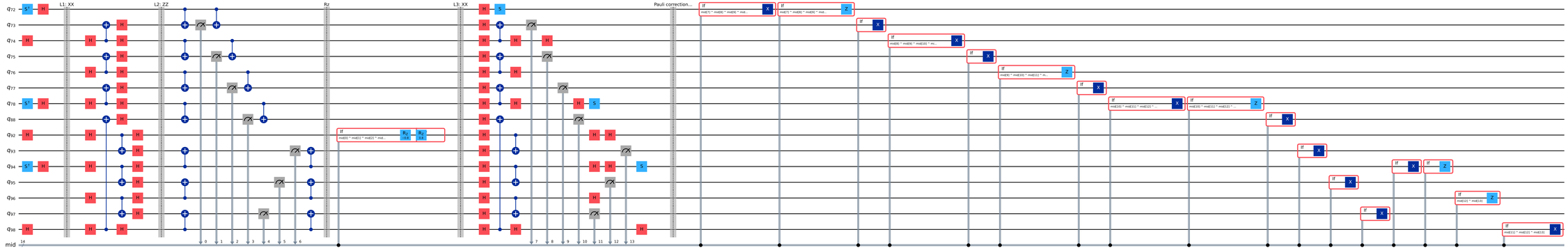


[Z25] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [21, 11, 12, 13, 14, 15, 25, 26, 27, 28, 38, 39, 49, 59, 69, 79, 89, 88, 98, 97, 87, 86, 96, 106, 105] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 48, 2q gates 48 (+0 vs. untranspiled from routing)

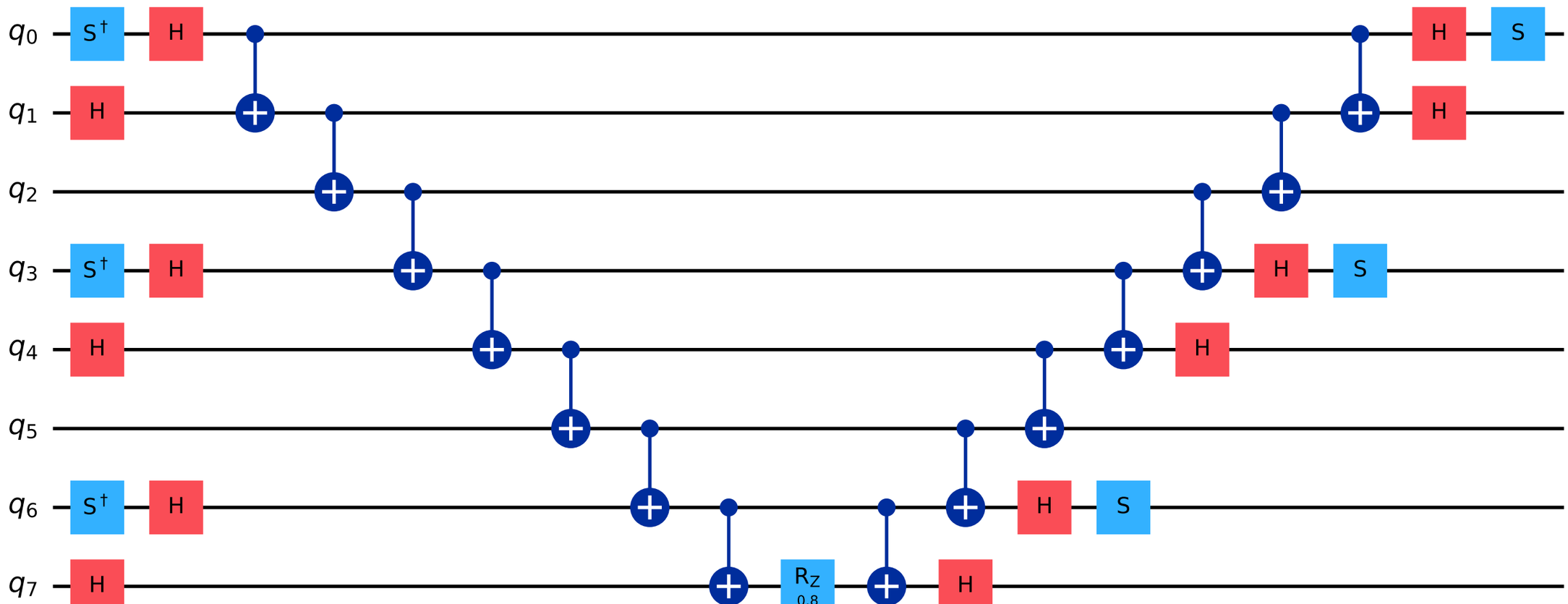



[Z25] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [10, 30, 50, 70, 90, 92, 72, 52, 32, 12, 14, 34, 54, 74, 94, 96, 76, 56, 36, 16, 18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 96
0 1 2 3 4 5 6 7 8 9
0 9 10 19 20
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
1 8 11 18 21
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
2 7 12 17 22
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
3 6 13 16 23
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
4 5 14 15 24
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

[XYZXYZXY] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 4, 14 mid-circuit measurements, logical→physical [72, 74, 76, 78, 98, 96, 94, 92]

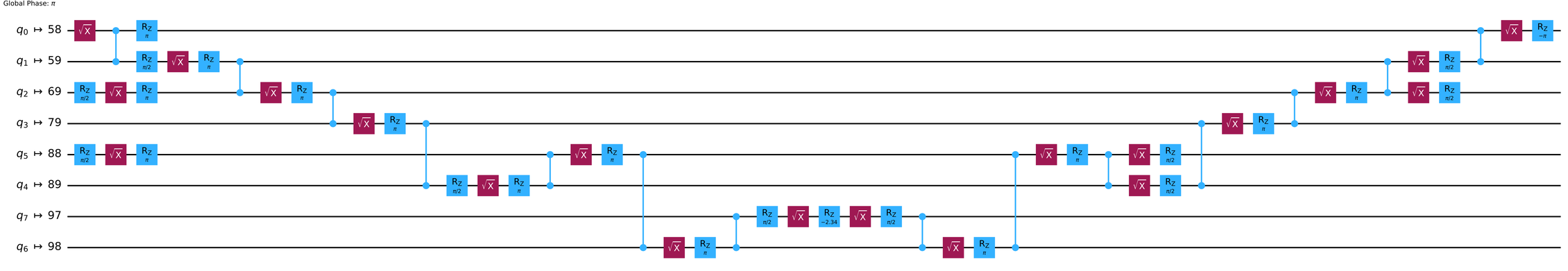


[XYZXYZXY] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14



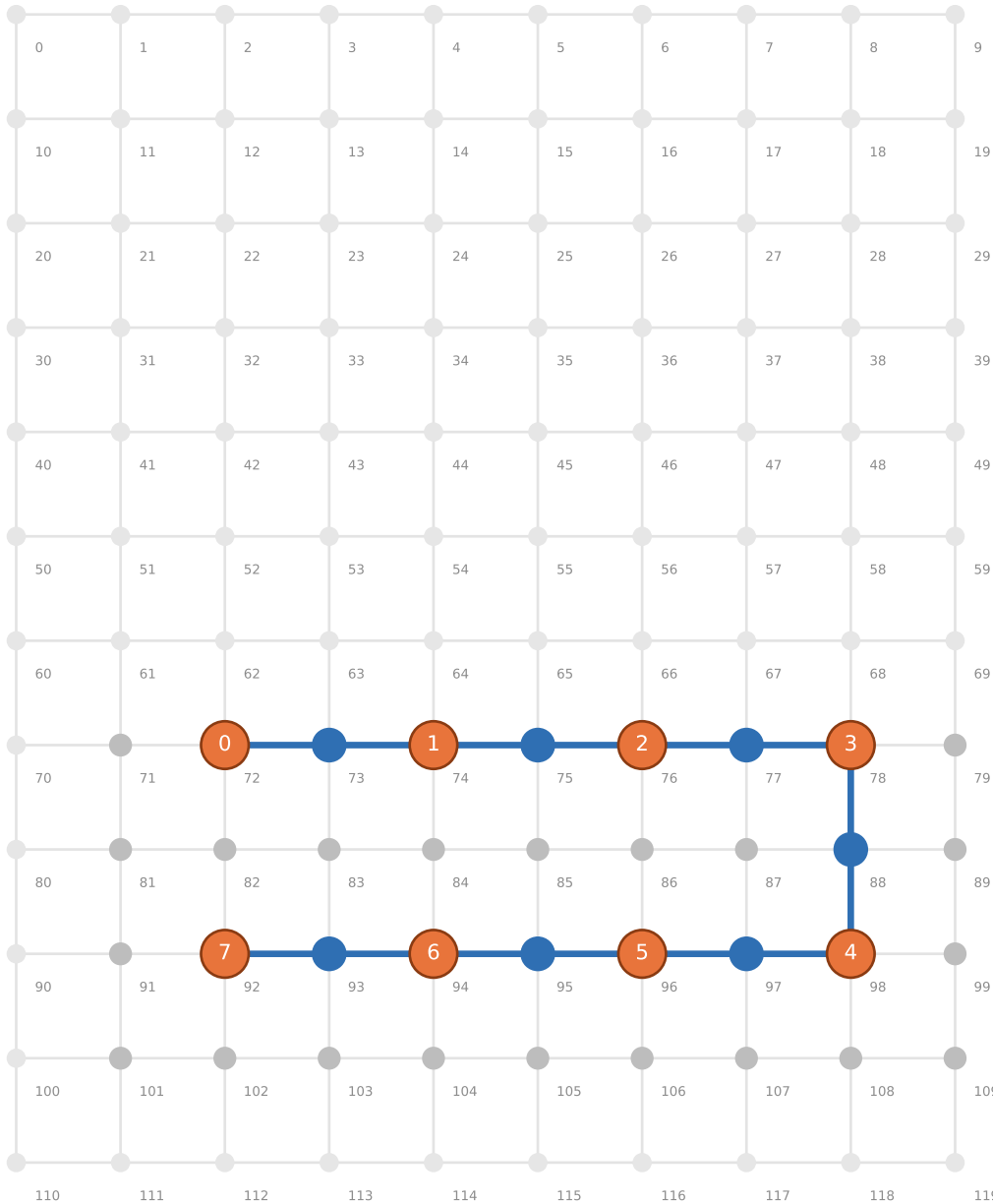
[XYZXYZXY] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:62 rz:62 cz:28 if_else:18 measure:14 x:1)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7 (8 logical qubits)
2q depth 4, total depth 31

[XYZXYZXY] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:25 sx:20 cz:14)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14, total depth 48

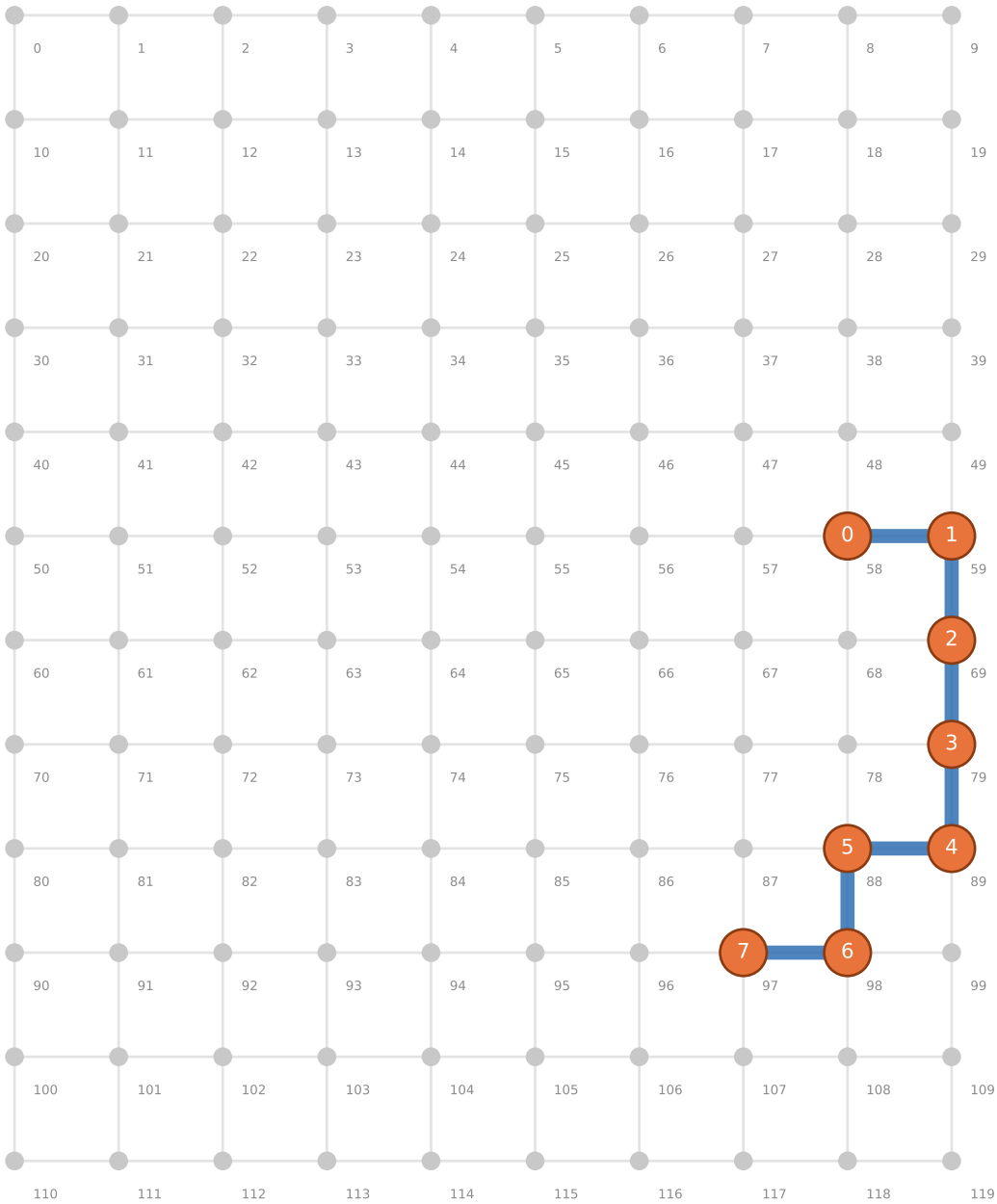


[XYZXYZXY] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

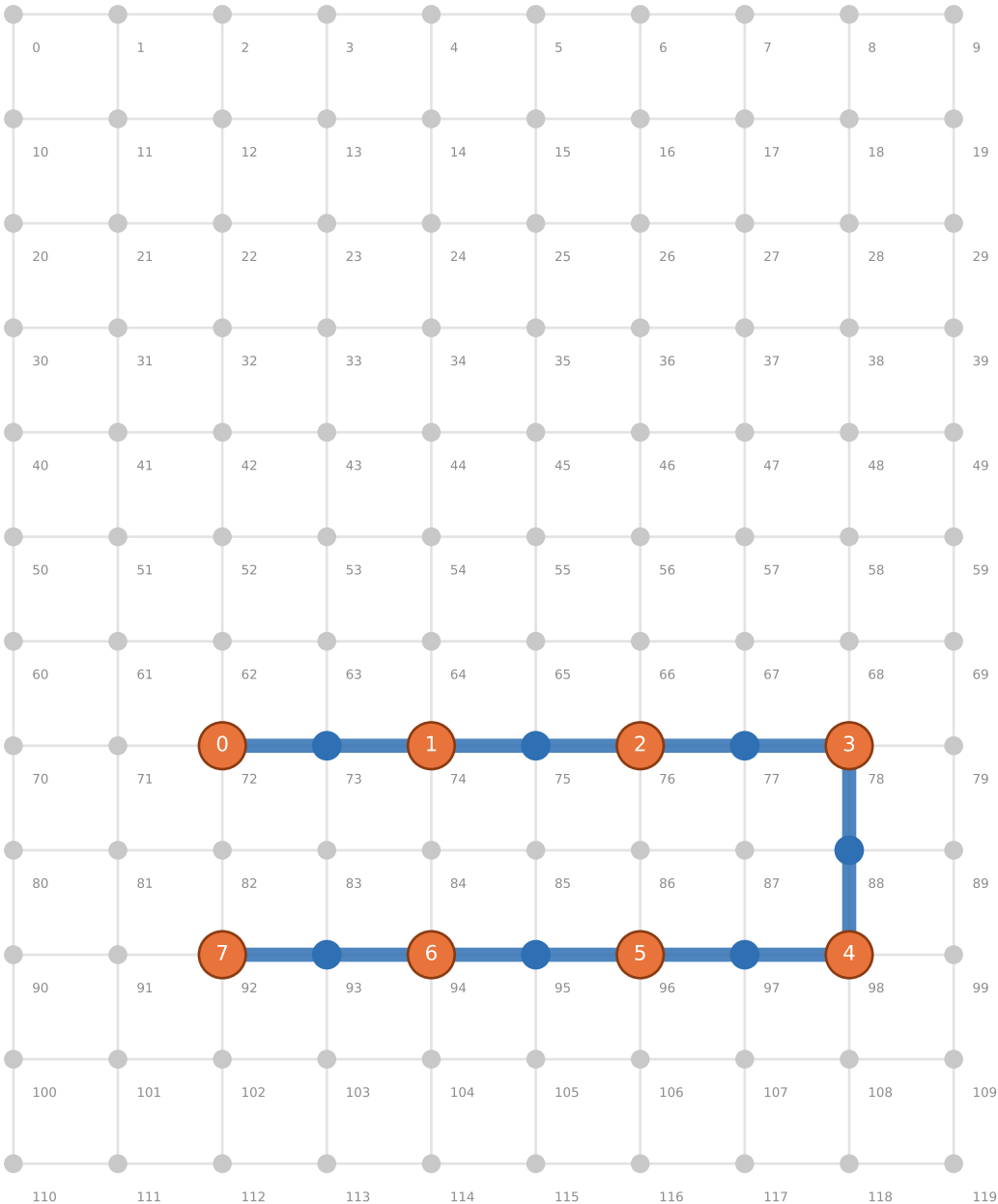
term 0: Y0 X1 Z2 Y3 X4 Z5 Y6 X7
chain length 15, 7 ancillas, 0 |0> fillers



[XYZXYZXY] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [58, 59, 69, 79, 89, 88, 98, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)

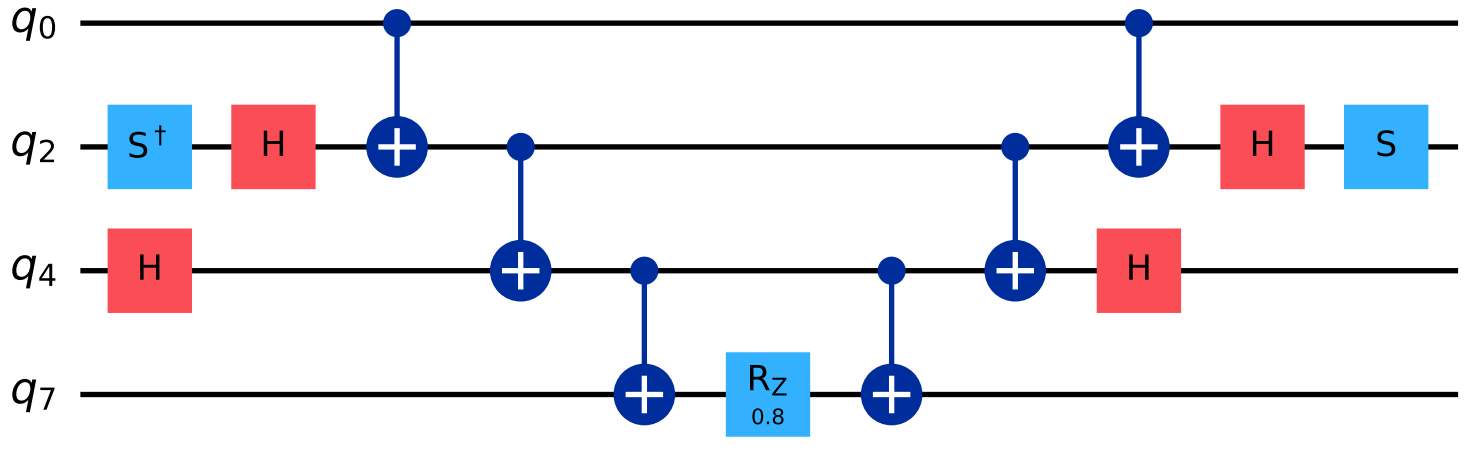


[XYZXYZXY] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 28

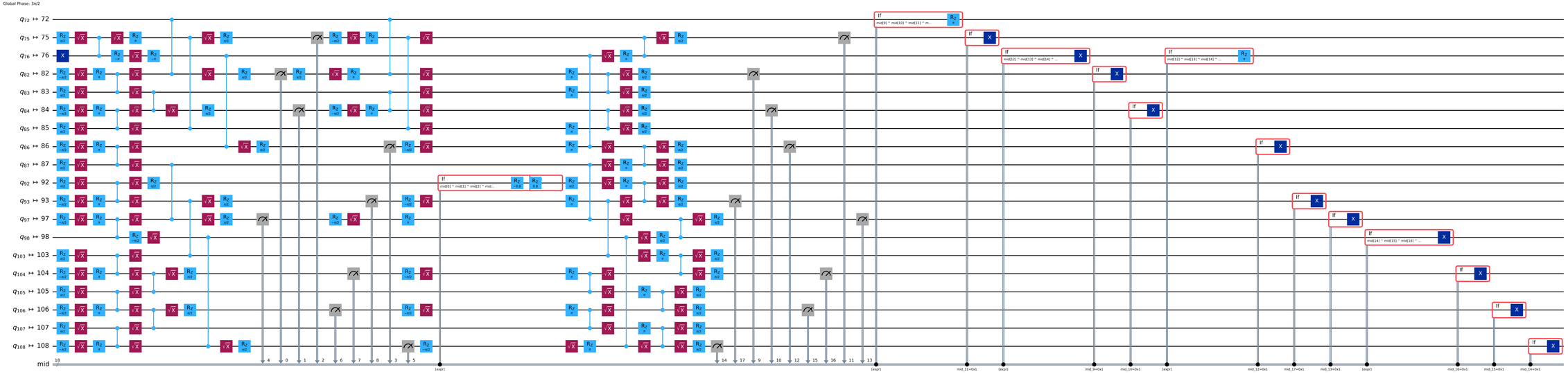


[ZIIXIYIZ_gapped] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 4, 18 mid-circuit measurements, logical→physical [72, 74, 76, 78, 98, 96, 94, 92]

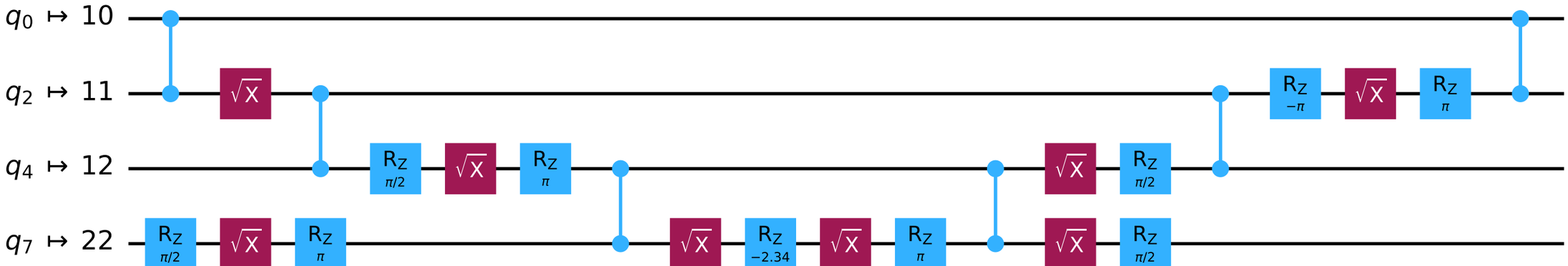
[ZIIXIYIZ_gapped] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 6

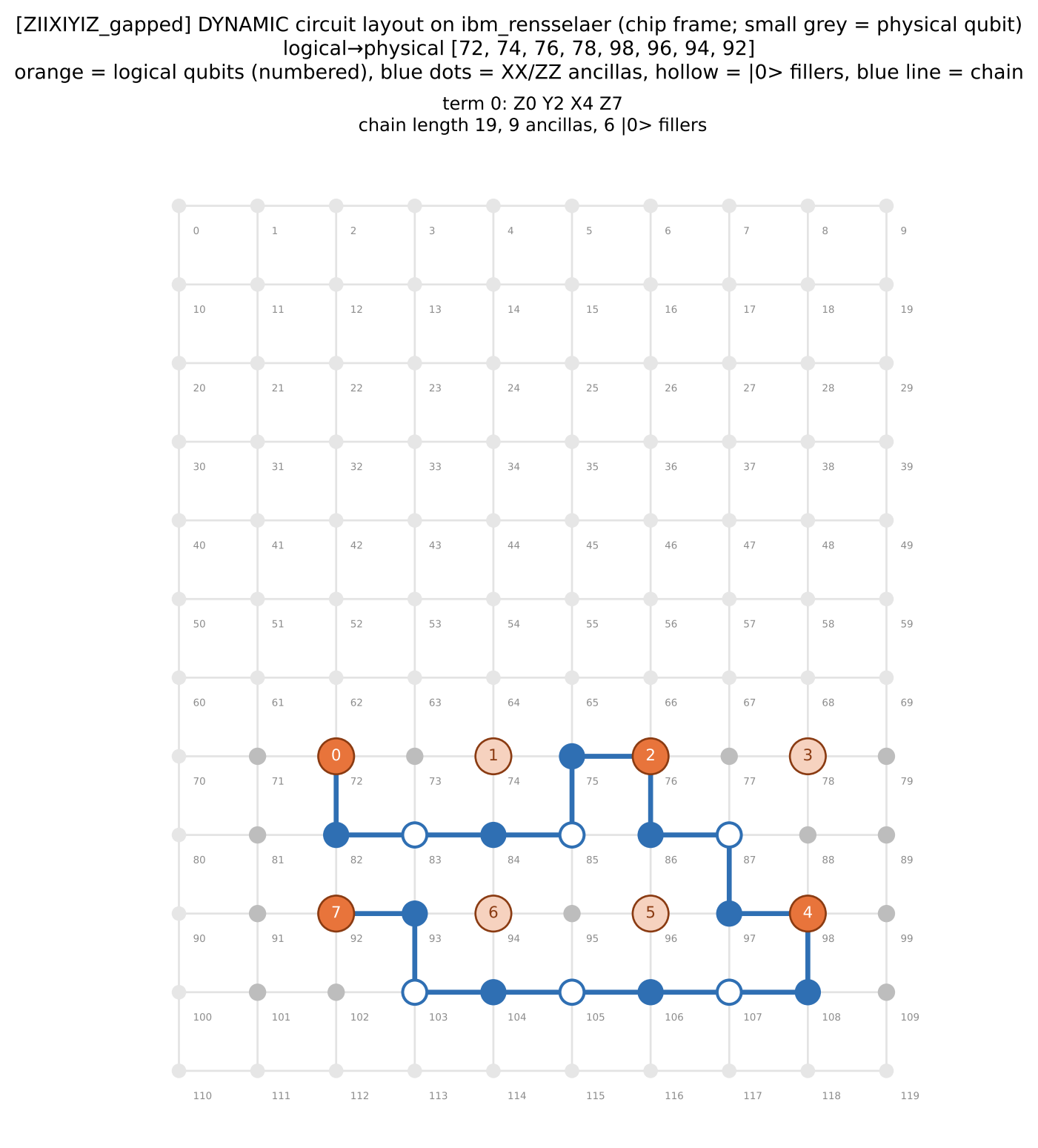


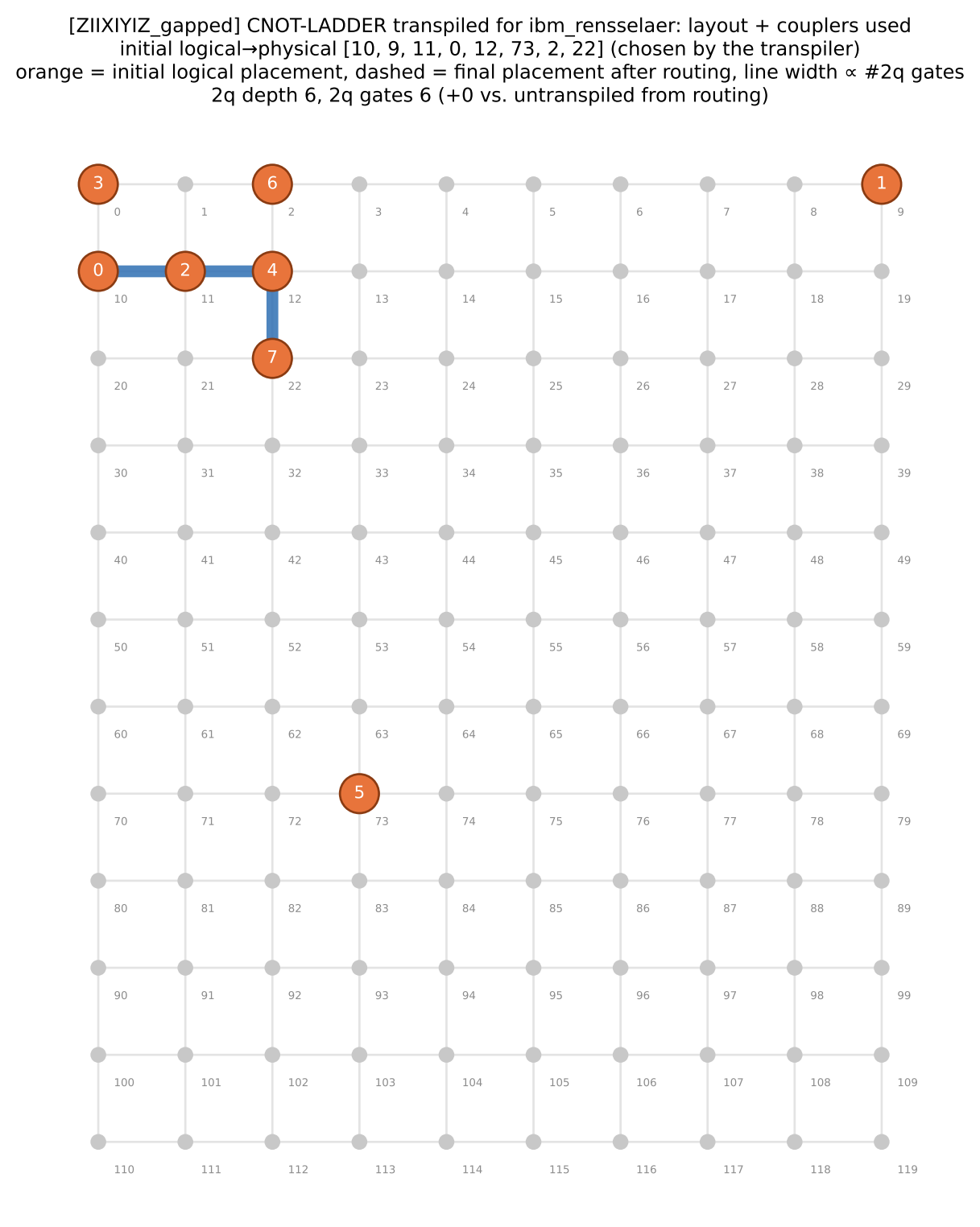
[ZIIXIYIZ_gapped] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:86 rz:83 cz:36 measure:18 if_else:14 x:1)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 4, total depth 26



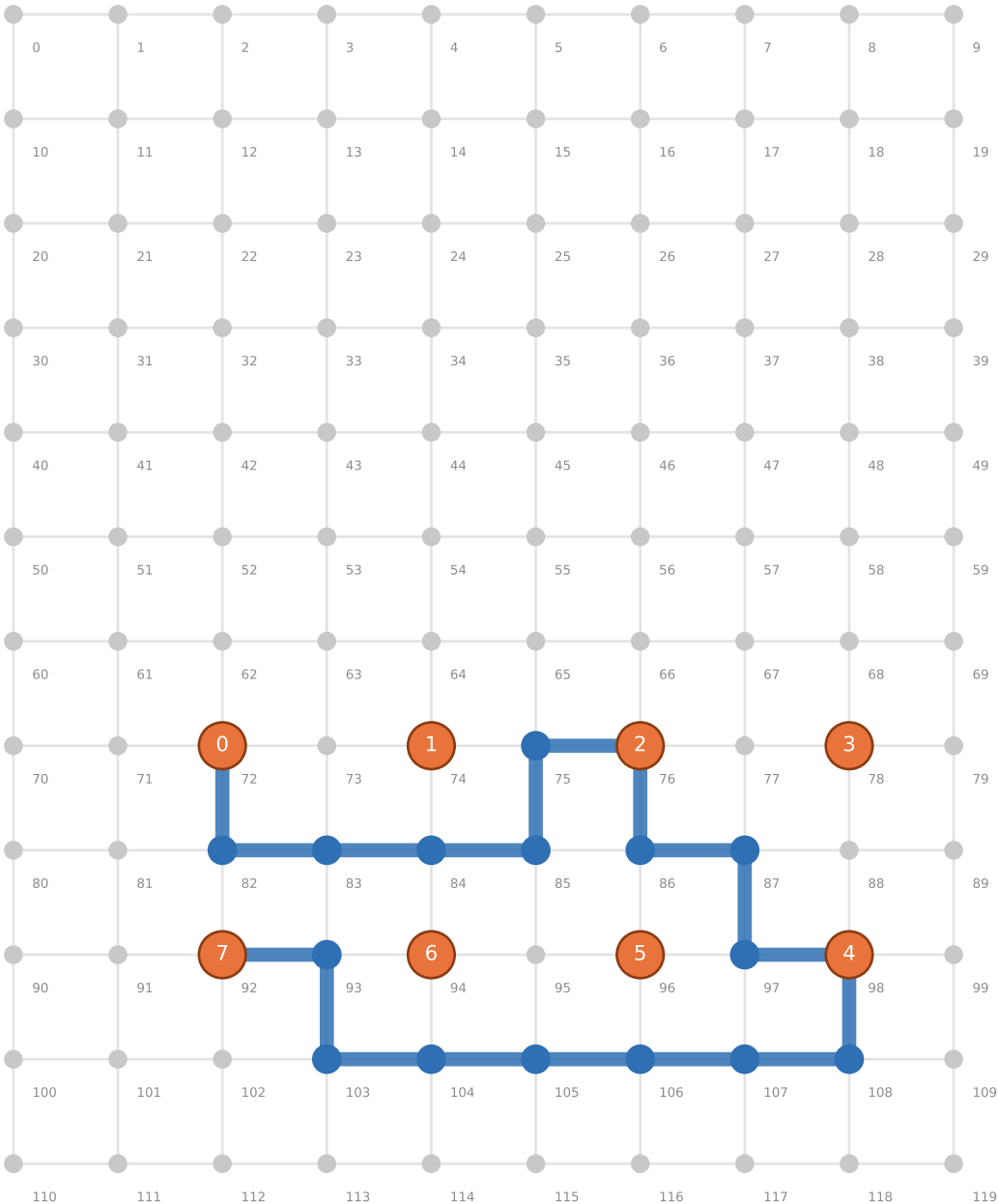
[ZIIXIYIZ_gapped] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:10 sx:8 cz:6)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
Global Phase: 0
2q depth 6, total depth 19



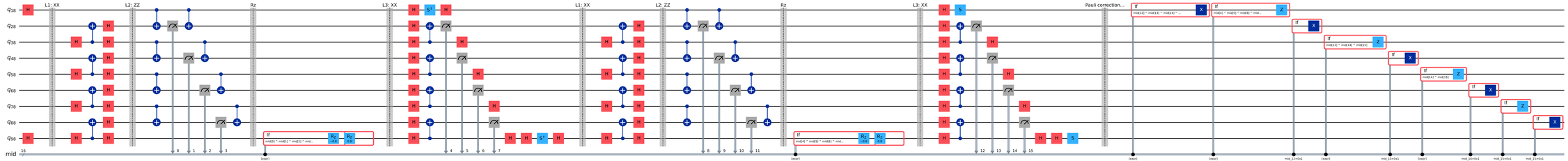

[ZIIXIYIZ_gapped] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Y2 X4 Z7
chain length 19, 9 ancillas, 6 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
3
7
6
5
4


[ZIIXIYIZ_gapped] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [10, 9, 11, 0, 12, 73, 2, 22] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
3
6
1
0
2
4
7
5
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

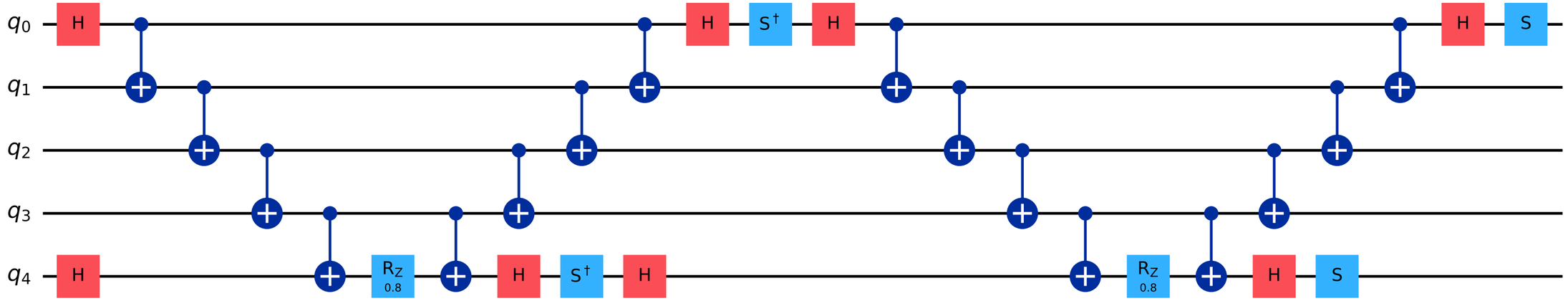
[ZIIXIYIZ_gapped] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 4, 2q gates 36



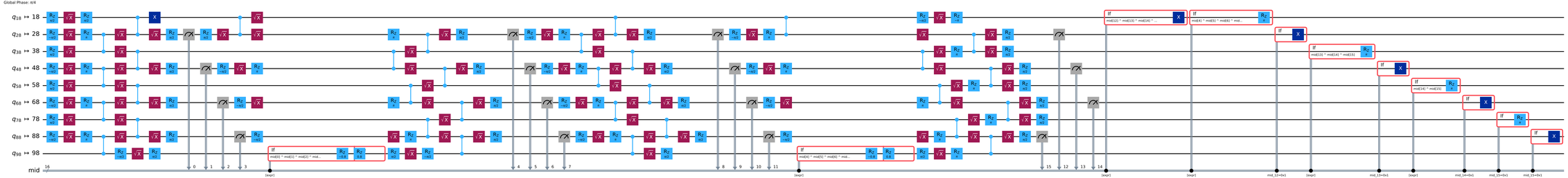
[XZZZX+YZZZY] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 8, 16 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



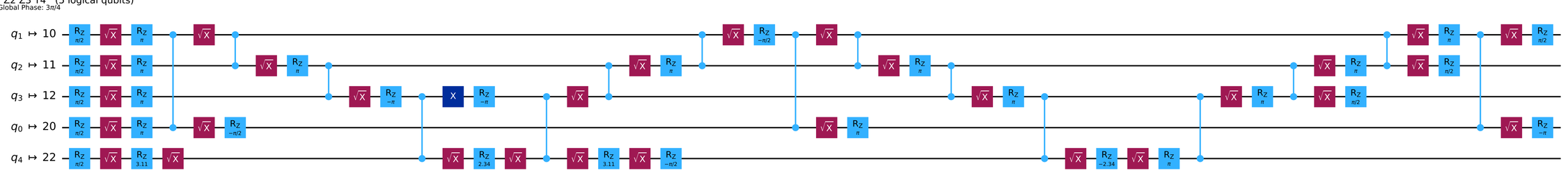
[XZZZX+YZZZY] CNOT-LADDER circuit (untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 16



[XZZZX+YZZZY] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: sx:73 rz:68 cz:32 measure:16 if_else:11 x:1)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 - Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 8, total depth 43

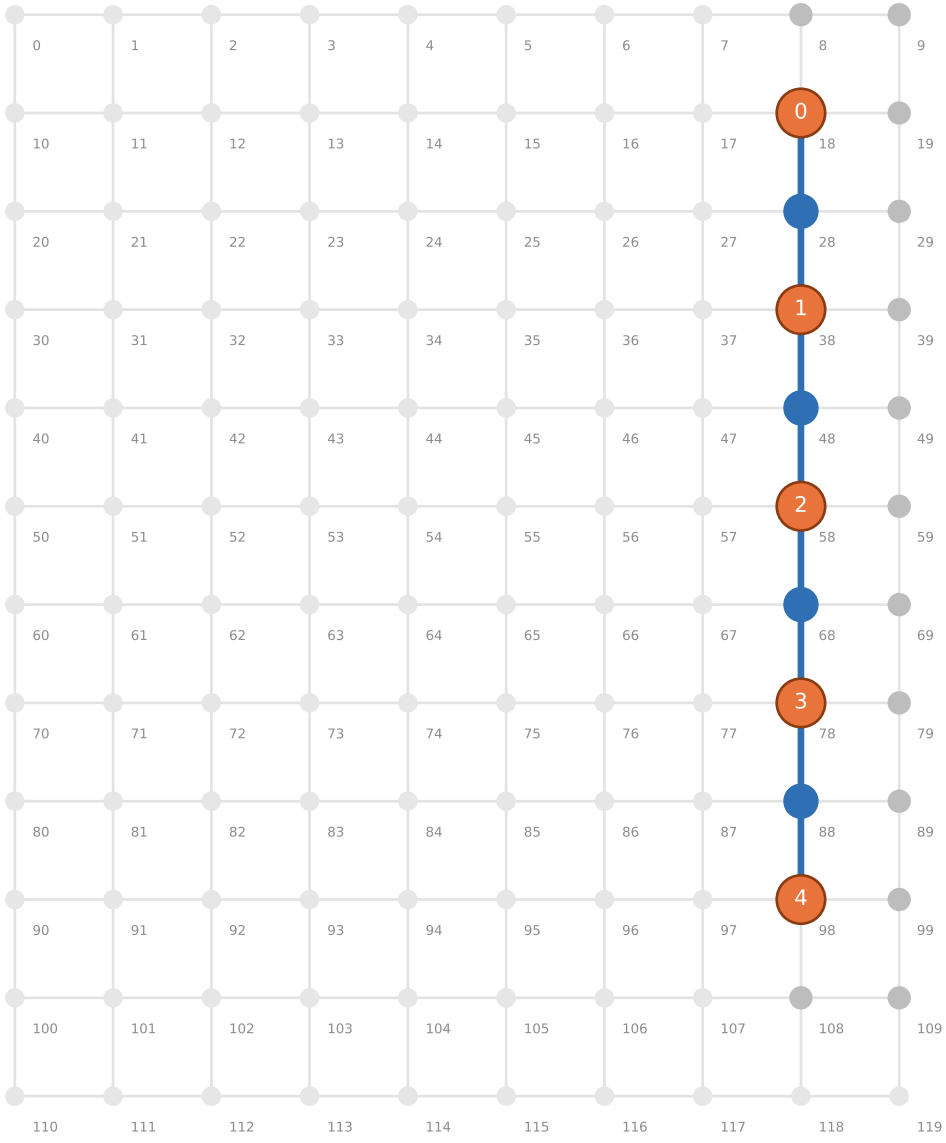


[XZZZX+YZZZY] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:31 sx:30 cz:15 x:1)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4 (5 logical qubits)
2q depth 15, total depth 48

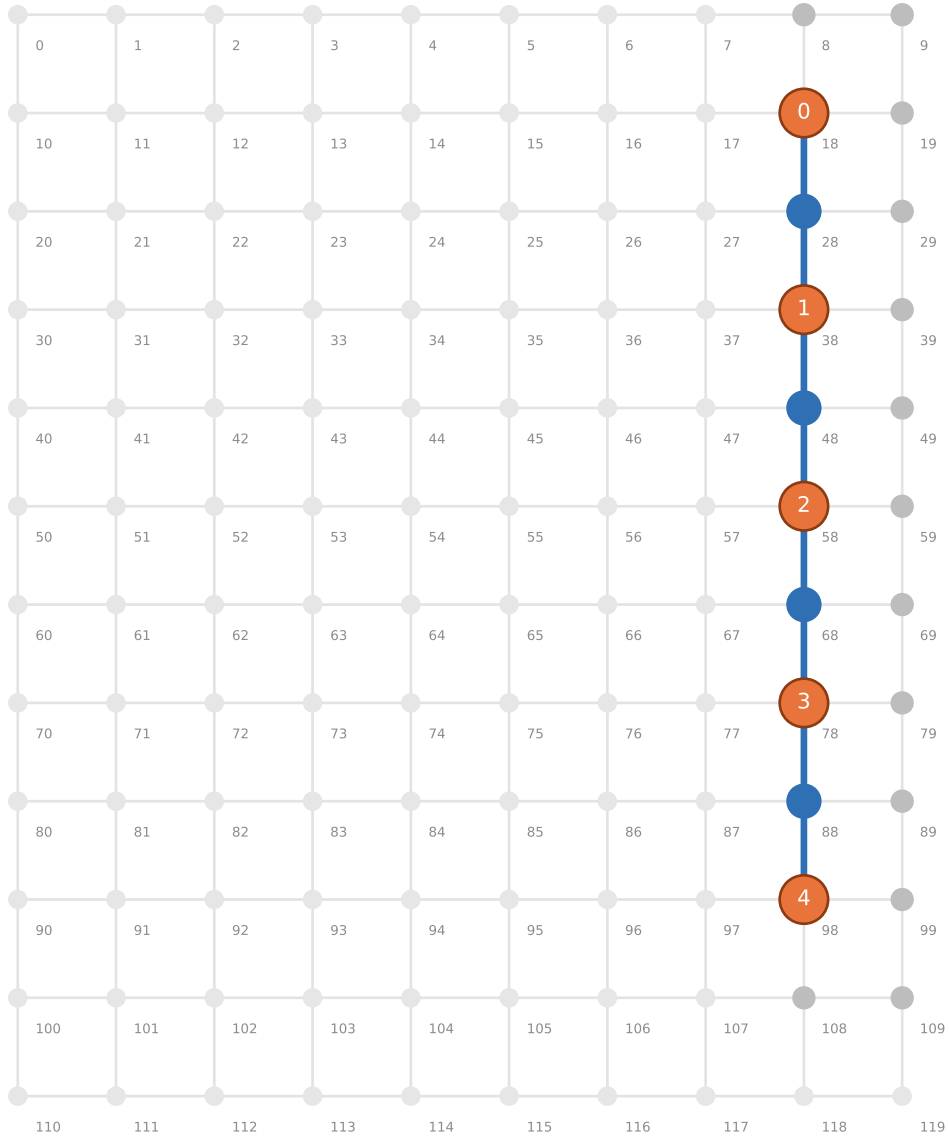


[XZZZX+YZZZY] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

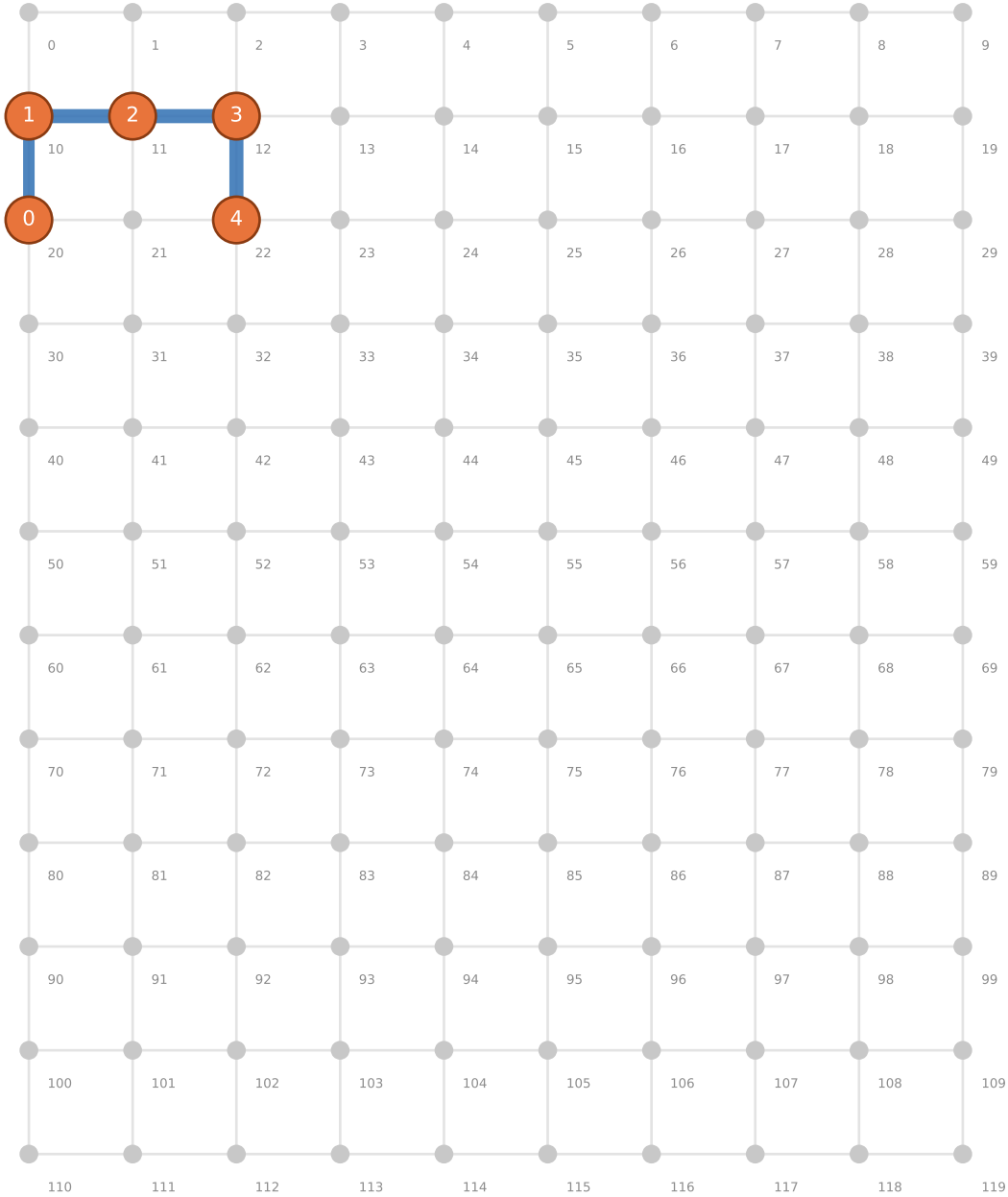
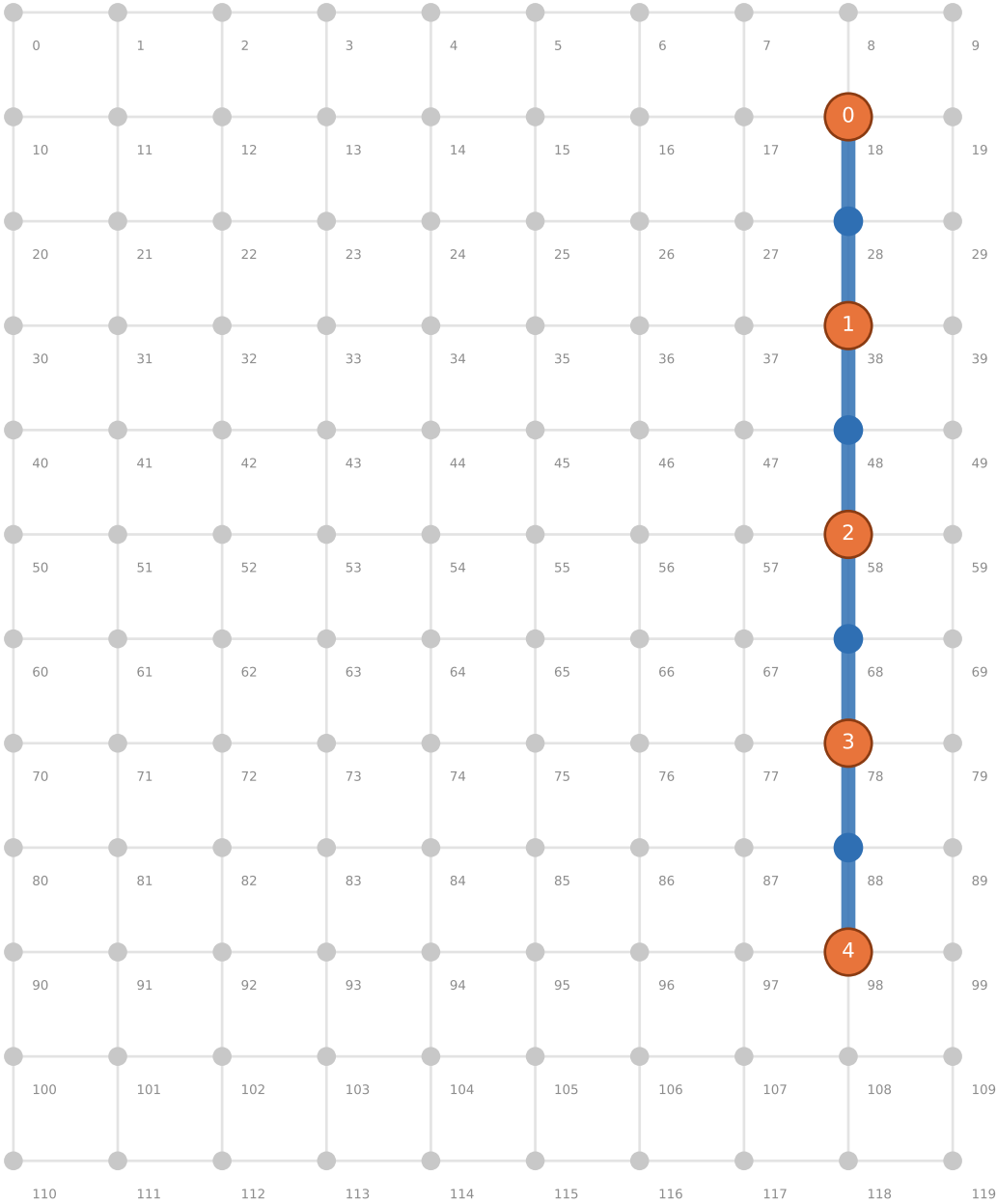
term 0: X0 Z1 Z2 Z3 X4
chain length 9, 4 ancillas, 0 |0> fillers



term 1: Y0 Z1 Z2 Z3 Y4
chain length 9, 4 ancillas, 0 |0> fillers



[XZZZX+YZZZY] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [20, 10, 11, 12, 22] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 15, 2q gates 15 (-1 vs. untranspiled from routing)



[XZZZX+YZZZY] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = ancillas/fillers, line width ∝ #2q gates
2q depth 8, 2q gates 32